

Deep Learning for Healthcare: Trends in Lightweight and Efficient Solutions

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Abstract

Deep learning has revolutionized modern healthcare, driving breakthroughs in medical imaging, disease detection, genomics, and personalized medicine. However, the substantial computational and memory demands of state-of-the-art models limit their adoption in real-world clinical environments, particularly in resource-constrained settings. This survey presents a comprehensive overview of lightweight and efficient deep learning approaches designed to address these challenges. We examine architectural innovations such as MobileNet, EfficientNet, ShuffleNet, GhostNet, and MobileViT, highlighting their design principles and the trade-offs between accuracy and computational complexity. We also review model compression techniques, including pruning, quantization, knowledge distillation, low-rank factorization, and neural architecture search, as complementary strategies for reducing computational costs while maintaining diagnostic performance. The paper explores the deployment of lightweight and efficient models across diverse healthcare domains, including disease detection and classification, remote monitoring, clinical decision support, reconstruction and restoration of medical images, genomics and precision medicine. Finally, we analyze emerging trends and identify open challenges related to model scalability, hardware–software co-optimization, and accuracy–efficiency balance. This work underscores the critical role of efficiency in advancing scalable, sustainable, and accessible AI-driven healthcare across global and clinical contexts.

Keywords:

Healthcare, Artificial Intelligence, Deep Learning, Lightweight and Efficient Models, Model Compression.

1. Introduction

In recent years, deep learning (DL) has emerged as a transformative force in healthcare, enabling breakthroughs in medical image analysis, disease detection, genomic research, and personalized medicine [1]. However, state-of-the-art DL models are often computationally intensive, memory-demanding, and energy-consuming, making them impractical for deployment in real-world environments where resources are limited [2]. Hospitals in rural or under-resourced areas, point-of-care devices, and wearables cannot afford the computational infrastructure required by conventional DL models. This limitation has created a pressing need for lightweight and efficient DL models that balance accuracy with deployability.

Lightweight and efficient DL models are designed to drastically reduce parameters, latency, and energy requirements while maintaining diagnostic accuracy [3]. Their efficiency enables deployment on mobile devices, bedside monitors, and even wearables, ensuring that intelligent healthcare systems are accessible at the patient’s side in real time. Such solutions are particularly valuable in remote and resource-constrained environments where traditional large-scale models are infeasible. For instance, rapid disease detection, high-quality medical image reconstruction, and continuous patient monitoring can now

be performed on compact devices, reducing reliance on cloud servers and expensive GPUs. This paradigm shift not only lowers costs but also reduces response times, which is critical for life-saving applications.

The importance of lightweight and efficient models extends beyond computational benefits. They pave the way for sustainable AI systems by minimizing energy consumption, which is vital given the increasing ecological footprint of large-scale DL models [4]. Furthermore, by democratizing access to advanced AI tools, these models contribute to equitable healthcare delivery across different socioeconomic and geographic contexts. This survey positions lightweight and efficient architectures and model compression techniques such as model pruning, quantization, knowledge distillation, low-rank factorization, and neural architecture search as central to the future of AI in healthcare, laying the foundation for scalable, robust, and universally accessible intelligent systems.

The increasing research attention to this domain is illustrated in Fig. 1, which shows the rapid growth of publications focusing on lightweight and efficient DL for healthcare from 2018 to mid-2025. This trend highlights the recognition of efficiency as a critical enabler for real-world clinical deployment. Fig. 2 provides a schematic overview of the organization of this survey, mapping the core approaches and applications covered in the following sections.

The remainder of this paper is structured as follows. Sec-

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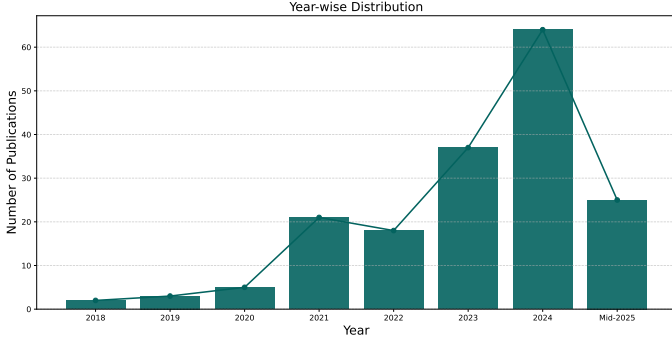


Figure 1: Number of papers published on lightweight and efficient DL in healthcare year-wise. The figure shows a clear upward trend in research activity from 2018 to 2024, indicating growing interest in lightweight and efficient techniques for DL in healthcare.

tion 2 reviews lightweight and efficient architectures by design, while Section 3 covers model compression techniques. Section 4 discusses applications of lightweight and efficient DL in disease detection and classification, and Section 5 highlights the role of lightweight and efficient DL in remote monitoring. Section 6 examines its role in clinical decision support systems, and Section 7 focuses on medical image reconstruction and restoration. Section 8 explores the role of lightweight and efficient DL in genomics and precision medicine. Section 9 provides a comprehensive discussion of key trends, trade-offs, and implications, while Section 10 outlines open challenges and future research directions. Finally, Section 11 concludes the paper.

2. Lightweight and efficient architectures by design

This section provides an overview of lightweight and efficient DL models designed for mobile and edge computing. It is structured into dedicated subsections, each focusing on a prominent family of lightweight and efficient architectures and their key technical innovations. Fig. 3 details the core features of each architecture.

2.1. MobileNet

MobileNet [5] is a family of lightweight and efficient convolutional neural networks (CNNs) designed for mobile devices and resource-constrained environments. The key innovation of MobileNet is the use of depthwise separable convolutions, which reduce the number of parameters and computations compared to standard convolutions while maintaining competitive accuracy.

MobileNetV1 [5] introduced depthwise separable convolutions, which factorize a standard convolution into two simpler operations: a depthwise convolution (DW) that applies a single filter to each input channel and a pointwise (1×1) convolution that combines the resulting outputs. It also introduced the concepts of a width multiplier and a resolution multiplier to further reduce the computational cost. MobileNetV2 [6] introduced inverted residual blocks with linear bottlenecks. These blocks expand the number of channels with a 1×1 convolution, apply DW, and then project the features back to a lower dimension

with another 1×1 convolution. Short-cut connections between bottleneck layers were also added to improve the information flow. MobileNetV3 [7] further optimized the architecture by incorporating squeeze-and-excitation (SE) modules for channel attention and utilizing neural architecture search (NAS) to find the optimal network structure to balance latency and accuracy. MobileNetV4 [8] is an evolution of its predecessors, merging inverted bottleneck, ConvNeXt, feed-forward network and an additional extra depth-wise variant.

2.2. EfficientNet

EfficientNet [9] is a CNN architecture and scaling method that uniformly scales all dimensions of depth, width, and resolution using a compound coefficient. This balanced scaling approach leads to better accuracy and efficiency compared to traditional scaling methods that focus on only one dimension.

The base model, EfficientNet-B0, was designed using neural architecture search and incorporates mobile inverted bottleneck convolutions (MBConv), which are variants of depthwise separable convolutions, along with SE modules and the Swish activation function. EfficientNet-Lite [10] is a more lightweight version of EfficientNet, specifically designed for mobile devices by using ReLU6 activation functions and removing SE blocks.

2.3. ShuffleNet

ShuffleNet [11] is an extremely computation-efficient CNN architecture designed specifically for mobile devices with limited computing power. It utilizes two main operations: pointwise group convolution and channel shuffle. Group convolution reduces computational cost by applying convolution only within specific channel groups, and the channel shuffle operation helps to mix features across different groups, improving the information flow.

ShuffleNetV1 [11] introduced pointwise group convolutions and channel shuffle operations, while ShuffleNetV2 [12] focused on optimizing for direct speed metrics rather than just floating-point operations per second (FLOPs). It introduced a channel split operation and moved the channel shuffle operation to a different location within the block to improve efficiency by reducing memory access cost (MAC).

2.4. SqueezeNet

SqueezeNet [13] is a small CNN architecture designed as a more compact replacement for larger networks like AlexNet. It achieves AlexNet-level accuracy on ImageNet with significantly fewer parameters. The basic building block of SqueezeNet is “fire module”, which consists of a squeeze layer followed by an “expand” layer that has a mix of 1×1 and 3×3 convolutional filters. The key design strategies of SqueezeNet include (1) using 1×1 filters instead of 3×3 filters to reduce the number of parameters; (2) decreasing the number of input channels to 3×3 filters. This is done using squeeze layers, which consist only of 1×1 convolutional filters, and (3) downsampling late in the network to have larger activation maps in the convolution layers, which can lead to higher accuracy with the same number of parameters.

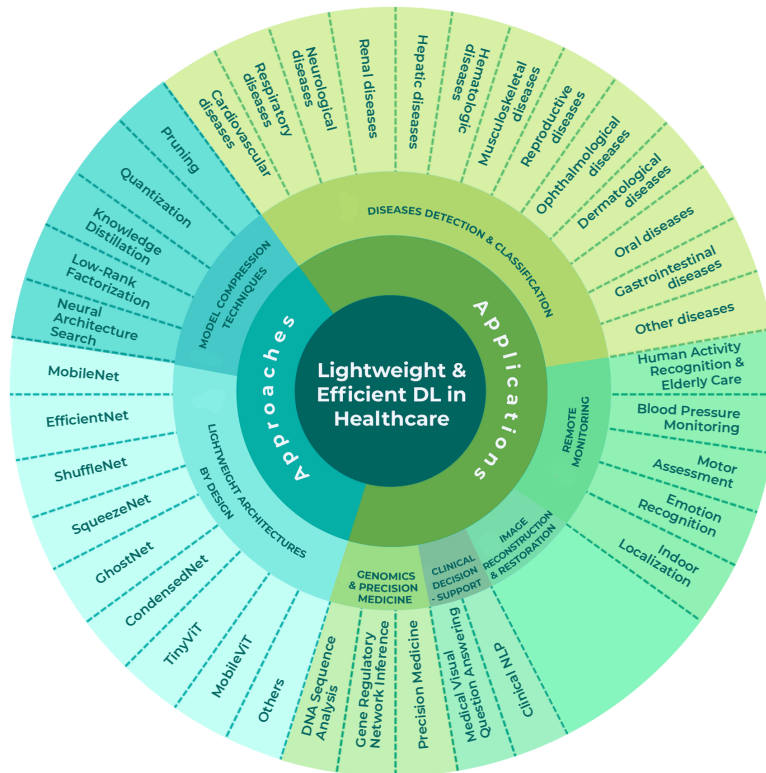


Figure 2: Schematic overview of lightweight and efficient DL for healthcare, focusing on approaches and applications. Approaches cover lightweight architectures by design and model compression techniques. Applications include disease detection & classification (across organ systems), remote monitoring, image reconstruction & restoration, clinical decision-support, and genomics & precision medicine.

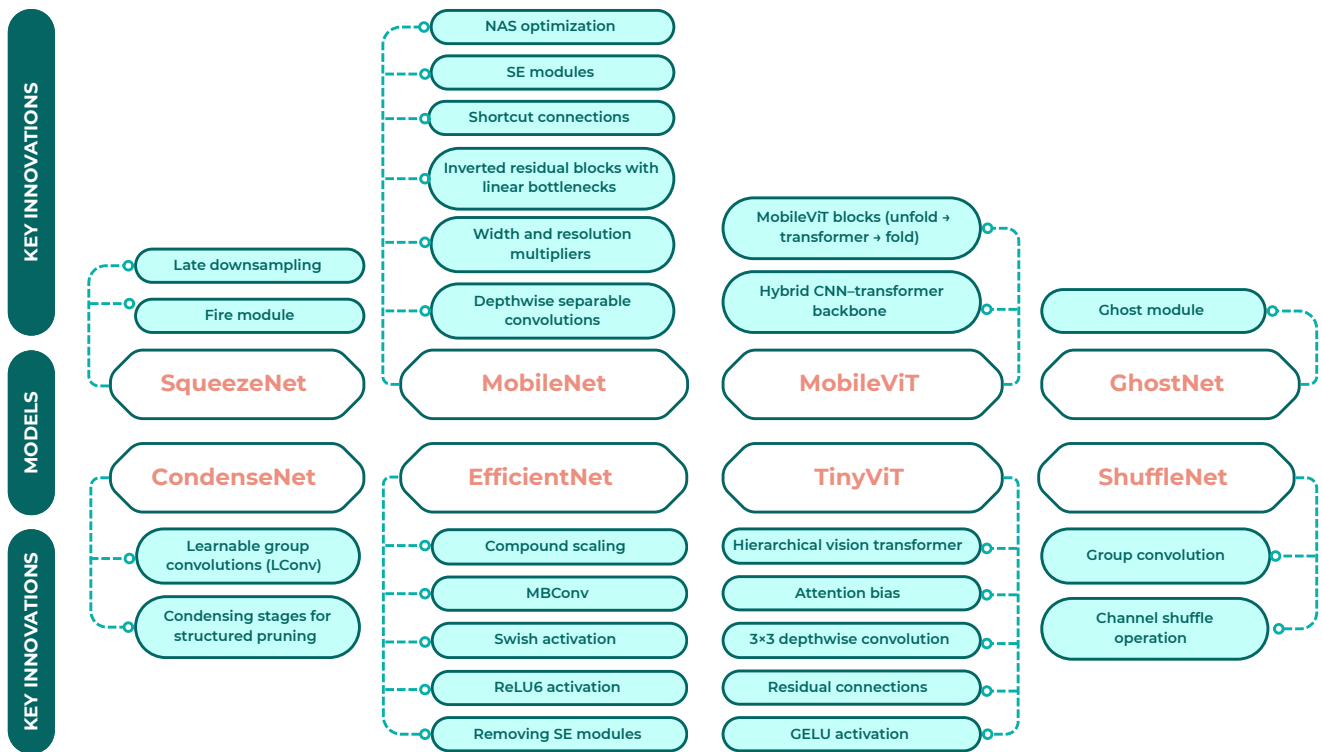


Figure 3: Overview of key architectural features and innovations across eight different lightweight and efficient DL models.

2.5. GhostNet

GhostNet [14] is a lightweight neural network architecture that aims to generate more features from cheap operations. It introduced the “ghost module” to generate more feature maps using a small set of original feature maps. For each original convolutional layer, a ghost module first uses a few cheap linear operations to generate many ghost feature maps. These ghost feature maps are then combined with the original feature maps to serve as the output of the ghost module. This approach significantly reduces the number of parameters and computational cost while maintaining comparable performance.

2.6. CondenseNet

CondenseNet [15] streamlines DenseNet for mobile efficiency by introducing a learnable group convolution (LConv). During training, the 1×1 layers are sparsified and iteratively pruned in “condensing” stages so that inputs are clustered into groups, and after training, an index/reordering step converts them into regular group convolutions for fast inference, leading to a more efficient network compared to standard CNNs.

2.7. TinyViT

TinyViT [16] is a lightweight, general-purpose hierarchical vision transformer (ViT) with four stages designed to be mobile-friendly. TinyViT incorporates attention biases and a 3×3 DW between the attention and multi-layer perceptron (MLP) layers to capture local information efficiently. The architecture also uses residual connections, GELU activation functions, and a combination of BatchNorm and LayerNorm normalization layers. The model is scaled down using a set of contraction factors applied to the embedded dimensions, number of blocks, window sizes, and expansion ratios.

2.8. MobileViT

MobileViT [17] is a lightweight hybrid backbone that combines CNNs’ efficiency and inductive biases with transformers’ ability to model global context for mobile vision. It alternates MobileNetV2-style inverted residual convolutional blocks with MobileViT blocks: local $3\times 3/1\times 1$ convolutions encode features, which are then unfolded into non-overlapping patches, processed by a transformer to capture long-range dependencies, and finally folded back and fused via convolutions.

2.9. Additional lightweight architectures

MnasNet [18] was discovered using neural architecture search and specifically designed for mobile vision tasks by balancing accuracy and latency. SqueezeNext [19] is an improvement over SqueezeNet that maintains the concept of ResNet shortcuts and decomposes 3×3 kernels into two low-rank kernels (3×1 and 1×3), further reducing the parameters. ShiftNet [20] proposed a shift operation as an efficient alternative to depthwise separable convolution, aiming to reduce memory movement. AdderNet [21] replaced multiplication operations with addition operations to reduce computation requirements.

MobileOne [22] is a mobile-first CNN that trains with over-parameterized, multi-branch depthwise–pointwise blocks (including reparameterizable skip/ 1×1 branches) and then structurally reparameterizes them into a simple, branch-free stack at inference to cut memory-access costs and enable wider layers.

2.10. Key trade-off highlights

Table 1 provides a comprehensive comparison of several lightweight neural network architectures designed for efficient deployment on resource-constrained devices like mobile phones and edge computing systems. It highlights model complexity and accuracy on ImageNet for popular architectures. The table reveals inherent trade-offs between accuracy and efficiency. For instance, while SqueezeNet achieves an accuracy of 57.5%, which is significantly lower than other models like TinyViT (84.8%), the model size is only 1.2M parameters. Notably, newer hybrid architectures like MobileViT and TinyViT demonstrate how combining convolutional and transformer-based approaches can achieve high accuracy while maintaining relatively low computational costs.

The comparison also illustrates the evolution of efficiency techniques in neural network design, from early methods like depthwise separable convolutions (MobileNet) and fire modules (SqueezeNet) to advanced strategies such as compound scaling (EfficientNet) and learned group convolution (CondenseNet). Table 1 serves as a valuable reference for selecting an appropriate architecture based on specific deployment constraints, whether prioritizing minimal model size or high accuracy.

Fig. 4 illustrates the key trade-offs between accuracy, computational complexity, and parameter count, which can be categorized as follows: (1) *Efficiency Leaders*: GhostNet and ShuffleNet V1/V2 stand out for their exceptional efficiency, achieving over 71% accuracy with very low computational complexity (under 300M FLOPs) and a small parameter count (3.5M); (2) *Accuracy Leaders*: EfficientNet-B7/V2 and TinyViT achieve the highest top-1 accuracy on ImageNet (over 84%), but this comes at a steep cost, requiring orders of magnitude more computation (tens of billions of FLOPs/MACs) and parameters (21M+); (3) *The Best Balance*: MobileNetV3 strikes a strong middle ground, offering high accuracy (75.2%) with low complexity (0.219G MACs) and a moderate parameter count (5.4M), making it a popular choice for on-device deployment; (4) *Parameter Efficiency*: SqueezeNet and CondenseNet are optimized for the smallest possible parameter count (1.2M and 2.9M, respectively), but this focus comes with a significant compromise on accuracy (57.5% and 71.0%); and (5) *The Modern Hybrid*: MobileViT demonstrates that combining CNNs and transformers can yield high accuracy (78.4%) with a relatively low parameter count (5.6M), though its computational complexity is higher than standalone CNNs.

3. Model compression techniques

DL models often contain millions of parameters, making them computationally expensive and difficult to deploy in

Table 1: Performance comparison of efficient neural network architectures for mobile and edge devices. The table shows parameter counts, computational complexity metrics (MACs/FLOPs), and ImageNet accuracy across multiple model families, highlighting the trade-offs between model size and computational requirements.

Architecture	Parameters	Complexity	Accuracy (ImageNet)
MobileNet	V1: 4.2M, V2: 6.9M, V3: 5.4M, V4: 3.8M	MACs → V1: 0.569G, V2: 0.585G, V3: 0.219G, V4: 0.200G	V1: 70.6%, V2: 74.7%, V3: 75.2%, V4: 73.8%
EfficientNet	B7: 66M, V2-Large: 120M, Lite4: 13M	FLOPs → B7: 37B, V2-L: 53B, Lite4: 2.64B	B7: 84.7%, V2-L: 85.7%, Lite4: 81.5%
ShuffleNet	V1: 3.4M, V2: 3.5M	FLOPs → V1: 292M, V2: 299M	V1: 71.5%, V2: 72.6%
SqueezeNet	1.2M	MACs → 0.837G	57.5%
GhostNet	5.2M	FLOPs → 141M	73.9%
CondenseNet	2.9M	FLOPs → 274M	71.0%
TinyViT	21M	MACs → 4.3G	84.8%
MobileViT	5.6M	FLOPs → 2.3M	78.4%

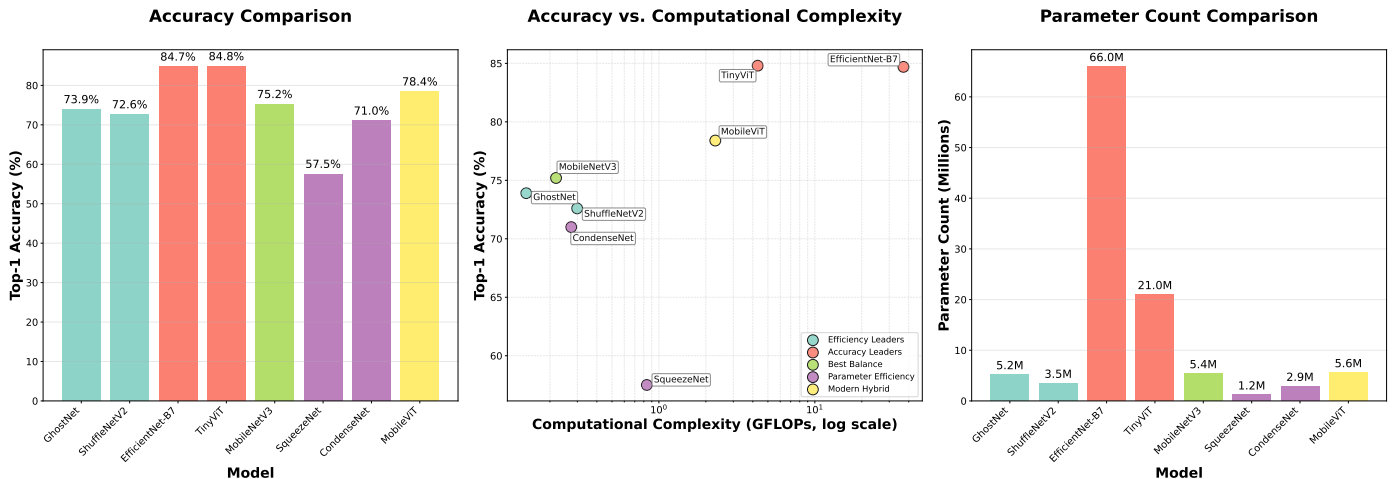


Figure 4: Comparison of accuracy, computational complexity, and parameter efficiency across eight different lightweight DL models. The left plot shows top-1 accuracy, the center plot shows the trade-off between accuracy and computational complexity, identifying EfficientNet-B7 and TinyViT as accuracy leaders, while GhostNet, ShuffleNetV2, and MobileNetV3 demonstrate strong efficiency. The right plot compares parameter counts. SqueezeNet and CondenseNet exhibit the smallest model sizes, whereas EfficientNet-B7 and TinyViT have significantly larger capacities, reflecting the trade-off between compactness and accuracy.

resource-constrained environments. Model compression techniques aim to reduce this complexity while preserving accuracy, using methods such as pruning, quantization, knowledge distillation, low-rank factorization, and neural architecture search. These approaches enable faster inference, lower memory usage, and more efficient deployment across a wide range of hardware platforms. Fig. 5 gives an overview of model compression techniques which will be discussed below.

3.1. Pruning

Pruning is a model compression technique that removes redundant or less important parameters (such as weights, neurons, filters, or even entire layers) from a neural network to reduce its size, computational cost, and memory use while maintaining accuracy. The key dimensions of pruning methods include what is pruned, when pruning occurs, and how importance is assessed. If done carefully, pruning can yield large savings in inference time and storage with minimal loss in performance; if done too aggressively, it risks hurting generalization or requiring expensive fine-tuning [23].

3.1.1. Types of pruning

There are two types of pruning: unstructured and structured. *Unstructured pruning* removes individual weights (matrix elements) anywhere in the network to reduce the total number of

non-zero weights [23]. It is formally posed as enforcing an ℓ_0 constraint on the weight set. It is commonly implemented via binary masks applied to each weight, setting unneeded weights to zero during training or standalone inference [24]. Unstructured pruning achieves high sparsity and is flexible without the need of many architectural changes. On the other hand, it may cause no or little latency improvement on most GPUs/CPUs or slow training due to masking overhead.

Structured pruning removes entire units from the network, such as neurons, channels, filters, blocks, attention heads, or even full layers, rather than individual weights, resulting in a model with dense, regular architecture that standard hardware and libraries can exploit [23]. The advantages of structured pruning include real, automatic speedups and memory savings during inference and training on GPUs or CPUs, without needing special sparse libraries or hardware. It also simplifies deployment pipelines since pruned models align with conventional layer structures. On the other hand, disadvantages include lower compression granularity and, therefore, less overall parameter reduction compared to unstructured pruning. Coarse pruning units may lead to greater accuracy degradation, making methods like norm- or sensitivity-based scoring and careful fine-tuning essential [23].

Recent structured pruning efforts have shifted focus from

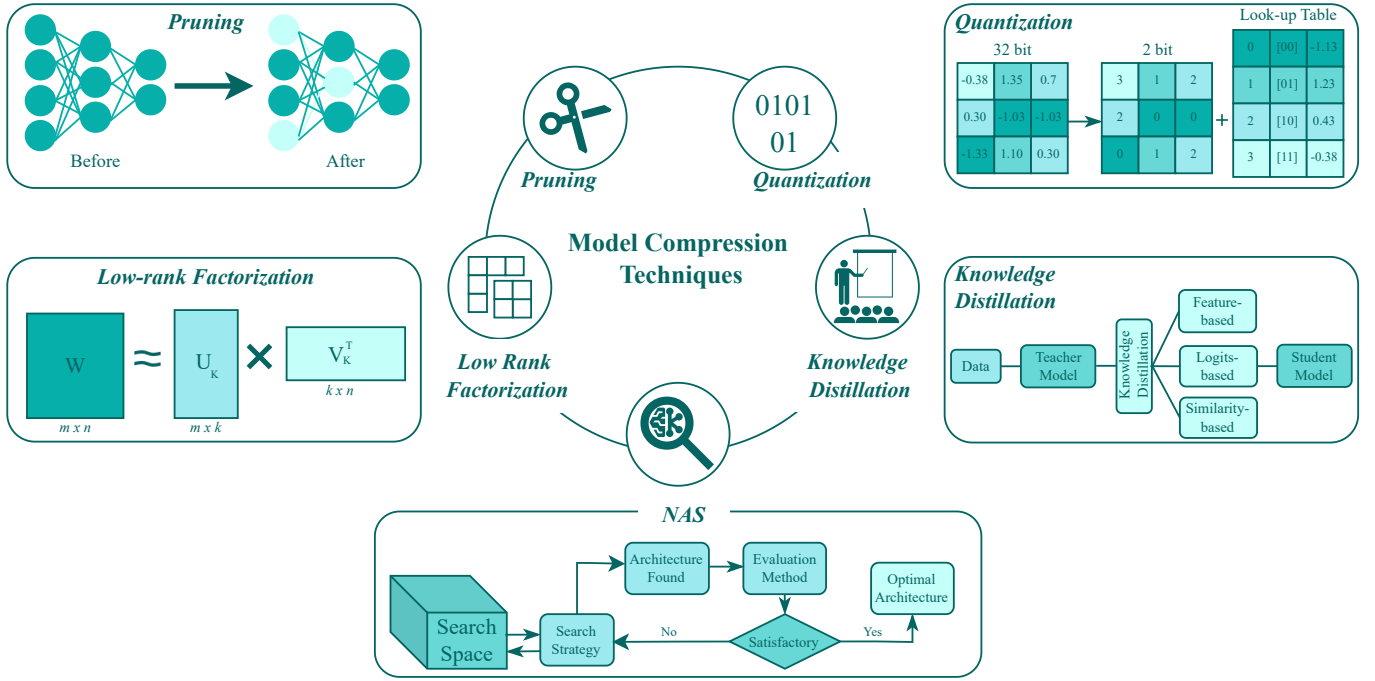


Figure 5: Overview of model compression techniques detailed in Section 3. Pruning removes redundant parameters within a network. Low-rank factorization leverages the low-dimensional redundancy in weight matrices. Quantization converts continuous weights and activations into a discrete set of representations. Knowledge distillation transfers knowledge from a teacher model to a student model. Neural architecture search automates the network design considering more efficient ways.

pruning individual filters or neurons to eliminating larger structural units, such as layers, blocks, or channel groups, for more effective deployment acceleration. For instance, [25] found that removing entire residual layers or transformer blocks yields significantly better latency and throughput gains than pruning individual filters, demonstrating that drop-in layer/block pruning can cut over 75% of FLOPs with little to no accuracy loss.

3.1.2. Key pruning techniques

Some of the pruning techniques are: magnitude pruning, one-shot pruning, iterative pruning, train-time pruning, and post-training pruning.

Magnitude pruning is the simplest and most widely used unstructured pruning method. Each weight is scored by its absolute value, and those closest to zero are removed, either all at once (one-shot) or gradually (iterative) during training, since small-magnitude weights are assumed to have minimal impact on model performance [23]. Iterative magnitude pruning, performed over multiple epochs, is often more effective in practice and tends to preserve accuracy better than one-shot pruning. The main advantages are simplicity, easy implementation, and strong empirical results even on large models. However, magnitude pruning is unaware of a weight’s actual influence on loss, which can lead to suboptimal selections. Thus, there are enhancements like gradient magnitude pruning [26] or more advanced methods like magnitude attention-based dynamic pruning [27], which use the magnitude scoring throughout training and deliver better results.

One-shot pruning refers to removing a set of parameters or

structures from a neural network in a single pruning step, rather than through multiple iterative prune–fine-tune cycles [23]. Typically, the process involves identifying low-importance weights in a pretrained model, pruning them all at once, and then performing a single round of fine-tuning to recover any lost accuracy. This approach reduces complexity and computational cost compared to iterative methods, making it faster and simpler to implement. However, because the pruning decision is made in one pass, it often leads to more aggressive and less nuanced removal of weights, which can result in larger accuracy drops than iterative approaches. Despite that, one-shot pruning is especially useful for quick compression and deployment scenarios where multiple retraining rounds are impractical.

In contrast to one-shot pruning, *iterative pruning* involves gradually removing small subsets of weights or structures, typically in a sequence of prune–fine-tune cycles, rather than all at once. Starting from a trained model, you remove a fraction of the least important weights (e.g., those with the smallest magnitude) and then fine-tune the model to recover accuracy. This prune and retrain cycle is repeated, while increasing sparsity each time, until the desired compression level is achieved. By distributing pruning across multiple stages, the network better adapts, often yielding higher accuracy at high sparsity levels compared to one-shot methods [28]. However, the downside is added computational cost from repeated training rounds, which iterative variants like ICE-Pruning aim to reduce via optimized fine-tuning schedules and efficient step choices [29].

Train-time pruning integrates the pruning process directly into model training, rather than applying it after training or be-

fore initialization [23]. As training proceeds, low-importance weights, filters, or neurons are dynamically removed, either based on magnitude or learned saliency, enabling the network to flexibly adapt its architecture. This approach, also known as dynamic sparse training, maintains a constant sparsity budget while jointly optimizing weights and structure, avoiding the need for full dense training followed by pruning [30].

The major advantages of train-time pruning include reduced training compute and memory requirements, faster deployment, and more efficient sparse models. However, challenges remain in choosing effective pruning and growth criteria, managing training stability, and coping with the lack of sparse-aware hardware and software support during training [23].

Post-training pruning refers to applying pruning after a model has been fully trained, usually by ranking and removing low-importance weights or structures, and then optionally fine-tuning the resulting sparse network to restore accuracy [23]. This approach enables model compression without modifying the original training pipeline, making it practical for compressing pre-existing models. Advantages include immediate size reduction and suitability for scenarios where retraining from scratch is not feasible. However, it leads to a nontrivial accuracy loss, especially at high sparsity, which may require a dedicated fine-tuning step or layer-wise calibration to recover accuracy [31, 32, 33].

3.2. Quantization

Quantization is a core model compression method that transforms continuous (usually 32-bit floating-point) weights and activations into a discrete set of lower-bit representations. This reduces model size, memory bandwidth, and arithmetic costs, enabling efficient deployment on edge devices and faster inference [34]. Quantization is a non-differentiable technique, where a continuous value $x \in \mathbb{R}$ is mapped to a discrete set of values, as follows:

$$q(x) = \text{round}\left(\frac{x}{\Delta}\right) \cdot \Delta, \quad (1)$$

where Δ is the quantization step size. Quantization considers several aspects: precision, quantizer design, symmetry in mapping, and training strategy. Below is a summary of each aspect.

3.2.1. Precision

There are three main types of quantization: binary, ternary, and low-bit. To meet the diverse computational and deployment demands across edge devices, cloud services, and large-scale AI systems, researchers and developers have explored these three types, each offering distinct trade-offs in precision, efficiency, and hardware compatibility. Binary quantization reduces weights to two values, $\{-1, +1\}$. This yields maximum compression ($\sim 32\times$ compared to FP32) and inferential speed, and enables ultra-efficient bitwise operations, which is ideal for resource-constrained or embedded systems such as microcontrollers and edge devices. However, the extreme loss of precision often leads to a significant accuracy drop (around 10%), making it unsuitable for complex tasks without careful retraining [35, 34].

Given the two values in binary quantization, it becomes challenging to perform all the required neural network operations. Thus, researchers introduced ternary quantization by adding a zero state $\{-1, 0, +1\}$, improving expressiveness and incorporating sparsity. This way, ternary quantization has a compression rate of $\sim 16\times$ compared to FP32, making it less lightweight than binary, but with lower accuracy losses. [35, 36].

Lastly, low-bit quantization techniques, such as int4 and int8, are widely adopted because they are simpler to deploy and offer a balance between efficiency and performance compared to ternary and binary [37, 35]. These methods are dominant in mobile, cloud, and large language models inference. Regardless, lower bits (< 4) can still entail a noticeable accuracy loss and offer less extreme compression than binary/ternary. For instance, LLM.int8() [38] demonstrates that 8-bit weight quantization can support inference of 175B-parameter transformers “without any performance degradation”.

3.2.2. Quantizer design

The mapping of the full-precision values to the compressed values occurs through mainly two design options: uniform and non-uniform. Uniform quantization divides the full range of floating-point values into equally sized intervals and maps each value to the nearest integer level, making it hardware-friendly and straightforward to implement [35, 34]. In contrast, non-uniform quantization uses variable-sized intervals, based on statistical distributions or learned clustering, to allocate more precision where values are dense, offering finer granularity but with increased implementation complexity and reduced hardware compatibility [37]. Due to the latter reasons, uniform quantization remains widely used.

3.2.3. Symmetry in mapping

In uniform quantization, one chooses between symmetric and asymmetric mappings based on the distribution of weights or activations. Let $x \in \mathbb{R}$ be a floating-point value, $b \in \mathbb{N}$ the bit-width, and $q_{\min}, q_{\max} \in \mathbb{Z}$ the integer bounds. Let $\alpha, \beta \in \mathbb{R}$ denote the floating-point clipping range (minimum and maximum), $s > 0$ the scale, $z \in \mathbb{Z}$ the zero-point (offset), $x_{\text{int}} \in \{q_{\min}, \dots, q_{\max}\}$ the quantized integer, and $\hat{x} \in \mathbb{R}$ the de-quantized value.

General (asymmetric) uniform quantization uses an affine map from $[\alpha, \beta]$ to $[q_{\min}, q_{\max}]$:

$$s = \frac{\beta - \alpha}{q_{\max} - q_{\min}}, \quad (2)$$

$$z = \text{round}\left(-\frac{\alpha}{s}\right) + q_{\min}, \quad (3)$$

$$x_{\text{int}} = \text{clamp}\left(\text{round}\left(\frac{x}{s} + z\right), q_{\min}, q_{\max}\right), \quad (4)$$

$$\hat{x} = s(x_{\text{int}} - z). \quad (5)$$

Symmetric quantization assumes the clipped float range is centered at zero, i.e., $\alpha = -\beta$ and fixes the zero-point to $z = 0$ (for signed integers).

Symmetric mapping is hardware-friendly and common for weight tensors or zero-mean activations. However, for highly skewed/non-negative activations, asymmetric mapping with

$\alpha \neq \beta$ and $z \neq 0$ yields a tighter fit to the data distribution and typically lower quantization error. While this improves representational accuracy, especially for non-uniform data, it increases implementation complexity due to additional computation for zero-point alignment. As a result, the choice between symmetric and asymmetric quantization should be guided by the data characteristics: symmetric is computationally efficient and suited for balanced distributions, while asymmetric is more accurate for skewed data but incurs greater overhead.

3.2.4. Training strategy: Post-training quantization vs. quantization-aware training

Quantization-aware training (QAT) is a fine-tuning approach used to mitigate the loss in model performance that typically follows quantization, especially in low-bit settings. It involves retraining the model while simulating quantized weights and activations during the forward and backward passes, allowing the model to adapt to the effects of quantization [35, 34]. A critical component of QAT is handling the non-differentiability of the quantization operation, which is commonly addressed using the Straight-Through Estimator (STE). STE approximates the gradient of the quantization function by treating it as an identity function during backpropagation. Despite its simplicity, STE has been shown to work effectively even in extreme low-bit regimes such as binary quantization, and its theoretical justification has been supported by studies like [39]. Alternatively, non-STE techniques like ProxQuant [40], AdaRound [41], and QIT [42], offer promising directions, though they introduce additional tuning complexity.

While QAT consistently achieves high accuracy after quantization and is especially valuable for long-lived deployment scenarios, its primary disadvantage is the computational overhead as it requires hundreds of training epochs. This makes it less suitable for models with short deployment lifespans or limited training budgets [35, 34]. To mitigate these risks, recent QAT methods are shifting from full retraining to parameter-efficient, blockwise, and low-rank adaptation techniques. These approaches drastically reduce compute and memory overhead while maintaining, or even improving, quantization performance, especially implemented on LLMs.

On the other hand, *post-training quantization (PTQ)* is a technique used to quantize a pre-trained neural network without retraining. It is especially valuable when training data is unavailable, sensitive, or when fast deployment is required. PTQ typically uses a small set of calibration data, or even no data in extreme zero-shot scenarios, to determine quantization parameters such as scale and zero-point. Recently, many leading quantization studies focus on PTQ, a trend driven by its efficiency and practicality over retraining-based QAT [34, 35]. PTQ methods also respect data privacy and dramatically reduce deployment time, making them highly appealing for real-world settings. From the recent studies, FIMA-Q [43] introduces a quantization loss for ViTs by leveraging an efficient diagonal and low-rank Fisher information matrix. Unlike earlier methods that used simple Hessian or diagonal approximations, FIMA-Q acknowledges the interdependencies among output units and

connects this calculation to KL divergence enabling context-aware clipping and reconstruction.

3.3. Knowledge distillation

While pruning and quantization act directly on a model’s parameters, these techniques can struggle when aggressively applied, as they often introduce accuracy degradation that is difficult to recover. Knowledge distillation (KD), on the other hand, does not perform direct parameter manipulation and transfers the learned representations of a large teacher model into a smaller student model. This allows the student to inherit high-level generalization ability from the teacher while maintaining a much lower parameter count [44, 45].

3.3.1. Sources of knowledge

In KD, there are different sources where the student learns from the teacher. The first is the *logit-based distillation*. Logit-based distillation transfers knowledge from a teacher to a student model by aligning their output logits, which contain predictive information [45]. Early approaches focused on applying softmax with temperature to produce soft labels and using KL divergence as the loss [46]. Over time, many methods have refined this idea by normalizing, smoothing, or perturbing logits, and by decoupling the distillation loss to better capture class distributions [47, 48, 49]. While logit distillation is simple and effective, it is limited to final-layer outputs and supervised tasks, often missing insights from intermediate representations. This creates the need to distill intermediate features and similarities from the teacher.

The second source is *feature-based distillation*, which transfers knowledge from the intermediate layers of a network. These layers capture multi-level representations that provide information leading to the final output, making them particularly valuable for representation learning tasks. The idea, first introduced in [50], is to align the features of the student with those of the teacher by minimizing their differences. Over time, methods have emerged that match attention maps [51], neuron selectivity patterns [52], or activation boundaries [53]. Others distill pre-activation and channel-wise features [54, 55] or align feature maps across layers [56, 57]. More recent work focuses on transforming or masking feature maps to better align with the teacher [58], or applying advanced mechanisms such as diffusion models [59] and attention modules [60]. In general, feature-based distillation provides richer knowledge than logits alone, but its effectiveness depends on how features and layers are selected. It gets more challenging when teacher and student networks have different architectures, since mismatched feature sizes or representation gaps can reduce performance.

Unlike logit or feature distillation, *similarity-based methods* transfer knowledge by focusing on the relationships between features or instances rather than their exact values. Early work such as FSP [61] computed inner products between teacher features and minimized their squared differences as a cost function, while [62] proposed distilling relative dissimilarities across representations. Later studies extended this idea to measure similarities among logits [63] and across multiple teachers [64].

Broadly, methods differ by the level of similarity they consider: instance/sample-level [65], feature/channel-level [66, 67], or class-level [68, 69].

Together, these methods equip the student with a higher-order understanding of structural relationships in the teacher network, enabling richer knowledge transfer than using raw logits or features alone.

3.3.2. Distillation schemes

A key design choice in KD is the training scheme, which determines how the teacher and student interact. Based on architecture and training mode, schemes are generally divided into offline, online, and self-distillation. In the offline setting, the teacher is a larger pre-trained model; in the online setting, teacher and student are trained together; and in self-distillation, both roles are played by the same network. Each approach serves a different purpose: offline is preferred when a strong pre-trained teacher is available, online is suitable when such a teacher is not accessible, and self-distillation is important in self-supervised contexts without labeled data [44, 45].

The *offline approach*, first introduced in [46], is the most common and consists of pre-training a large teacher model and then transferring its knowledge to the student while the teacher remains fixed. This method is widely adopted because of its simplicity, but it has limitations: it requires significant computational resources to load the teacher, and the student’s performance is inherently constrained by the teacher’s capabilities. Furthermore, any biases from the teacher may also be inherited by the student [47, 69].

Unlike offline distillation, *online methods* train teacher and student simultaneously in a single stage. Early work [70] showed that using a group of students as a virtual teacher removes the need for a pre-trained model, while [71, 72] applied ensembles of multi-branch logits for on-the-fly guidance.

Self-distillation is a variant of online distillation where there is no external teacher; instead, the same model supervises itself [73, 74, 75, 76]. Typically, a teacher version of the model is first trained and then used to guide a new model initialized with a different seed [73], or to supervise subsequent epochs [75]. Other methods partition the network into sections [74], add shallow classifiers for layer-wise supervision [77], or provide theoretical insights connecting self-distillation to label smoothing [78, 79].

3.3.3. Distillation algorithms

Aside from the previously mentioned schemes and sources of knowledge, there are several algorithms applied in the context of KD. We will briefly introduce the following algorithms: 1) attention-based, 2) adversarial, 3) multi-teacher, 4) cross-modal, 5) graph-based, and 6) adaptive.

Starting with (1) *attention-based algorithms*, they transfer rich information from the teacher to the student by leveraging attention maps, which highlight the most important regions of the teacher’s feature space [51]. Instead of matching predictions directly, this approach enables the student to focus on mimicking the critical parts of the input, improving efficiency and interpretability.

Adversarial distillation (2) integrates knowledge transfer into an adversarial framework, inspired by the training dynamics of Generative Adversarial Networks (GANs) [80]. This approach enhances robustness by either generating synthetic training data when the original data is unavailable, introducing a discriminator to better align teacher and student outputs, or applying adversarial training to minimize discrepancies under perturbed samples [81, 82, 83, 84, 85].

Multi-teacher distillation (3) extends the traditional teacher–student setup by training the student with guidance from several teachers instead of one. The idea is that combining knowledge from diverse teachers strengthens the student’s generalization ability, but it also raises the challenge of how to effectively aggregate and balance these different sources [46, 86]. To address this, some works adaptively weight or select teachers so that the student can learn without being overwhelmed by conflicting or redundant signals [87, 88].

Cross-modal distillation (4) tackles the problem where one modality has rich annotations while another has few. By transferring supervision from the stronger modality to the weaker one, the student can improve learning even if that modality is absent at inference [89]. This setup enables effective alignment between modalities, ensuring that the student can benefit from complementary information across domains [90, 91].

Graph-based distillation (5) leverages graph structures to transfer knowledge by modeling relationships between features or instances as vertices and edges, rather than just comparing them in pairs. This allows capturing both local and global dependencies, making the transfer of knowledge richer and more structured, especially when handling complex tasks or multiple teachers [65, 92]. With the rise of Graph Neural Networks (GNNs), graph-based distillation has been applied both to improve teacher performance and to compress large GNNs into lightweight models suitable for real-time use [44].

Adaptive distillation (6) focuses on dynamically adjusting the distillation process instead of using a fixed setup, so that the transfer adapts to the quality and reliability of the teacher’s outputs. This can include modifying loss definitions, weighting samples differently, or pruning teacher components to better match the student [93]. For example, adaptive losses can prevent incorrect teacher predictions from misleading the student, while dynamic weighting methods adjust importance based on sample difficulty or relevance [94].

3.4. Low-rank factorization

Compared to pruning and quantization, which operate by eliminating or simplifying weights, or distillation, which trains new models, the strength of low-rank factorization lies in structurally leveraging the inherent low-dimensional redundancy of layer weight matrices [95]. It directly reduces model complexity without retraining from scratch and often integrates smoothly with existing architectures, making it a practical and scalable approach to maintaining speed and efficiency without a full redesign. Low-rank factorization offers a distinct middle ground among the previously mentioned techniques: it compresses and accelerates models by restructuring existing weights with less manual tuning and architectural change

and without the sparsity constraints [95, 23]. Recognizing this, the field has evolved from classical fixed-rank decompositions to diverse and adaptive variants, including fixed-rank singular value decomposition (SVD)/truncated SVD, adaptive-rank SVD, joint optimization approaches, differentiable or mask-based rank selection, and low-rank adaptation methods [96, 97].

3.4.1. SVD approaches:

One of the ways to factorize matrices is by using the traditional SVD approach [98]. In model compression, we could apply fixed-rank, truncated, and adaptive SVD. The SVD of a weight matrix $W \in \mathbb{R}^{m \times n}$ is expressed as:

$$W = U\Sigma V^\top, \quad (6)$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices containing the left and right singular vectors, and Σ is a diagonal matrix of singular values arranged in descending order. On the other hand, *Truncated SVD (TSVD)* approximates W by retaining only the top k singular values and corresponding singular vectors:

$$\tilde{W} = U_k \Sigma_k V_k^\top, \quad (7)$$

where $U_k \in \mathbb{R}^{m \times k}$, $\Sigma_k \in \mathbb{R}^{k \times k}$, and $V_k \in \mathbb{R}^{n \times k}$.

By choosing $k \ll \min(m, n)$, TSVD substantially reduces both the number of parameters, because one stores only U_k , Σ_k , and V_k instead of the full matrix, and the computational workload during inference, since multiplication by $\tilde{W}$ requires fewer operations.

In neural network compression, TSVD is typically applied post-training to large weight matrices, such as those in fully connected or projection layers. It achieves compression and inference acceleration without necessitating a full retraining of the model. However, TSVD involves a trade-off between compression efficiency and model accuracy. Selecting a too small k can significantly degrade performance, so a careful balance must be struck [97].

Adaptive-rank SVD [99] improves on the traditional fixed-rank or truncated SVD by selecting the rank dynamically for each layer, using metrics like spectral entropy or energy distribution of singular values to determine how many components to retain. This ensures each layer preserves a sufficient portion of its informational content, leading to more effective layer-wise compression while better maintaining model accuracy.

3.4.2. Joint optimization approaches

Unlike the conventional two-stage pipeline of factorization followed by fine-tuning, joint optimization methods tackle low-rank compression and model learning simultaneously by solving a unified optimization problem [96, 97, 100]. This avoids the mismatch between factorized weights and task adaptation seen in sequential methods. A general formulation is:

$$\min_{U, V, r} \mathcal{L}_{\text{task}}(UV^\top) + \lambda \cdot C(r), \quad (8)$$

where $\mathcal{L}_{\text{task}}$ is the network’s original task loss, $UV^\top$ represents the compressed weight matrix W_c , r is the rank, $C(r)$ is a compression cost, and λ balances compression against task perfor-

mance. To connect weight factorization error with loss degradation, first-order Taylor expansion is used:

$$\Delta \mathcal{L} \approx \nabla_W \mathcal{L} \cdot \Delta W, \quad (9)$$

where $\Delta W = W_c - W$ is the compression error and $\nabla_W \mathcal{L}$ reflects the sensitivity of the loss to changes in the weights. This shows that small errors in the compressed weights directly affect model loss, which motivates the following constrained optimization problem:

$$\begin{aligned} \min_{W_c, r} \quad & \text{CompressionCost}(r) \\ \text{s.t.} \quad & \mathcal{L}(W_c) - \mathcal{L}(W) \leq \epsilon, \end{aligned} \quad (10)$$

where ϵ bounds allowable performance degradation [100]. In practice, these formulations enable “lossless” compression, where accuracy drop is negligible or even improved due to implicit regularization.

3.4.3. Low-rank adaptation methods

Parameter-Efficient Fine-Tuning methods allow fine-tuning large models by learning low-rank updates instead of updating full weight matrices. Some of the key variants are as follows:

LoRA (Low-Rank Adaptation) [101]: Given a pretrained weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA learns a low-rank update $\Delta W = BA$, with $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$, where $r \ll \min(d_{\text{out}}, d_{\text{in}})$. The adapted layer computes:

$$y = W_0 x + \Delta W x = W_0 x + B(Ax). \quad (11)$$

This approach keeps W_0 frozen and introduces no additional inference overhead once merged.

Quantized LoRA (QLoRA) [102] combines LoRA with aggressive quantization of the base model to 4-bit NF4 format. It preserves performance by: 1) quantizing W_0 to 4-bit using the NormalFloat (NF4) format, 2) applying double quantization, and 3) using paged optimizers to manage GPU memory spikes during training. This enables fine-tuning of massive models (e.g., 65B parameters) on a single 48 GB GPU while preserving nearly full-precision performance.

Weight-decomposed LoRA (DoRA) decomposes W_0 into a magnitude and a normalized direction [103]:

$$W_0 = m \cdot \frac{V}{\|V\|_c}. \quad (12)$$

Only the direction V is updated via $\Delta V = BA$, while m is learned separately. This decoupling aims to match full fine-tuning’s capacity more closely without adding inference overhead as only the low-rank path and magnitude terms are trained.

Dynamic LoRA [104] trains adapters that support multiple ranks simultaneously, where lower ranks form prefixes of higher-rank updates, enabling flexible rank selection at deployment. It does that by training one adapter to work across a range of ranks instead of a single fixed rank.

Adaptive LoRA distributes rank budgets across layers adaptively, pruning or expanding per-layer rank based on importance scores [105]. The update is parameterized in SVD form and singular values are pruned/expanded during training so that critical layers get higher effective rank and less important layers get lower rank [105].

3.5. Neural architecture search

Neural Architecture Search (NAS) has emerged as a promising approach to automate the design of neural networks. Instead of manually crafting architectures, NAS algorithms explore a space of possible models and identify those that achieve strong performance for a given task. Unlike previously mentioned model compression techniques that reduce redundancy in existing models, NAS can design architectures from scratch that are inherently efficient. A key driver of the success of the current DL methods is the availability of high-performing neural architectures. NAS pushes this further by automating the architecture design step, much as DL previously automated feature engineering [106].

The benefits of NAS are two fold. First, it reduces reliance on human intuition and trial-and-error in model design, which can be both time-consuming and biased. Second, NAS methods have already demonstrated architectures that surpass human-designed counterparts across various benchmarks, including large-scale tasks like ImageNet [107], diverse datasets [108], and resource-constrained scenarios [109]. NAS is typically structured around three fundamental components [106]. The first is search spaces, which are the collection of candidate architectures that the NAS algorithm can choose from. These spaces may range from thousands to 10^{20} possibilities. Incorporating prior knowledge can simplify the search but may also introduce bias that limits novelty. The second is search strategies which is the the optimization process used to explore the search space and identify promising architectures. The third component is performance estimation strategies, which are techniques used to estimate model performance without fully training each candidate. Examples include learning curve extrapolation and weight-sharing approaches, which accelerate the search by avoiding repeated full training runs.

NAS can be expressed as the following optimization problem. Given a search space $\mathcal{S}$, dataset $\mathcal{D}$, a training procedure $\mathcal{T}$, the objective is to find an architecture $m \in \mathcal{S}$ that achieves the highest validation accuracy when trained on $\mathcal{D}$ using procedure $\mathcal{T}$ within the constrained time or computation budget [106]:

$$\begin{aligned} \min_{m \in \mathcal{S}} \quad & \mathcal{L}_{\text{val}}(w^*(m), m; \mathcal{D}, \mathcal{T}) \\ \text{s.t.} \quad & w^*(m) = \arg \min_w \mathcal{L}_{\text{train}}(w, m; \mathcal{D}, \mathcal{T}), \end{aligned} \quad (13)$$

where $\mathcal{L}_{\text{val}}$ and $\mathcal{L}_{\text{train}}$ denote validation and training losses and w are the model parameters trained for a given architecture m .

As mentioned, the first component of NAS is search spaces. The search space is a central element of NAS and defines the set of possible architectures that the algorithm can explore. Most existing search spaces are tailored to specific tasks and are often inspired by successful manually-designed architectures [106]. Designing the search space involves an important trade-off. A small, highly engineered search space can make the search more efficient but risks embedding human biases, limiting the discovery of novel designs. On the other hand, very large search spaces with primitive building blocks require more computa-

tion but increase the potential for uncovering new architectures [106].

Several key terms are commonly used to describe components of search spaces [106, 109]: 1) Operation/primitive: the smallest building unit of a search space, typically represented as a triplet of activation, operation, and normalization. 2) Layer: often used in chain-structured or macro-level spaces to denote the same as an operation or primitive. 3) Block/Module: a sequential arrangement of layers, frequently used in chain-structured or macro-level search spaces. 4) Cell: a directed acyclic graph (DAG) of operations, common in cell-based search spaces, where the maximum number of operations is usually fixed.

3.5.1. Types of search spaces

There are several types of search spaces in NAS that demonstrate different exploration approaches. *Macro search spaces* define an entire architecture at once, either as a DAG of operations or through macro-level hyperparameters. In the DAG-based variant, each node corresponds to an operation, while the topology itself is also a search variable [110, 111]. For example, the NASBOT CNN space [111] allows choices among convolution, pooling, and fully connected layers with limited depth. Macro spaces are flexible and expressive, enabling the discovery of highly novel architectures. However, their large design space makes them computationally expensive to search, often requiring significant resources.

Chain-structured search spaces represent architectures as sequential stacks of layers, typically inspired by popular backbones. These spaces can vary kernel sizes, expansion ratios, or convolutional patterns, as seen in ProxlessNAS [112] and DASH [108]. The simplicity of this type makes them easy to implement, fast to evaluate, and well-suited for building on proven manual designs. Nevertheless, the limited structural variety restricts the chances of finding truly innovative architectures.

Cell-based search spaces are among the most widely used in NAS, motivated by the observation that SOTA CNNs often rely on repeated structures such as residual blocks in ResNets [113]. Instead of searching the full architecture, the search space is restricted to relatively small cells, which are then stacked repeatedly to form the overall network. In this formulation, the cells define the micro-architecture, while the outer network skeleton (macro-structure) remains fixed.

Cell-based search significantly reduces search complexity, making it computationally feasible while still delivering high-performing architectures. However, while they define a very large search space, the performance variance across architectures is relatively small [114], suggesting limited expressive power. Additionally, by confining the search to small cells, these spaces can overlook higher-level connections, reducing the chance of discovering fundamentally novel architectures. Recent work has thus argued for combining micro-level search with macro-level exploration.

Hierarchical search spaces extend beyond a single design level by organizing motifs at multiple levels, where higher-level

motifs are DAGs of lower-level motifs [109, 106]. A common setup combines macro-level hyperparameters with cell- or chain-structured designs, as in MnasNet [18], which uses MobileNetV2 as its backbone. Hierarchical spaces are highly expressive, enabling more diverse and complex architectures compared to flat representations. They also improve efficiency by structuring the search in a multi-level manner. The added complexity makes them harder to design and computationally more challenging to search.

3.5.2. Search Strategies

The second component of NAS is search strategies. Despite many NAS approaches being computationally expensive, there are several attempts that construct more efficient approaches [109]. In this part of the survey, we will focus on the efficient approaches.

Firstly, *reinforcement learning (RL)* has been widely adopted in NAS since the search process can be naturally modeled as a Markov Decision Process. While early approaches like NAS-Net [113] were effective, they were also computationally expensive due to the repeated training of sampled models. To improve efficiency, later works focused on reducing search costs by simplifying the controller’s predictions or combining search phases. For example, FPNets [115] limited the Recurrent Neural Network (RNN) controller to predict only architectural hyperparameters (e.g., filter size, stride) while fixing the macro structure, thereby cutting down the search space.

Secondly, *evolutionary algorithms* provide another common approach in NAS by evolving a population of candidate models through mutation and selection [109]. To improve efficiency, several works have adapted this method to hardware-aware settings. For instance, TinyNAS [116] first tailors the search space to resource-constrained devices and then applies an evolutionary process within this reduced space, greatly reducing computation while ensuring accuracy.

Lastly, *gradient-based methods* have become particularly attractive in NAS because they drastically cut-down search costs by enabling weight sharing across candidate architectures [109]. Instead of training each model separately, a supernet is trained where shared edges represent different architectural options. Efficient approaches like ProxylessNAS [112] and FBNet [117] combine accuracy and latency terms in their loss functions, allowing the search to directly optimize for both performance and hardware efficiency.

4. Lightweight and efficient DL for disease detection and classification

DL has transformed disease detection and classification by enabling automated, and accurate interpretation of complex medical data such as images, signals, and electronic health records (EHRs). However, the high computational cost and memory requirements of SOTA architectures often limit their use in real-world healthcare, particularly in low-resource settings and on mobile or wearable platforms. To address this

gap, researchers are increasingly developing lightweight and efficient DL models that preserve diagnostic accuracy while drastically reducing parameters, inference time, and energy consumption. Such models are vital for scaling AI to diverse healthcare environments. The following subsections highlight advances in this direction across different categories of diseases.

4.1. Cardiovascular diseases

Cardiovascular diseases (CVDs) encompass a broad range of conditions affecting the heart and blood vessels, including coronary artery disease, arrhythmia, myocardial infarction, valvular disorders, and structural abnormalities. Recent research has focused on developing lightweight and efficient DL frameworks that maintain diagnostic accuracy while enabling deployment on portable and resource-limited medical devices. A summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 2.

4.1.1. Coronary artery disease

Coronary Artery Disease (CAD) is one of the most prevalent CVDs worldwide and remains a leading cause of morbidity and mortality. Early detection is critical for effective treatment, yet traditional diagnostic methods such as angiography and advanced imaging are resource-intensive and often impractical in low-resource or mobile healthcare environments. Lightweight DL models have therefore been explored to provide accurate, efficient, and deployable solutions for CAD detection across diverse modalities, including ECG, heart sound analysis, and MRI. Abdusalomov et al. [118] proposed a lightweight DL model for CAD detection in athletes by modifying the VGG-16 architecture with ResNet-inspired residual blocks. These residual connections improve gradient flow and feature extraction without increasing computational cost, while selective filter adjustments in deeper layers substantially reduce model complexity. The resulting architecture demonstrated strong performance, underscoring its potential for real-time deployment in resource-constrained settings such as sports clinics and mobile diagnostic systems. Zhang et al. [119] introduced a comprehensive AI-assisted portable system for early CAD diagnosis that integrates a custom-built heart sound sensor with an on-board executable model. At its core, the system leverages TYNet, a compact CNN trained through KD from a VGG-16 teacher model, thereby maintaining diagnostic accuracy while drastically reducing computational requirements. TYNet is further optimized through 8-bit quantization, attention mechanisms, double-branch convolutions, and depthwise separable layers, ensuring efficiency and reliability for embedded deployment on portable devices. Iqbal et al. [120] developed a lightweight CNN for real-time CAD classification from MRI, optimized through time complexity analysis and hyperparameter tuning. Their model achieves inference times as low as 8ms per image while matching the accuracy of a CNN–Random Forest ensemble. To enhance clinical applicability, the framework integrates explainable AI techniques such as Grad-CAM and layer-wise relevance propagation, providing interpretable predictions that

improve transparency and foster clinical trust in diagnostic decisions.

4.1.2. Arrhythmia

Arrhythmia, encompassing irregular heart rhythms such as atrial fibrillation and other conduction abnormalities, is a major risk factor for stroke and sudden cardiac arrest. Continuous ECG monitoring for arrhythmia detection often depends on resource-constrained wearables and mobile devices, motivating the development of lightweight and efficient DL models capable of real-time inference. Hybrid and attention-based architectures have been widely explored to achieve compact yet expressive representations for ECG analysis. Alamatsaz et al. [121] proposed a lightweight CNN–LSTM hybrid for automated arrhythmia detection from ECG signals, achieving 98.24% accuracy across eight arrhythmia types while incorporating SHAP-based interpretability, enabling explainable and reliable clinical decision support. Meng et al. [122] introduced the fusing transformer for dynamic ECG heartbeat classification, replacing standard self-attention with LightConv attention to significantly reduce parameters without sacrificing accuracy, demonstrating the feasibility of transformer-based ECG analysis for portable cardiac diagnostic tools. Jiang et al. [123] presented DCETEN, a transformer-based ECG classification framework that integrates efficient channel attention and depthwise separable convolutions for compact feature extraction. By further incorporating hand-crafted RR interval features and pruning, DCETEN achieved 99.84% accuracy and a 99.67% F1-score on the MIT-BIH dataset, showing strong potential for embedded ECG diagnostic systems.

KD has become a prominent strategy for transferring knowledge from complex teacher networks to compact student models suitable for clinical deployment. Wong et al. [124] developed an ultra-efficient edge model for ECG-based cardiac disease detection, where a student network 45× smaller than its teacher maintained comparable diagnostic accuracy and robustness to adversarial perturbations. Their approach enables secure and reliable ECG analysis for real-time precision health monitoring on edge devices. An et al. [125] designed a single-lead ECG monitoring system trained via KD from a multi-lead teacher model, achieving efficient arrhythmia classification validated on the Chapman–Shaoxing dataset and deployable on microcontroller-based wearable devices. Faraone et al. [126] combined KD and fixed-point quantization to transfer knowledge from a CNN–RNN teacher to a CNN–GRU student, yielding a 38% speed-up and 43% memory reduction while maintaining high accuracy for low-power atrial fibrillation detection on wearable system-on-chips (SoCs). Beyond static compression, Wang and Sun [127] proposed OTCD, an energy-efficient online classifier for time-series health data that combines drift detection, latent-space normalization, and feature distillation to sustain diagnostic accuracy under changing physiological conditions in long-term ECG monitoring.

Model compression through pruning and quantization has also been extensively applied to real-time ECG analysis. Xiaolin et al. [128] introduced a multistage pruning approach that reduced parameters by 60% and runtime complexity by over

60% while preserving classification accuracy for arrhythmia detection on edge hardware. Gu et al. [129] developed a shallow CNN combining quantization and fixed-point arithmetic for real-time heart rate anomaly detection on wearable devices, providing an efficient solution for continuous cardiac health monitoring. Ribeiro et al. [130] further demonstrated a quantized 1D-CNN trained directly on raw ECG signals from the MIT-BIH dataset, achieving efficient inference on ARM Cortex processors while maintaining high diagnostic performance, confirming its feasibility for homecare-oriented continuous arrhythmia screening.

Hardware-aware optimization has further enhanced efficiency in embedded cardiac diagnostic systems. Lu et al. [131] built an unstructured sparse CNN accelerator that skips zero-weight multiplications using compressed storage and tile-first dataflow, achieving 98.99% accuracy and 48% efficiency gains on FPGA implementations of wearable ECG classifiers. Ran et al. [132] designed a large-scale multilabel CNN co-optimized with pruning, quantization, and FPGA acceleration, enabling the detection of 26 arrhythmia types from over 200,000 ECG recordings for continuous homecare-based cardiac monitoring.

NAS has also been explored to generate efficient model topologies tailored for cardiovascular monitoring. Song et al. [133] introduced NAS-PPG, a PPG-based heart rate estimation framework that applies NAS with hyperparameter tuning to identify an optimal CNN–LSTM configuration, achieving a 75× reduction in parameters and a 26% improvement in MAE across ISPC, BAMI, and PPG-DaLiA datasets, thus enabling energy-efficient cardiac tracking on wearable sensors. Mukhopadhyay et al. [134] proposed EQ-NAS, a Deep Q-Learning-based NAS framework that generates pareto-optimal ECG models constrained to TFLite operations. The resulting architectures fit within 256 KB of flash memory, allowing continuous atrial fibrillation screening on ultra-low-power microcontrollers used in medical-grade wearables.

Physiological knowledge has also been integrated into lightweight architectures to enhance interpretability and robustness in clinical settings. Lu et al. [135] developed KecNet, which integrates a Sinc-convolutional layer and rhythm variation coefficient to learn interpretable rhythm-domain features from ECG signals. Trained on MIT-BIH, KecNet achieved high diagnostic accuracy with fewer parameters and superior noise robustness, supporting reliable arrhythmia screening in mobile and ambulatory healthcare environments.

4.1.3. Myocardial infarction

Myocardial Infarction (MI), commonly known as a heart attack, remains one of the leading causes of mortality worldwide. Early detection through ECG analysis is crucial for timely intervention, yet conventional diagnostic approaches often rely on resource-intensive infrastructure and expert interpretation. Lightweight and efficient DL models have therefore emerged as promising solutions for rapid, accurate, and accessible MI diagnosis on portable and Internet of Medical Things (IoMT) devices, enabling effective deployment in emergency care, remote monitoring, and low-resource clinical environments. Cao et al. [136] introduced ML-Net, a multi-channel

lightweight CNN specifically designed for efficient MI detection from 12-lead ECGs. The model incorporates a Channel Block (CB) framework to compress parameters and minimize computational overhead. The CB integrates four key stages: an initial 1D convolution with max-pooling for coarse feature extraction, squeeze convolutions for parameter reduction, multi-scale convolutions for hierarchical ECG feature learning, and depthwise-pointwise convolutions to further lower complexity. Trained on standard ECG datasets, ML-Net achieved strong diagnostic accuracy while remaining highly efficient, demonstrating suitability for real-time MI classification on portable and IoMT platforms used in point-of-care and telecardiology settings. Borah et al. [137] proposed HeartCare, a KD framework for detecting and localizing multi-label MI from 12-lead ECG signals. The framework employs a CNN-based teacher-student structure trained on the PTB-XL dataset to classify five super-diagnostic MI classes and was further validated on external datasets including SPH, CPSC 2018, and an Indian patient cohort. Through KD, HeartCare effectively distilled the representational power of the teacher into a compact student model, achieving up to 94.25% accuracy and a micro-averaged ROC of 0.95. These results highlight strong generalization across diverse patient populations and hospital systems while maintaining computational efficiency for deployment on wearable and edge-based ECG diagnostic devices.

4.1.4. Cardiac valvular conditions

Cardiac valvular diseases, including conditions such as aortic stenosis, mitral regurgitation, and mitral valve prolapse, contribute significantly to cardiovascular morbidity and require early detection to guide timely intervention. Traditional diagnosis through ECG and auscultation is resource-intensive and demands specialist expertise, motivating the development of lightweight DL models capable of leveraging phonocardiogram (PCG) signals for accessible, real-time screening across diverse healthcare settings. Shuvo et al. [138] introduced CardioXNet, a lightweight convolutional recurrent neural network for classifying five cardiac conditions directly from raw PCG signals. CardioXNet combines parallel CNN branches with SE modules for efficient feature extraction and employs a residual Bi-LSTM to capture temporal dependencies without requiring handcrafted features. The model achieved strong classification accuracy while maintaining a low parameter count and computational footprint, making it suitable for mobile and point-of-care heart sound diagnostics. Zhu et al. [139] proposed LEP-CNet, a compact CNN-based architecture optimized for low-power PCG classification. The model incorporated group convolutions, depthwise separable convolutions, and feature concatenation to reduce computational operations while preserving diagnostic precision. LEP-CNet enables efficient real-time heart sound analysis on wearable and IoMT platforms, supporting continuous monitoring of valvular health. Al-Issa and Alqudah [140] presented a lightweight hybrid framework integrating 1D-CNN and LSTM networks for automated classification of valvular conditions, including normal, aortic stenosis, mitral regurgitation, mitral stenosis, and mitral valve prolapse. The CNN blocks use 1D convolution, batch normaliza-

tion, ReLU activation, and max-pooling layers for hierarchical feature extraction, followed by an LSTM module that models sequential dependencies in heart sound patterns. This hybrid design improves diagnostic accuracy over single-architecture baselines while substantially reducing model complexity, enabling real-time inference in resource-constrained clinical environments.

4.1.5. Cardiac segmentation

Cardiac image segmentation plays a critical role in quantifying heart structure and function, supporting tasks such as ventricular volume estimation, myocardial wall analysis, and surgical planning. However, conventional DL models for medical image segmentation are often computationally demanding and unsuitable for real-time or resource-limited clinical environments. Recent research has explored model compression strategies, such as pruning, to reduce computational cost while maintaining segmentation accuracy. Sreeram and Kainz [141] investigated the impact of pruning redundant operations in cardiac segmentation networks to enhance efficiency without significant performance degradation. Their approach achieved up to a 1148× reduction in model size with only about 4% performance loss, and in some cases, enabled CPU-based inference to outperform GPU execution speeds. These results demonstrate the potential of pruning-based optimization for adapting segmentation models to portable hardware, facilitating rapid and accessible cardiac image analysis in low-resource clinical and point-of-care settings.

4.1.6. Summary

Lightweight DL research for CVD detection and classification emphasizes compact architectures such as shallow CNNs, 1D-CNNs, CNN-LSTM hybrids, and transformer models with depthwise separable convolutions. While pruning, quantization, KD, and NAS-based methods have been explored in several studies, lightweight-by-design architectures remain the most commonly used approach for achieving high diagnostic accuracy with markedly reduced computational and memory requirements.

4.2. Respiratory diseases

Respiratory diseases such as COVID-19, pneumonia, tuberculosis, and lung cancer pose substantial global burdens, especially in settings with limited imaging infrastructure and specialist availability. Lightweight and efficient DL models have the potential to enable rapid, accurate, and deployable screening on portable, edge, and mobile/IoMT devices across clinical and remote environments. A consolidated summary of representative works, modalities, efficiency techniques, and performance is provided in Table 3.

4.2.1. COVID-19

COVID-19, caused by SARS-CoV-2, has had profound global health and socioeconomic impacts. Rapid and accurate diagnosis remains critical, yet access to PCR testing and advanced imaging is often limited. Lightweight and efficient DL

Table 2: Summary of lightweight and efficient DL models for CVD detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Compression Techniques used					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Abdusalomov et al. [118]	CAD	ICA images/videos	Modified VGG16	✓	✗	✗	✗	✗	✗	CADICA	Acc: 90.3%	1.2M parameters
Zhang et al. [119]	CAD	PCG	CNN enhanced with attention	✓	✗	✓	✓	✗	✗	Custom PCG data	Acc: 85.2%	52.16K parameters
Iqbal et al. [120]	CAD	MRI	Modified LeNet-5	✓	✗	✗	✗	✗	✗	CAD Cardiac MRI data from Kaggle	Acc: 99.35%	Model size: 1.94 MB
Alamatsaz et al. [121]	Arrhythmia	ECG	1D-CNN+LSTM	✓	✗	✗	✗	✗	✗	MIT-BIH, LTAF	Acc: 98.24%	Inference time: 5.127 ms
Meng et al. [122]	Arrhythmia	ECG	Fussing Transformer	✓	✗	✗	✗	✗	✗	CPSC 2020	Acc: 99.32%	Fewer params via LightConv
Jiang et al. [123]	Arrhythmia	ECG	Transformer with depthwise separable convolutions	✓	✓	✗	✗	✗	✗	MIT-BIH	Acc: 99.84%	25% reduction in model size
Wong et al. [124]	Arrhythmia	ECG	CNN	✓	✗	✗	✓	✗	✗	MIT-BIH	Acc: 98.1%	45× smaller than teacher
An et al. [125]	Arrhythmia	ECG	Compact CNN	✓	✗	✗	✓	✗	✗	Chapman-Shaoxing 12-lead ECG	Acc: 96.32%	1242.58× smaller than teacher
Faraone et al. [126]	Arrhythmia	ECG	CNN-GRU	✓	✗	✓	✓	✗	✗	CinC2017 challenge data	F1: 82.8%	138K parameters
Wang & Sun [127]	Arrhythmia	ECG/PPG	Temporal convolutional network	✓	✗	✗	✓	✗	✗	MIT-BIH, INCART	Acc +2.7–14.7% gain	Model size: 1 KB
Xiaolin et al. [128]	Arrhythmia	ECG	1D-CNN	✗	✓	✗	✗	✗	✗	MIT-BIH	Acc: 97.7%	60% sparsity
Gu et al. [129]	Arrhythmia	ECG	Shallow CNN	✓	✗	✓	✗	✗	✗	MIT-BIH	Acc: 97.69%	Inference time: 0.3 ms
Ribeiro et al. [130]	Arrhythmia	ECG	1D-CNN	✓	✗	✓	✗	✗	✗	MIT-BIH	Acc: 99.6%	Inference time: 7.65 ms
Lu et al. [131]	Arrhythmia	ECG	Sparse CNN Accelerator	✓	✓	✓	✗	✗	✗	Custom ECG data	Acc: 98.99%	70% sparsity
Ran et al. [132]	Arrhythmia	ECG	Multilabel CNN	✓	✓	✓	✗	✗	✗	200K ECG recordings	AUC: 94.4%	Inference time: 2.895 s
Song et al. [133]	Arrhythmia	PPG	CNN-LSTM	✗	✗	✗	✗	✓	✗	ISPC, BAMI, PPG-DaLiA	26% improvement in MAE vs SOTA	75× parameter reduction
Mukhopadhyay et al. [134]	Arrhythmia	ECG	EQ-NAS	✓	✗	✗	✗	✓	✗	PhysioNet/CinC 2017 challenge data	Acc: 81.84%	Model size: 106.76 KB
Lu et al. [135]	Arrhythmia	ECG	SincConv CNN	✓	✗	✗	✗	✗	✗	MIT-BIH	Acc: 99.31%	Inference time: 50.1 ms
Cao et al. [136]	MI	ECG	Multi-channel Lightweight CNN	✓	✗	✗	✗	✗	✗	PTB-XL	Acc: 96.65%	Model size: 45.75 MB
Borah et al. [137]	MI	ECG	CNN	✓	✗	✗	✓	✗	✗	PTB-XL, SPH, CPSC 2018, Indian cohort	Acc: 94.25%	Model size: 7.7 MB
Shuvo et al. [138]	Valvular disease	PCG	CRNN	✓	✗	✗	✗	✗	✗	PhysioNet/CinC 2016 challenge data	Acc: 86.57%	Model size: 7.96 MB
Zhu et al. [139]	Valvular disease	PCG	1D-CNN	✓	✗	✗	✗	✗	✗	PhysioNet/CinC 2016 challenge data	Acc: 93.1%	70.7% model size reduction
Al-Issa & Alqudah [140]	Valvular disease	PCG	1D-CNN+LSTM	✓	✗	✗	✗	✗	✗	PhysioNet/CinC 2016 challenge data	Acc: 90.65%	Inference time: 1.6985 ms
Sreeram & Kainz [141]	Cardiac segmentation	Ultrasound	DeepLabV3	✗	✓	✗	✗	✗	✗	EchoNet-Dynamic data	Dice: 0.90	Up to 1148× smaller

models for chest X-ray (CXR), CT, and multimodal signals provide cost-effective and deployable solutions for triage, point-of-care screening, and remote monitoring. Compact CXR CNNs have achieved strong accuracy under tight resource constraints. Hussein et al. [142] designed shallow CNNs with ELU activations, HeNormal initialization, and aggressive dropout to reach 98.55% (binary) and 96.83% (multiclass) accuracy for CXR-based COVID-19 detection in clinical triage. Karakanis and Leontidis [143] built extremely lightweight CNNs trained with conditional GAN augmentation to overcome limited data, surpassing a ResNet8 baseline for portable screening. Alqudah et al. [144] evaluated AOCT-Net, MobileNet, and ShuffleNet on CXRs with classical classifiers, consistently exceeding 98%

across metrics for bedside decision support. Nayak et al. [145] developed LW-CBRGPNet, which combines three convolution, batch normalization, and ReLU activation blocks with global average pooling and a fully connected layer, reaching 98.33% accuracy in multi-class classification while reducing training time and memory cost.

Pruning and quantization further compress imaging models for edge use. Sriwiboon [146] applied customized pruning and quantization to EfficientNetV2 and MobileNetV2 on CXR/CT, shrinking models to 142 KB and 70 KB, respectively, with negligible loss, supporting on-device diagnosis in mobile clinics.

KD boosts compact model performance while preserving efficiency. BabaAhmadi et al. [147] designed a dual-teacher sys-

tem where VGG19 and ResNet50V2 transfer knowledge to a MobileNetV2 student, achieving 98.8% accuracy and an F1-score of 99.1% with reduced computational cost.

NAS has been explored for CT-based COVID-19. He et al. [148] proposed EMARS, a multi-objective evolutionary NAS with accuracy, size, and robustness objectives (e.g., NSGA-III, MBConv, weight sharing), discovering lightweight CT architectures with interpretable CAM lesion localization suitable for rapid radiology workflows.

Beyond imaging, multimodal screening extends coverage to daily-life monitoring. Hassantabar et al. [149] combined wearable physiological sensors with smartphone questionnaires, using synthetic pretraining and a grow-and-prune DNN, to enable real-time COVID-19 screening in a mobile app, supporting community-level surveillance.

4.2.2. *Pneumonia*

Pneumonia remains a leading cause of morbidity and mortality, particularly among children and immunocompromised patients. Lightweight and efficient DL models have the potential to automate CXR screening on portable devices in low-resource environments. Nettur et al. [150] proposed a weighted-average ensemble of MobileNetV2 and NASNetMobile for pediatric CXR, outperforming individual models and heavier baselines (ResNet50, DenseNet201) for point-of-care pediatric triage.

RL-driven compression accelerates and streamlines deployment. Soni et al. [151] introduced HMC, a hybrid RL-based model compression framework designed for healthcare applications. HMC integrates model-based and model-free RL to overcome the long training times of traditional RL approaches. Evaluated on the NIH CXR Pneumothorax dataset for disease classification and the NLM Frontal dataset for view classification, HMC significantly reduced training time, improved generalization across datasets, and maintained strong compression performance. Its efficiency and adaptability make it suitable for deploying pneumonia and other disease classification models on resource-constrained medical devices.

4.2.3. *Tuberculosis*

Tuberculosis (TB) disproportionately affects low- and middle-income regions where radiologist access is limited. Lightweight and efficient DL on CXR/CT can deliver scalable, automated screening. Compression via pruning preserves accuracy while reducing compute. Too et al. [152] pruned a DenseNet-based CXR model, achieving 99% accuracy with 65.8% size reduction, enabling mobile TB screening in primary-care settings. Hybrid CNN–transformer designs capture local and global features efficiently. Tian et al. [153] developed a dual-branch CT model (CNN for local details, transformer for global context with skip connections), achieving fast training and high accuracy for radiology workflows.

4.2.4. *Lung cancer*

Lung cancer diagnosis spans CXR, CT, and histopathology, often requiring advanced imaging or pathology expertise. Lightweight and efficient DL can support early detection and subtype classification in clinical and edge environments.

Compact and hybrid architectures reduce overhead while maintaining accuracy. Agarwal et al. [154] proposed CNN-O-ELMNet, a hybrid framework that integrates a CNN with an optimized Extreme Learning Machine (O-ELM) enhanced by the imperialistic competitive algorithm (ICA) to overcome random initialization limitations. Validated on both CXR and CT scans, CNN-O-ELMNet accurately classified conditions including cancer, pneumothorax, and TB, while a preprocessing pipeline mapped Brixia scores into severity categories, supporting robust and interpretable assessments. Gao et al. [155] fine-tuned a MobileNetV2-based CT classifier with a custom softmax head to efficiently distinguish normal, pneumonia, and tumor cases for clinic-ready triage. Sakr [156] designed a shallow CNN for histopathology, achieving competitive accuracy with few parameters for near-real-time pathology support.

KD reduces complexity while retaining diagnostic performance. Pavel et al. [157] proposed a three-stage KD pipeline (teacher → teaching assistant → student) for CT-based non-small-cell lung cancer, combining cost-sensitive learning and shapley explanations for edge deployment and web-based inference. Alif et al. [158] distilled large CNNs into a compact student on LC25000 histopathology, maintaining strong subtype performance with Grad-CAM interpretability. Zheng et al. [159] distilled a swin teacher into a ConvNeXt student for fine-grained histology, surpassing SOTA baselines.

4.2.5. *Multiple pulmonary diseases*

Some works have targeted multiple pulmonary pathologies in one model for broad clinical utility. To enable efficient multi-disease diagnosis, several works have investigated compact CNN architectures. Khan et al. [160] benchmarked MobileNetV2, EfficientNet-B0, NASNetMobile, and ResNet50V2 across COVID-19, emphysema, pneumonia, TB, and normal. EfficientNet-B0 yielded the best balance for real-time clinical use. Sanida and Dasygenis [161] proposed a custom lightweight CNN with three convolutional blocks followed by batch Normalization, maxPooling, and compact fully connected layers for lung disease classification from CXR images. Their design achieved high accuracy while maintaining low computational cost, making it suitable for edge devices.

KD enhances generalization while keeping models light. Asif et al. [162] developed LWSE, a lightweight stacked ensemble combining MobileNet with a compact CNN, aggregated through a MLP for the detection of multiple chest infectious diseases from CXR and CT images. LWSE improved generalization across datasets while reducing computational overhead. Asham et al. [163] further advanced this line of work with a MobileNet-based student model enhanced by ConvMixer blocks and trained via KD from a MobileNet teacher to detect multiple pulmonary diseases from CXRs. Similarly, Kabir et al. [164] proposed a distillation-based framework combining multiple CXR datasets to train a large teacher network, which guides a lightweight student model, achieving an average accuracy of 97% with high precision and recall and outperforming conventional baselines. Galih et al. [165] employed a ViT–MobileNet teacher–student framework, achieving 93.07% accuracy on a CXR dataset that included COVID-19, bacterial

and viral pneumonia, and TB. These approaches demonstrate how distillation can balance efficiency and accuracy for multi-disease pulmonary diagnosis.

Quantization enables mobile deployment. Aldamani et al. [166] introduced LungVision, a lightweight diagnostic framework for nine lung diseases using pre-trained CNN backbones such as ResNet50, MobileNetV2, DenseNet, NASNetMobile, and EfficientNetB2. Through FP16, integer quantization, and QAT, their models achieved up to 75.6% size reduction with minimal accuracy loss, with EfficientNetB2 reaching an F1-score of 98.58%.

Acoustic analysis broadens screening beyond imaging. Wanasinghe et al. [167] developed a CNN model that integrates multiple acoustic features, including MFCCs, spectrograms, and chroma representations, to classify lung sounds as normal, crackle, or wheeze. Their system enables accurate, real-time detection of conditions such as asthma, COPD, and pneumonia, offering practical deployment potential in remote and resource-limited healthcare environments.

4.2.6. Other respiratory conditions

Beyond major pulmonary diseases such as lung cancer, pneumonia, TB, and COVID-19, a variety of other respiratory disorders including sleep apnea and conditions characterized by abnormal breath sounds also require timely detection to improve patient outcomes. Embedded device-based approaches have been explored for sleep apnea screening. Lin et al. [168] used Mel filter bank features with a CNN-LSTM, pruning and quantization to reduce parameters by 40.27% with 0.35% accuracy loss, and deployed on an STM32 ARM platform for in-home, contactless apnea screening across multiple apnea types. Lightweight models have been explored for respiratory anomaly detection. Acharya and Basu [169] combined Mel-spectrogram features with a CNN-RNN on ICBHI'17 under noisy conditions. Local log-weight quantization delivered 4× memory reduction with no performance loss, enabling long-term wearable monitoring of wheeze, crackle, and normal/combined sounds.

4.2.7. Summary

Lightweight DL research for respiratory disease detection and classification primarily adopts compact CNNs and efficient architectures such as MobileNet and EfficientNet variants. While pruning, quantization, KD, and NAS-based methods have been explored in several studies, lightweight-by-design models remain the most commonly used approach for delivering high diagnostic accuracy with substantially reduced computational cost.

4.3. Neurological diseases

Neurological diseases span brain tumors, alzheimer's disease, parkinson's disease, attention deficit hyperactivity disorder, cerebral palsy, epilepsy, and neuromuscular conditions. Recent research has emphasized lightweight and efficient DL models that preserve diagnostic performance while enabling deployment on IoMT platforms, wearables, mobile devices, and

resource-limited clinical systems. A consolidated summary of representative studies including their modalities, datasets, and efficiency techniques is provided in Table 4.

4.3.1. Brain tumors

Brain tumors represent one of the most challenging neurological conditions, with high morbidity and mortality linked to delayed detection and misdiagnosis. Lightweight and efficient DL models have been explored to enable fast, automated, and accessible brain tumor diagnosis and detection, particularly for deployment in IoMT platforms, mobile systems, and other resource-constrained clinical environments. Several works have focused on lightweight CNN and hybrid architectures for efficient MRI-based brain tumor detection. Ma and Zhang [170] proposed an end-to-end framework combining a CSPDarknet-based YOLOv4-CSP backbone with LSTM for diagnostic report generation, reducing redundancy via CSPNet, PANet modifications, and scale-invariant fusion for efficient clinical reporting. Khan et al. [171] designed a modified MobileNet with residual skip blocks to enhance feature propagation and reduce information loss, improving MRI tumor classification accuracy at low compute for real-time screening. Attention mechanisms have been integrated into lightweight models to boost diagnostic precision without significantly increasing parameters. Pacal et al. [172] augmented EfficientNetV2 with global and channel attention, improving feature weighting across MRI scans while remaining deployable on modest hardware. Muksimova et al. [173] introduced an attention-driven YOLOv5m with an efficient spatial attention module for brain tumor detection, supporting rapid, accurate identification in time-constrained clinical workflows. Pruning has also been applied to streamline brain tumor models. Sivakumar and Padmapriya [174] pruned weights up to 70% and neurons up to 10% in a five-layer CNN for MRI classification, cutting inference time while maintaining accuracy. Recent work has applied KD to improve the efficiency of deep models for brain tumor diagnosis and segmentation. Ananthathanavit et al. [175] distilled a ResNet18 teacher into a compact student retaining only the early residual block, achieving strong accuracy with far fewer parameters for MRI-based tumor diagnosis. Shi et al. [176] transferred SAM's segmentation capability to a U-Net student using a KD objective mixing Dice loss on annotations with KL divergence on SAM outputs, enabling SAM-level glioma segmentation at reduced model size for near-real-time use. Zheng and Kang [177] introduced Symmetrical-NAS to discover lightweight encoder-decoder networks and coupled it with KD and separable convolutions, achieving up to 90× FLOP reduction on BRATS with minimal Dice loss. Beyond KD and pruning, specialized lightweight segmentation networks further reduce computation. Wu et al. [178] proposed LightNet, a feed-forward, multi-path CNN with lightweight channel attention in the up-sampling stage, delivering competitive tumor segmentation while remaining mobile-deployable.

4.3.2. Alzheimer's disease

Alzheimer's disease (AD) is the most common neurodegenerative disorder, characterized by progressive cognitive decline

Table 3: Summary of lightweight and efficient DL for respiratory disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Hussein et al. [142]	COVID-19	CXR	Shallow CNN	✓	✗	✗	✗	✗	✗	COVID-19 radiography data	Acc: 96.83%	592.9K parameters
Karakanis & Leonidis [143]	COVID-19	CXR	Shallow CNN	✓	✗	✗	✗	✗	✗	COVID-19 pneumonia CXR data from kaggle	Acc: 98.7%	49.1K parameters
Alqudah et al. [144]	COVID-19	CXR	MobileNet	✓	✗	✗	✗	✗	✗	Cohen, SIRM, RSNA, CXR from Kaggle	Acc: 99.46%	Lightweight architecture
Nayak et al. [145]	COVID-19	CXR	Shallow CNN	✓	✗	✗	✗	✗	✗	CXR from figshare	Acc: 98.33%	Lightweight architecture
Sriwiboon [146]	COVID-19	CXR/CT	EfficientNetV2	✓	✓	✓	✗	✗	✗	Kaggle CXR, Kaggle COVID-19 Radiography, Kaggle Large COVID-19 CT scan slice, Mendeley Data Chest CT	Acc: 97.7%	0.07M parameters
BabaAhmadi et al. [147]	COVID-19	CXR	MobileNetV2	✓	✗	✗	✓	✗	✗	Covid-CXR, RSNA	Acc: 98.8%	95.3% reduction in parameters
He et al. [148]	COVID-19	CT	Shallow CNN	✓	✗	✗	✗	✓	✗	Clean-CC-CCII, MosMedData and COVID-CTset	Acc: 89.61%, 81.25%, 92.86% respectively	Model size: 3.39 MB
Hassantabar et al. [149]	COVID-19	Wearable data	Multi-layer perceptron	✓	✓	✗	✗	✗	✗	Data collected from San Matteo Hospital in Pavia, Italy	Acc: 98.1%	3.4x fewer parameters
Nettur et al. [150]	Pneumonia	CXR	MobileNetV2 + NASNetMobile	✓	✗	✗	✗	✗	✗	CXR collected at Guangzhou Women and Children's Medical Center	Acc: 98.63%	Lightweight architecture
Soni et al. [151]	Pneumonia	CXR	Inception-V3	✗	✓	✗	✓	✗	✗	NIH Pneumothorax	Acc: 79.14%	Model size reduction: 52.3%
Too et al. [152]	TB	CXR	DenseNet	✗	✓	✗	✗	✗	✗	CXR TB dataset	Acc: 99%	Model size reduction: 65.8%
Tian et al. [153]	TB	CT	CNN+Transformer	✓	✗	✗	✗	✗	✗	CT images provided by the First Hospital of Lanzhou University	Acc: 97.23%	16M FLOPS
Agarwal et al. [154]	Lung cancer	CXR/CT	Modified EfficientNetV2L	✓	✗	✗	✗	✗	✗	LIDC-IDRI	Acc: 97.7%	2.48K trainable parameters
Gao et al. [155]	Lung tumor	CT	Modified MobileNetV2	✓	✗	✗	✗	✗	✗	Chest CT	Acc: 99.6%	Lightweight architecture
Sakr [156]	Lung cancer	Histopath	Shallow CNN	✓	✗	✗	✗	✗	✗	LC25000	Acc: 99.5%	2.032M parameters
Pavel et al. [157]	Lung cancer	CT	Shallow CNN	✓	✗	✗	✓	✗	✗	NSCLC-Radiomics	Acc: 94.53%	25K parameters
Alif et al. [158]	Lung cancer	Histopath	Shallow CNN	✓	✗	✗	✓	✗	✗	LC25000	Acc: 92%	668.9K parameters
Zheng et al. [159]	Lung cancer	Histopath	ConvNeXt	✓	✗	✗	✓	✗	✗	Lung histopathology data	Acc: 85.64%	Lightweight architecture
Khan et al. [160]	Multiple lung diseases	CXR	Modified EfficientNet-B0	✓	✗	✗	✗	✗	✗	RSNA, NIAID, Guangzhou Medical Center, Padchest and NIH	Acc: 94.07%	16.38M parameters
Sanida & Dasygenis [161]	Multiple lung diseases	CXR	Shallow CNN	✓	✗	✗	✗	✗	✗	CXR data from 4 sources	Acc: 98.56%	Model size: 12.49 MB
Asif et al. [162]	Multiple lung diseases	CXR/CT	MobileNet + compact CNN ensemble	✓	✗	✗	✗	✗	✗	2 CXR and 1 CT datasets	Acc: 96.40%, 97.89%, 98.83%	Lightweight architecture
Asham et al. [163]	Multiple lung diseases	CXR	MobileNet + ConvMixer	✓	✗	✗	✓	✗	✗	COVID-19 Radiography data	Acc: 97.92%	0.63M parameters
Kabir et al. [164]	Multiple lung diseases	CXR	Shallow CNN	✓	✗	✗	✓	✗	✗	COVID-19 CXR data, TB CXR data	Acc: 96.08%	1M parameters
Galih et al. [165]	Multiple lung diseases	CXR	MobileNet	✓	✗	✗	✓	✗	✗	Lung disease data from Kaggle	Acc: 93.07%	Lightweight architecture
Aldamani et al. [166]	Multiple lung diseases	CXR	EfficientNetB2	✓	✗	✓	✗	✗	✗	Multi-source CXR dataset	F1: 98.58%	75.59% size reduction
Wanasinghe et al. [167]	Multiple lung diseases	Acoustic	Shallow CNN	✓	✗	✗	✗	✗	✗	ICBHI 2017 challenge data and another lung sound dataset available at Mendeley Data	Acc: 91.04%	Lightweight architecture
Lin et al. [168]	Sleep apnea	Acoustic	CNN+LSTM	✗	✓	✓	✗	✗	✗	Data collected at Shenzhen People's Hospital	Acc: 74.27%	40.27% parameters reduction
Acharya & Basu [169]	Resp. anomalies	Acoustic	CNN-RNN	✓	✗	✓	✗	✗	✗	ICBHI'17	Acc: 71.81%	4x memory reduction

and memory impairment. Early and accurate diagnosis is crucial for patient care, but PET, CSF biomarkers, and MRI inter-

pretation are resource-intensive. Lightweight MRI-based models can provide rapid, scalable diagnostic support in constrained

settings. El-Latif et al. [179] proposed a seven-layer CNN trained on a public MRI dataset with extensive augmentation to address imbalance, achieving 99.22% binary classification and 95.93% multi-class classification accuracy while keeping computational complexity low supporting practical AD triage in clinics with limited hardware.

4.3.3. Parkinson's disease

Parkinson's disease (PD) features motor symptoms such as tremors and freezing of gait (FoG), requiring timely, objective monitoring. Yi and Hwang [180] developed a wearable-oriented attention-based framework with pruning and dynamic quantization for FoG detection, compressing the model 29.9 $\times$ and attaining F1 score of 0.994, demonstrating real-time feasibility on embedded devices for continuous PD monitoring.

4.3.4. Attention deficit hyperactivity disorder

Attention deficit hyperactivity disorder (ADHD) assessment commonly relies on subjective reports. Lightweight DL models offer scalable screening from digital text. Karim et al. [181] distilled a large transformer into LastBERT, preserving accuracy and thereby enabling practical mental-health screening pipelines in constrained environments.

4.3.5. Cerebral palsy

Early screening of cerebral palsy (CP) through infant movement analysis is crucial for timely intervention but remains challenged by resource limitations. Tempel et al. [182] introduced a lightweight NAS framework guided by RL to jointly optimize network architectures and hyperparameters for early CP detection from infant movement data. Their approach leverages multi-stream convolutional architectures that process skeletal features, bone geometry, position, velocity, and acceleration extracted from videos of 557 infants across multiple countries. The RL-guided search identifies compact models that achieve high diagnostic accuracy while minimizing computational overhead, outperforming ensemble-based baselines. The resulting framework enables practical deployment in resource-limited clinical settings, supporting timely and scalable CP screening.

4.3.6. Epileptic seizures

Seizure detection is essential for safe, independent living but long-term EEG monitoring is cumbersome. Qiu et al. [183] proposed LightSeizureNet (LSN), a lightweight 1D-CNN designed for real-time seizure detection on ultra-low power wearable devices. The model integrates dilated convolutions, global average pooling, and kernel-wise structured pruning to reduce computational complexity while maintaining high diagnostic accuracy. Evaluated on the CHB-MIT dataset, LSN achieved up to 99.77% accuracy with only 3.7M MACs. Importantly, the model also provides clinically relevant interpretability by highlighting activated brain regions and frequency bands during seizures, enhancing its utility for medical decision-making. Baghersalimi et al. [184] developed a many-to-one KD framework for seizure detection on low-power IoT wearables. A

multi-modal teacher model, Res1DCNN, trained on ECG and two EEG channels from the EPILEPSIAE dataset, transfers knowledge to a lightweight ECG-only student model. Their approach, validated on 29 patients and over 4,600 hours of recordings, segments signals into balanced 3-second seizure/non-seizure windows. The student achieves teacher-level accuracy while drastically reducing energy use and computational complexity, enabling long-term deployment on edge platforms. By relying solely on ECG, the system avoids the stigma and discomfort of continuous EEG monitoring, making it practical for everyday seizure management.

4.3.7. Other neurological conditions

Beyond major neurological disorders such as brain tumor, AD, PD, and epilepsy, there are other conditions where lightweight DL models can enhance diagnosis and monitoring. Neuromuscular disorders, for instance, often require continuous assessment of electromyogram (EMG) signals to evaluate muscle activity. Traditional EMG analysis can be computationally demanding and unsuitable for real-time monitoring, underscoring the need for efficient, wearable-friendly AI solutions. Choi [185] proposed a lightweight EMG signal classification framework specifically tailored for wearable applications. The model employs a compact 1D-CNN combined with efficient feature extraction using maximal overlap discrete wavelet transform. Deployed entirely on low-cost FPGAs without reliance on external memory, it enables real-time, low-power inference while maintaining high accuracy.

4.3.8. Summary

Lightweight DL research for neurological disease detection and classification primarily relies on compact and efficient architectures such as shallow CNNs, 1D-CNNs, MobileNet and EfficientNet variants, GhostNet-based designs, and lightweight U-Net derivatives. Although pruning, quantization, KD, and NAS-based methods appear in several works, the majority of studies achieve efficiency through lightweight-by-design architectures tailored for low-compute environments.

4.4. Renal diseases

Renal diseases span kidney abnormalities, e.g. stones, cysts, tumors, and bladder disorders. Recent efforts emphasize lightweight and efficient DL models that preserve diagnostic performance while enabling deployment on point-of-care devices, SoC-based scanners, and resource-limited clinical settings. A consolidated summary of representative studies including modalities, datasets, and efficiency techniques is provided in Table 5.

4.4.1. Kidney abnormalities

Kidney abnormalities, including stones, cysts, and tumors, represent a major clinical burden and are commonly diagnosed using CT imaging. Accurate and timely detection is critical to prevent obstruction, infection, and malignancy, but manual reading is time-consuming and resource-intensive. Lightweight and efficient DL models can automate classification and detection with minimal compute, enabling use in point-of-care

Table 4: Summary of lightweight and efficient DL for neurological disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Ma & Zhang [170]	Brain tumor	MRI	YOLOv4-CSP + LSTM	✓	✗	✗	✗	✗	✗	Brain tumor data from kaggle	Acc: 97%	Lightweight architecture
Khan et al. [171]	Brain tumor	MRI	Modified MobileNet	✓	✗	✗	✗	✗	✗	BR35H binary, Figshare multi-class data	Acc: 99.83% (binary), 96.95% (multi-class)	3.5M parameters
Pacal et al. [172]	Brain tumor	MRI	Modified EfficientNetV2	✓	✗	✗	✗	✗	✗	Brain Tumor MRI Data Kaggle	Acc: 99.76%	14.41M parameters
Muksimova et al. [173]	Brain tumor	MRI	Modified YOLOv5m	✓	✗	✗	✗	✗	✗	Brain tumor MRI data from figshare	mAP50: 91.5%	22.1M parameters
Sivakumar & Padmapriya [174]	Brain tumor	MRI	CNN	✗	✓	✗	✗	✗	✗	BRATS 2013 data	Acc: 90%	Inference time: 15 s
Anantathanavit et al. [175]	Brain tumor	MRI	Modified ResNet18	✓	✗	✗	✓	✗	✗	Brain tumor MRI data	Acc: 98.1%	Lightweight architecture
Shi et al. [176]	Brain tumor	MRI	U-Net	✓	✗	✗	✓	✗	✗	Glioma dataset from the First Affiliated Hospital of Zhengzhou University	Acc: 87.37%	7.76M parameters
Zheng & Kang [177]	Brain tumor	MRI	Modified ResUnet	✓	✗	✗	✓	✓	✗	BRATS2018 data	Dice: 83.33%	0.20M parameters
Wu et al. [178]	Brain tumor	MRI	Modified Ghost-Net	✓	✗	✗	✗	✗	✗	BRATS2015 data	Dice: 84%	0.834M parameters
El-Latif et al. [179]	AD	MRI	Shallow CNN	✓	✗	✗	✗	✗	✗	Kaggle Alzheimer's data	Acc: 95.93%	Lightweight architecture
Yi & Hwang [180]	PD	Wearable sensors	Attention-based CNN-GRU hybrid	✗	✓	✓	✗	✗	✗	Daphnet Freezing of Gait data, Multimodal data	F1: 99.4%	29.9× parameter reduction
Karim et al. [181]	ADHD	Text	LastBERT	✓	✗	✗	✓	✗	✗	GLUE benchmark	Acc: 85%	29M parameters
Tempel et al. [182]	CP	Video	CNN-GCN	✗	✗	✗	✗	✓	✗	Infant CP data	Acc: 90.6%	0.621M parameters
Qiu et al. [183]	Epilepsy	EEG	1D-CNN	✓	✓	✗	✗	✗	✗	CHB-MIT scalp EEG data	Acc: 99.77%	113.8K parameters
Baghersalimi et al. [184]	Epilepsy	EEG/ECG	Res1DCNN	✓	✗	✗	✓	✗	✗	EPILEPSIAE data	Acc: 89.98%	inference time: 0.720 s
Choi [185]	Neuromuscular	EMG	1D-CNN	✓	✗	✓	✗	✗	✗	sEMG for Basic Hand movements data	Acc: 95%	inference time: 520 ms

settings. Karthikeyan et al. [186] introduced a lightweight end-to-end CNN for CT-based classification of stones, cysts, and tumors. The shallow design minimizes preprocessing and computation while achieving strong accuracy, supporting assisted reading for radiologists and integration into low-cost, portable CT workflows. Asif et al. [187] proposed StoneNet, a MobileNet-derived CNN with depthwise separable convolutions for kidney stone detection in CT. Using global average pooling, batch normalization, and dropout, StoneNet attains 97.98% accuracy with fast training and inference, making it suitable for real-time CT triage in busy clinical environments.

4.4.2. Bladder disorders

Bladder disorders, including urinary retention and dysfunction, are often assessed through bladder volume measurement, which typically requires specialized equipment and trained personnel. Portable, automated solutions powered by lightweight DL models can enable point-of-care urinary assessment, improving accessibility in low-resource settings and supporting real-time monitoring. Cho et al. [188] proposed a lightweight multitask DL framework integrated into a SoC portable ultrasound device for automatic bladder volume estimation. Their model, based on MobileNetV2 with quantizable SE blocks, jointly performs bladder detection and segmentation on sagittal and transverse ultrasound images. Achieving over 99% classi-

fication accuracy and a Dice score of 0.954, the system further estimated bladder volume within $\pm 6\%$ error when validated on phantom data. Its efficient design and low-power deployment demonstrate strong potential for real-time, point-of-care urinary assessment.

4.4.3. Summary

Lightweight DL research for renal disease detection and assessment primarily relies on lightweight-by-design architectures such as shallow CNNs, MobileNet variants and quantization.

4.5. Hepatic diseases

Hepatic diseases, encompassing both malignant and benign conditions, require rapid and reliable interpretation of histopathology and imaging for effective diagnosis and management. Recent work emphasizes lightweight and efficient DL that preserves accuracy while enabling deployment on edge/SoC hardware and in resource-limited settings. A consolidated summary of representative studies including modalities, datasets, and efficiency techniques is provided in Table 6.

4.5.1. Cholangiocarcinoma

Cholangiocarcinoma is often diagnosed late because clinical signs are non-specific. Histopathology is the diagnos-

Table 5: Summary of lightweight and efficient DL for renal disease detection and assessment.

Study	Category	Modality	Model	Lightweight by Design	Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Karthikeyan et al. [186]	Kidney abnormalities	CT	Shallow CNN	✓	✗	✗	✗	✗	✗	CT kidney data	Acc: 99%	Lightweight architecture
Asif et al. [187]	Kidney stones	CT	Modified MobileNet	✓	✗	✗	✗	✗	✗	CT image data collected at Elazig Fethi Sekin City Hospital, Turkey	Acc: 97.98%	3.3M parameters
Cho et al. [188]	Bladder disorders	Ultrasound	Modified MobileNetV2	✓	✗	✓	✗	✗	✗	Bladder images obtained using SoC-based portable ultrasound system	Dice: 0.954	0.97M parameters

tic gold standard but can be slow and variable across readers. Chakrabarti and Rao [189] proposed CholangioNet, a lightweight CNN for automated cholangiocarcinoma recognition in histopathology slides. Using a compact stack of convolution–pooling blocks followed by dense layers, the model achieves 98.4% validation accuracy while remaining small and fast, supporting rapid, consistent pathology triage in both specialized and low-resource settings.

4.5.2. Hepatocellular carcinoma

Hepatocellular carcinoma (HCC) is the most prevalent primary liver malignancy. Accurate automated lesion segmentation is critical for early intervention, treatment planning, and response assessment, however, existing models are often too computationally heavy for real-time deployment. Xu et al. [190] introduced EdgeSeg, an FPGA-oriented encoder–decoder with a Split-Attention-Shuffle Bottleneck backbone optimized for edge deployment. After training on GPU, the network is quantized to 8-bit and compiled with Vitis AI for the Kria KV260 board, delivering accurate CT-based HCC segmentation with low memory, low power, and minimal latency, well suited to portable and bedside scenarios.

4.5.3. Liver segmentation

Accurate liver segmentation from CT scans is essential for surgery and therapy planning, though 3D models remain computationally demanding for real-time use. Xu et al. [191] proposed an efficient real-time framework based on KD with a growing teacher assistant network. By ensembling soft targets from the teacher and intermediate assistant snapshots, a compact 3D student achieves up to a 2% Dice improvement over naive students while using only ~2% of the teacher’s parameters and reaching 13 ms inference, supporting time-critical intraoperative workflows.

4.5.4. Summary

Lightweight DL efforts in hepatic disease detection and classification leverage compact CNN architectures, streamlined encoder–decoder architectures, quantization, and KD to enable accurate performance under strict computational constraints.

4.6. Hematological diseases

Hematological diseases, including malignant and infectious conditions, manifest through alterations in blood cell morphology and distribution. Automated analysis of peripheral blood images using DL enables scalable and objective diagnosis, yet practical deployment remains limited by computational demands. Recent work has therefore focused on lightweight and efficient models that preserve accuracy while supporting deployment on mobile, edge, and IoMT devices particularly valuable in resource-constrained settings. A consolidated summary of representative studies including modalities, datasets, and efficiency techniques is provided in Table 7.

4.6.1. Leukemia

Leukemia comprises malignancies such as acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), and B-cell ALL (B-ALL). Conventional workflows, e.g. bone marrow biopsy and manual smear review, are invasive and time-intensive. Lightweight CNNs on microscopic blood images enable rapid and scalable triage. Batool et al. [192] adapted EfficientNet-B3 with depthwise separable convolutions, MB-Conv blocks, and inverted residual connections to capture fine-grained smear features while reducing compute, memory, and latency. Das et al. [193] built a MobileNetV2 feature extractor using depthwise separable convolutions with tunable width multipliers, paired with an SVM head. Evaluated on multiple public datasets, their framework demonstrated strong accuracy with minimal computational requirements, making it practical for real-time leukemia screening. Prakash et al. [194] introduced a YOLOv8 pipeline augmented with residual convolutional block attention mechanism and depthwise separable layers for fast, accurate stage detection. Khan et al. [195] used deep convolutional generative adversarial network based augmentation followed by a Faster R-CNN detector with a MobileNetV2 backbone and dual attention modules, yielding robust leukocyte detection and subtype classification in real-time settings. Hosseini et al. [196] designed a mobile application powered by optimized lightweight CNNs, including EfficientNetB0, MobileNetV2, and NASNet Mobile, combined with K-means clustering. Their final model achieved high accuracy, sensitivity, and specificity for B-ALL classification. By enabling non-invasive screening directly from blood smear images, this application provides a portable and cost-effective al-

Table 6: Summary of lightweight and efficient DL models for hepatic disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Chakrabarti & Rao [189]	Cholangiocarcinoma	Histopath	Shallow CNN	✓	✗	✗	✗	✗	✗	Kaggle's cholangiocarcinoma based histopathology data	Acc: 98.4%	Lightweight architecture
Xu et al. [190]	HCC	CT	Asymmetric encoder-decoder segmentation network	✓	✗	✓	✗	✗	✗	Abdomen CT images of HCC collected at Sir Run Shaw Hospital	Dice: 60.62%	0.587M parameters
Xu et al. [191]	Liver segmentation	CT	Shallow 3D U-Net	✓	✗	✗	✓	✗	✗	Liver CT data, MICCAI-SLiver challenge data	Dice: 84.8%	30K parameters

ternative to invasive diagnostic procedures such as bone marrow biopsies.

4.6.2. Lymphoma

Lymphoma diagnosis relies on histopathology, which is laborious and requires expertise. Gupta et al. [197] proposed ReducedFireNet, a squeezeNet-inspired lightweight CNN using four fire modules with max-pooling and dropout, reaching 96.88% accuracy on a three-class malignant lymphoma set while occupying only 0.39 MB and 0.201 GFLOPs well suited to IoMT devices.

4.6.3. Infectious diseases

Infectious diseases such as malaria, viral infections, and other conditions that manifest in blood cell morphology remain a major global health burden, particularly in low- and middle-income countries. Manual blood smear examination by trained specialists is the gold standard for diagnosis but is labor-intensive, time-consuming, and difficult to scale in resource-limited settings. Lightweight DL models provide an efficient alternative by enabling rapid, automated, and portable screening of infectious diseases directly from microscopic images. Mei et al. [198] presented efficient model (EMO)-6M, a compact hybrid CNN-transformer with KD for reactive lymphocyte recognition, combining local/global cues with strong generalization under tight resource budgets. For malaria, Nettur et al. [199] proposed UltraLightSqueezeNet, trimming fire modules and using parameter-efficient 1×1 and 3×3 convolutions to cut parameters by up to $54\times$ versus SqueezeNet1.1 while retaining high accuracy, ideal for edge deployment.

4.6.4. Summary

Lightweight DL research for hematological disease detection and classification primarily relies on lightweight-by-design architectures such as Modified MobileNetV2, EfficientNet variants, lightweight YOLO-based detectors, and SqueezeNet-style modules. KD and quantization appear only minimally.

4.7. Ophthalmological diseases

Ophthalmological diseases such as retinal disorders and eyelid tumors have seen extensive research in lightweight and efficient DL models, enabling diagnosis from Optical Coherence

Tomography (OCT), fundus, and pathological images with reduced computational demands. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is provided in Table 8.

4.7.1. Retinal vessel segmentation

Retinal vessel segmentation plays a crucial role in the early diagnosis of eye diseases such as diabetic retinopathy, glaucoma, and hypertensive retinopathy. Manual delineation of retinal vasculature is labor-intensive and prone to variability, while conventional DL methods often require large models that are computationally expensive. Lightweight segmentation frameworks provide a pathway toward fast, accurate, and deployable retinal analysis in both clinical and point-of-care settings. Aurangzeb et al. [200] introduced the first adaptation of ColonSegNet [201], originally developed for polyp detection, to the task of retinal vessel segmentation. Their model is a lightweight encoder-decoder architecture enhanced with residual blocks, SE modules, and multi-scale skip connections to capture fine vascular structures. Validated on DRIVE, CHASE_DB1, and STARE datasets, the framework achieved high segmentation accuracy with only $\sim 5M$ trainable parameters, significantly reducing computational overhead while preserving robustness across multiple benchmarks.

4.7.2. Retinal disease classification

Retinal diseases such as diabetic retinopathy, age-related macular degeneration, and choroidal neovascularization are leading causes of vision loss worldwide. OCT and fundus imaging play a central role in diagnosis, but manual interpretation is time-intensive and requires trained specialists. Lightweight DL models offer the potential for automated, efficient, and interpretable retinal disease classification that can be deployed on portable and edge devices for use in resource-limited settings. Sunija et al. [202] proposed a lightweight six-layer CNN for classifying OCT images into four categories: normal, diabetic macular edema, drusen, and choroidal neovascularization. Their model employs aggressive downsampling and weight sharing, reducing parameters to just $\sim 6.9\%$ of ResNet-50 while maintaining expert-level performance. Grad-CAMs were further incorporated to localize disease-specific features, enhancing interpretability and supporting clinical adoption. Bhandari et al. [203] introduced

Table 7: Summary of lightweight and efficient DL for hematological disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Batool et al. [192]	Leukemia	Blood smear	Modified EfficientNet-B3	✓	✗	✗	✗	✗	✗	C_NMC_2019	Acc: 99.31%	11.09M parameters
Das et al. [193]	Leukemia	Blood smear	Modified MobileNetV2	✓	✗	✗	✗	✗	✗	ALLIDB1, LIDB2, AL-ASH database	Acc: 99.39%, 98.21%, 97.73% respectively	Inference time: 21.1 s, 64.9 s, 15.18 s respectively
Prakash et al. [194]	Leukemia	Blood smear	Modified YOLOv8	✓	✗	✗	✗	✗	✗	Leukemia smear images obtained from RoboFlow	mAP: 98.4%	Inference time: 3.5 ms
Khan et al. [195]	Leukemia	Blood smear	Modified MobileNetV2	✓	✗	✗	✗	✗	✗	PBC, LISC, and Raabin-WBC benchmark data	Acc: 99.83%, 99.35%, 99.60% respectively	Lightweight architecture
Hosseini et al. [196]	Leukemia	Blood smear	MobileNetV2	✓	✗	✗	✗	✗	✗	B-ALL smear data collected at several Tehran hospitals	Acc: 100%	3.538M parameters
Gupta et al. [197]	Lymphoma	Histopath	Modified SqueezeNet	✓	✗	✓	✗	✗	✗	Malignant lymphoma data	Acc: 96.88%	Model size: 0.391 MB
Mei et al. [198]	Infectious	Blood smear	EMO-6M	✓	✗	✗	✓	✗	✗	Peripheral blood smear data collected at Zhongnan Hospital of Wuhan University	Acc: 93.55%	6.1M parameters
Nettur et al. [199]	Malaria	Blood smear	Modified SqueezeNet1.1	✓	✗	✗	✗	✗	✗	NIH Malaria	Acc: 94.76%	54x fewer parameters

a highly compact CNN for OCT-based retinal disease diagnosis, designed with only ~0.98 million parameters and 0.24 GFLOPs to ensure feasibility in resource-constrained settings. Their model demonstrated strong accuracy across multiple disease classes and robustness against adversarial attacks. By incorporating explainable AI tools such as LIME and SHAP, the framework provides transparent predictions and enhances clinical trust, addressing one of the key barriers to deploying AI in ophthalmology. Gavel and Azad [204] presented a lightweight classification framework optimized for on-device deployment using the Messidor dataset. Leveraging MobileNet and EfficientNet-Lite architectures with structured pruning and 8-bit quantization, their system reduced model size and inference time while maintaining diagnostic accuracy. Tested on Raspberry Pi 4, the approach achieved real-time retinal disease classification, demonstrating practical usability in low-resource healthcare environments.

4.7.3. Diabetic retinopathy

Diabetic Retinopathy (DR) is one of the most common complications of diabetes and a leading cause of preventable blindness worldwide. Early detection through retinal fundus imaging is critical for timely treatment, but conventional diagnostic techniques are resource-intensive and require significant human expertise. Manual grading of DR severity further adds to the workload and is subject to inter-observer variability. In contrast, lightweight and efficient DL models provide a scalable solution by automating DR detection and classification, enabling accurate screening even in resource-constrained clinical environments. Al Shafi et al. [205] proposed a transformer-based KD framework for DR detection using the APTOS 2019 fundus image dataset. Their method addresses class imbalance with cost-sensitive learning and offline augmentation, while employing a MobileViTV2 transformer as the teacher to trans-

fer knowledge to a lightweight GoogLeNet student via KL-divergence. By enhancing data diversity and generalization, the framework improves accuracy and robustness, particularly for minority DR stages. To support clinical adoption, the authors integrated SHAP-based explainable AI, enabling transparent and interpretable predictions of DR severity. Islam et al. [206] introduced a lightweight DR classification pipeline optimized for deployment in resource-limited settings. Their approach consists of three stages: image preprocessing, KD, and hyperparameter optimization. Preprocessing includes Gaussian denoising, ROI selection, unsharp masking, and gamma correction to enhance fundus image quality across APTOS and IDRiD datasets. A hybrid teacher model combining ResNet152V2 and a Swin Transformer distills knowledge into a compact Xception-based student network augmented with a customized depthwise-separable convolutional block attention module. The final student, at only 102 MB, achieves 100% binary and 99.04% multi-class accuracy on APTOS and 98.05% binary and 94.17% multi-class accuracy on IDRiD, significantly outperforming eight state-of-the-art classifiers.

4.7.4. Eyelid tumors

Eyelid tumors, including basal cell carcinoma, meibomian gland carcinoma, and cutaneous melanoma, represent a significant subset of ocular malignancies. Accurate diagnosis from histopathological images is challenging due to the fine-grained nature of tumor features, and traditional DL models can be computationally expensive for routine clinical use. Lightweight DL frameworks provide an opportunity to deliver accurate and efficient eyelid tumor classification suitable for deployment in real-world settings. Huang et al. [207] proposed MAC-ResNet, a DL framework for eyelid tumor classification that integrates multiscale attention, cropping, residual nested structures, and focal loss to address the challenges of fine-grained pathological

images. To reduce computational cost, MAC-ResNet serves as a teacher model in a KD framework, transferring knowledge to a lightweight ResNet50 student. Validated on the ZLE dataset, the student model achieved high accuracy across multiple eyelid tumor subtypes while significantly lowering inference cost. Coupled with U-Net-based lesion localization, this approach improved diagnostic reliability and interpretability, making it clinically practical for efficient and trustworthy deployment.

4.7.5. Summary

Lightweight DL research for ophthalmological disease detection and classification primarily relies on shallow CNNs, efficient CNN backbones and KD.

4.8. Dermatological diseases

Dermatological diseases such as melanoma and other malignant or infectious skin conditions have seen extensive progress in lightweight and efficient DL, enabling diagnosis from dermoscopic, clinical, and histopathological images with reduced computational demands. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 9.

4.8.1. Skin cancer

Skin cancer, including melanoma and other malignant skin lesions, is one of the most common cancers worldwide. Early and accurate detection is vital for reducing morbidity and mortality, yet manual diagnosis from dermoscopic or histopathological images is labor-intensive and prone to inter-observer variability. Lightweight DL models provide an opportunity to deliver scalable, efficient, and accurate skin lesion classification and segmentation, enabling real-time deployment on portable devices for point-of-care diagnostics. Several works have focused on lightweight CNN architectures for skin cancer detection. Baig et al. [208] proposed Light-Dermo, a ShuffleNet-based CNN enhanced with SE blocks, achieving high accuracy on HAM10000, ISIC-2019, and ISIC-2020 datasets with low computational cost. Asif et al. [209] introduced SKINC-NET, a streamlined CNN with dense layers, batch normalization, and LeakyReLU activations, which effectively classified seven lesion types on HAM10000 while minimizing trainable parameters. Hoang et al. [210] combined entropy-based weighting with lesion segmentation and a wide-ShuffleNet classifier, reducing model complexity while outperforming state-of-the-art methods on HAM10000 and ISIC-2019.

Hybrid models have also been explored. Gamil et al. [211] combined principal component analysis (PCA)-based feature reduction, EfficientNet-B0 for deep feature extraction, and an AdaBoost ensemble for improved classification accuracy on DermIS and ISIC datasets. Alche et al. [212] introduced EfficientARL, which incorporates an Attention Residual Learning mechanism into EfficientNet-B0, adding spatial attention with minimal overhead to boost melanoma detection. Ding et al. [213] presented HI-MViT, an explainable and lightweight model combining MobileViT with improved MV2 blocks and depthwise separable convolutions. Validated on ISIC and PH²

datasets, HI-MViT delivered strong performance while providing interpretability through class activation mapping. Yuan et al. [214] proposed UCM-Net, a hybrid CNN-MLP segmentation model that replaced heavy convolutional blocks with efficient MLP modules, achieving high segmentation accuracy with reduced parameters.

A large body of work has focused on KD to optimize skin cancer models. Thaker and Asif [215] distilled knowledge from a MobileNet teacher into a compact CNN student, halving model size while boosting PR-AUC and ROC-AUC for malignant lesion detection. Saha et al. [216] employed a dual-loss distillation strategy and distilled knowledge from large teacher networks like ResNet50, DenseNet169, VGG16/19 into a 0.295M-parameter student, achieving 93.24% accuracy on HAM10000 and 84.13% on ISIC-2019. Islam et al. [217] proposed a fusion-teacher distillation framework combining ViT, ResNet152V2, and ConvNeXtBase, distilling knowledge into a lightweight 0.16M-parameter student that achieved 98.94% accuracy on Kaggle and 98.75% on HAM10000. Kabir et al. [218] proposed a multi-teacher distillation framework combining DenseNet161 and ResNet50, distilling knowledge into TinyStudent, a compact CNN that achieved competitive performance with over 77× fewer parameters, further refined through model souping and deployed via a Django-based web application. Khan et al. [219] developed a ResNet50-based distillation framework producing a distilled student network that was 15× smaller than EfficientNet-B0 while outperforming it across precision, recall, and F1. Wang et al. [220] introduced SSD-KD, a self-supervised diverse distillation framework transferring intra- and inter-instance relational knowledge to a MobileNetV2 student, achieving 85% accuracy on ISIC-2019. Zeng et al. [221] further advanced this with DSP-KD, a dual-stage progressive distillation method incorporating masked generative distillation and adaptive logit scaling, achieving 92.83% accuracy on ISIC-2019 with ShuffleNetV2 student. Zhang and Lu [222] proposed a framework for skin lesion segmentation, introducing masked and multi-head boundary feature distillation modules to improve boundary awareness, achieving strong segmentation performance on ISIC-2016 and PH² datasets.

Pruning strategies have also been applied. Medhat et al. [223] developed an IMP-based lightweight AlexNet that progressively prunes low-importance weights, reducing RAM usage and inference time while maintaining accuracy across PAD-UFES-20, MED-NODE, and PH² datasets. Xiang et al. [224] proposed a hierarchical weight pruning strategy for lightweight CNNs, applied to 11-class skin lesion classification on HAM10000. By pruning smallest and largest weights with layer-specific thresholds, followed by retraining, their method improved average accuracy by 1.88% across eight models and boosted SqueezeNet's Top-1 accuracy by 7.2%, making it highly effective for mobile and embedded deployment.

4.8.2. Nail melanoma

Nail melanoma is a rare but aggressive form of skin cancer that often presents late due to its subtle visual cues and misdiagnosis as benign nail disorders. Early detection is critical

Table 8: Summary of lightweight and efficient DL for ophthalmological disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Aurangzeb et al. [200]	Retinal vessel segmentation	Fundus	Modified Colon-SegNet	✓	✗	✗	✗	✗	✗	DRIVE, CHASE_DB1, STARE	Acc: 96.59%, 97.31%, 97.19% respectively	5M parameters
Sunija et al. [202]	Retinal disease classification	OCT	Shallow CNN	✓	✗	✗	✗	✗	✗	Retinal OCT data	Acc: 99.69%	6.9% of ResNet-50 parameters
Bhandari et al. [203]	Retinal disease classification	OCT	Shallow CNN	✓	✗	✗	✗	✗	✗	Kaggle's OCT data	Acc: 94.29%	0.98M parameters
Gavel & Azad [204]	Retinal disease classification	Fundus	Modified EfficientNet-Lite	✓	✓	✓	✗	✗	✗	Messidor data	Acc: 61.79%	Model size: 4 MB
Al Shafi et al. [205]	Diabetic retinopathy	Fundus	GoogLeNet	✓	✗	✗	✓	✗	✗	APTOS 2019	Acc: 78%	6.62M parameters
Islam et al. [206]	Diabetic retinopathy	Fundus	Modified Xception	✓	✗	✗	✓	✗	✗	APTOS 2019, IDRiD 2018	Acc: 99.04%, 94.17% respectively	Model size: 102 MB
Huang et al. [207]	Eyelid tumors	Histopath	ResNet50	✓	✗	✗	✓	✗	✗	ZLett data	Acc: 94.06%	Lightweight compared to the teacher

to improve patient outcomes, yet manual diagnosis from dermoscopic or clinical images remains challenging. Lightweight DL models optimized for edge deployment can support timely, accessible, and accurate nail melanoma screening. Hussain et al. [225] introduced the Nail-Melanoma-350 dataset alongside a transfer learning-based workflow for nail melanoma detection. Their pipeline integrates image preprocessing, augmentation, fine-tuning, and model compression. Seven pre-trained CNN architectures, including VGG19, ResNet, Inception, and MobileNet, were systematically evaluated, each achieving over 95% accuracy by extracting key nail features such as color, shape, and texture. To ensure real-world usability, the models were quantized and optimized for deployment on edge and mobile devices, enabling resource-efficient yet accurate nail melanoma diagnosis.

4.8.3. Viral skin diseases

Viral skin diseases such as monkeypox and chickenpox present with visually similar lesions that are difficult to differentiate clinically, especially in early stages. Rapid and accurate diagnosis is essential to prevent outbreaks and improve patient outcomes. However, traditional DL models for skin disease recognition are often computationally intensive, limiting their use in resource-constrained settings. Lightweight DL architectures provide practical solutions for scalable and efficient diagnosis of these viral skin conditions. Almars [226] proposed DeepGenMon, a lightweight attention-based CNN framework optimized through genetic algorithm-driven hyperparameter tuning. By combining attention mechanisms with CNNs, DeepGenMon enhanced both efficiency and interpretability, achieving up to 98.5% accuracy in classifying monkeypox, chickenpox, and melasma. With its fast inference and low resource consumption, the model demonstrated strong potential for real-time deployment. Biswas and Tešić [227] introduced BinaryDNet53, a binarized CNN for monkeypox classification. Their model integrates binarized weights and biases, reduced cross stage partial blocks, feature fusion, image complexity scoring, and a large-margin SVM classifier. These optimizations minimize computational and power requirements while achiev-

ing state-of-the-art accuracy, outperforming traditional architectures such as ResNet and DarkNet. The design makes BinaryDNet53 particularly well-suited for deployment on low-power, resource-constrained devices. Raha et al. [228] developed a resource-aware monkeypox diagnosis framework leveraging the Extended Monkeypox Skin Images Dataset (EMSID), which includes eight visually similar skin conditions. A high-capacity ConvNeXt-B teacher model transfers knowledge to a lightweight SqueezeNet student via soft-label distillation. Their approach improved classification accuracy from 89.6% to 91.9% while retaining interpretability. With its compact design and reliance on lightweight architectures, the framework supports deployment in edge-based diagnostic systems.

4.8.4. Summary

Lightweight DL research for dermatological disease detection and classification relies on shallow CNNs, ShuffleNet variants, MobileNet variants, EfficientNet variants, compact UNets, KD, pruning and quantization.

4.9. Musculoskeletal diseases

Research on musculoskeletal diseases, including bone tumors and diverse spinal disorders, has seen notable progress with lightweight and efficient DL models that support rapid, reliable analysis on portable and resource-limited devices. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 10.

4.9.1. Osteosarcoma

Osteosarcoma is the most common primary malignant bone tumor, primarily affecting adolescents and young adults. Early and accurate diagnosis is essential to guide treatment planning and improve patient outcomes, but histopathological assessment of bone biopsy samples is time-consuming and requires specialized expertise. Lightweight DL models embedded into diagnostic devices can facilitate rapid and accessible osteosarcoma detection, particularly in point-of-care and low-resource clinical environments. Liu et al. [229] proposed CAMobileNetV3, a lightweight model integrated into an intelli-

Table 9: Summary of lightweight and efficient DL for dermatological disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Baig et al. [208]	Skin cancer	Dermoscopy	Modified ShuffleNet	✓	✗	✗	✗	✗	✗	HAM10000, ISIS-2019, ISIC-2020	Acc: 99.14%	Lightweight architecture
Asif et al. [209]	Skin cancer	Dermoscopy	Shallow CNN	✓	✗	✗	✗	✗	✗	HAM10000	Acc: 98.54%	Model size: 7.7 MB
Hoang et al. [210]	Skin cancer	Dermoscopy	Modified ShuffleNet	✓	✗	✗	✗	✗	✗	HAM10000, ISIC2019	Acc: 86.33%, 82.56% on two respectively	79× fewer parameters than VGG19
Gamil et al. [211]	Skin cancer	Dermoscopy	Modified EfficientNetB0	✓	✗	✗	✗	✗	✗	DermIS, ISIC	Acc: 93%, 91%	Lightweight architecture
Alche et al. [212]	Skin cancer	Dermoscopy	Modified EfficientNetB0	✓	✗	✗	✗	✗	✗	ISIC 2017	Acc: 87.33%	Lightweight architecture
Ding et al. [213]	Skin cancer	Dermoscopy	Modified MobileViT	✓	✗	✗	✗	✗	✗	ISIC2017, ISIC2018, PH ²	Acc: 89.5%, 93.2%, 91.4% respectively	Lightweight architecture
Yuan et al. [214]	Skin cancer	Dermoscopy	Modified U-Net	✓	✗	✗	✗	✗	✗	ISIC2017, ISIC2018, PH ²	mDice: 87.66 ± 0.221%, 88.48 ± 0.109%, 94.42 ± 0.124% respectively	Model size: 1.19 MB
Thaker & Asif [215]	Skin cancer	Dermoscopy	Shallow CNN	✓	✗	✗	✓	✗	✗	HAM10000	ROC-AUC: 0.85	2× smaller
Saha et al. [216]	Skin cancer	Dermoscopy	Shallow CNN	✓	✗	✗	✓	✗	✗	HAM10000, ISIC2019	Acc: 88.89%, 83.42% respectively	0.29M parameters
Islam et al. [217]	Skin cancer	Dermoscopy	Shallow CNN	✓	✗	✓	✓	✗	✗	HAM10000, Kaggle skin cancer data	Acc: 98.75%, 98.94% respectively	Model size: 469.77 KB
Kabir et al. [218]	Skin cancer	Dermoscopy	Shallow CNN	✓	✗	✗	✓	✗	✗	HAM10000	Acc: 85.7%	0.35M parameters
Khan et al. [219]	Skin cancer	Dermoscopy	Shallow CNN	✓	✗	✗	✓	✗	✗	ISIC	Acc: 91.7%	0.26M parameters
Wang et al. [220]	Skin cancer	Dermoscopy	MobileNetV2	✓	✗	✗	✓	✗	✗	ISIC2019	Acc: 84.6%	3.5M parameters
Zeng et al. [221]	Skin cancer	Dermoscopy	ShuffleNetV2	✓	✗	✗	✓	✗	✗	ISIC2019	Acc: 92.83%	1.09M parameters
Zhang & Lu [222]	Skin cancer	Dermoscopy	EfficientViT	✓	✗	✗	✓	✗	✗	ISIC2016, PH ²	Dice: 89.51%, 88.56% respectively	0.24M FLOPS
Medhat et al. [223]	Skin cancer	Dermoscopy/non-dermoscopy	AlexNet	✗	✓	✗	✗	✗	✗	PAD-UFES-20, MED-NODE, PH ²	Acc: 97.62%, 96.79%, 96.75% respectively	Inference time: 0.45, 0.28, 0.3 minutes respectively
Xiang et al. [224]	Skin cancer	Dermoscopy	MnasNet	✓	✓	✗	✗	✗	✗	HAM10000	Acc: 83.5%	Model size: 12.5 MB
Hussain et al. [225]	Nail melanoma	Dermoscopy	ResNet152V2	✗	✗	✓	✗	✗	✗	Nail-Melanoma-350	Acc: 89%	Model size: 59.3 MB
Almars [226]	Viral skin diseases	Images	Attention-based CNN + genetic algorithm	✓	✗	✗	✗	✗	✗	Two monkeypox datasets from kaggle	Acc: 98.5% on dataset 1 and 98.2% on dataset 2	Inference time: 2.9764s on dataset 1 and 2.1753 s on dataset 2
Biswas & Tešić [227]	Viral skin diseases	Images	DarkNet53	✗	✗	✓	✗	✗	✗	MSLD v2.0	Acc: 85.78%	0.37M parameters
Raha et al. [228]	Viral skin diseases	Images	SqueezeNet	✓	✗	✗	✓	✗	✗	EMSID	Acc: 91.89%	0.7M parameters

gent microscope for osteosarcoma diagnosis from cellular microscopy images. Their approach embeds a coordinate attention mechanism into MobileNetV3, enhancing feature representation while preserving a compact design. CA-MobileNetV3 achieved high classification accuracy and significantly outperformed heavier architectures in terms of efficiency. Its seamless integration into a custom-built microscope enables real-time, on-device classification, accelerating diagnostic workflows and supporting practical clinical use.

4.9.2. Vertebra segmentation

Vertebra segmentation from CT scans is a critical step in spinal analysis for diagnosing fractures, deformities, and degenerative conditions, as well as for surgical planning. Manual segmentation is highly labor-intensive and time-consuming, while conventional DL models for 3D medical imaging are often computationally expensive, making them impractical for real-time or resource-constrained deployment. Lightweight DL frameworks offer a promising solution to balance accuracy and efficiency. Serrador et al. [230] proposed a two-step KD framework for individual vertebra segmentation in CT images. Their approach employs a high-capacity 3D U-Net teacher to guide a shallower student model through two distillation strategies: soft

output distillation and logit matching. By transferring structural and probabilistic knowledge, the student achieved a dice score improvement from 75.78% to 84.70%, while requiring significantly fewer parameters and faster inference time.

4.9.3. Lumbar spondylosis

Lumbar spondylosis, a degenerative spine condition associated with aging, is characterized by disc degeneration, bone spurs, and reduced spinal flexibility. It is a leading cause of chronic lower back pain and disability worldwide. Radiographic evaluation remains the primary diagnostic tool, but manual interpretation of lumbar X-rays can be challenging and time-intensive. Lightweight DL models offer a practical pathway to automate lumbar spondylosis detection while ensuring efficiency and accessibility in low-resource healthcare environments. Nguyen et al. [231] proposed a KD framework for lumbar spine X-ray classification, employing EfficientNet-B4 as the teacher and MobileNetV2 as the student. Their method, trained on 2,000 lumbar spine images, achieved 91.0% accuracy with strong precision and recall, significantly improving the performance of the compact student model. By transferring representational power from the high-capacity teacher to the lightweight MobileNetV2, the approach enables reliable, resource-efficient

lumbar spondylosis detection suitable for deployment in constrained clinical environments.

4.9.4. Spinal disorders

Spinal disorders, such as scoliosis, spinal stenosis, and degenerative disc disease, are common musculoskeletal conditions that can significantly impact quality of life. Accurate diagnosis often relies on X-ray or MRI, but manual interpretation is time-consuming and subject to inter-observer variability. Lightweight DL models offer a pathway to automated, efficient, and accessible spinal disorder analysis that can be deployed in both hospital and mobile healthcare environments. Qing and Qi [232] proposed a lightweight spine X-ray image segmentation framework that distills knowledge from an enhanced U-Net teacher into a Mobile U-Net student. The teacher network integrates residual connections to improve gradient flow and feature learning, as well as a channel attention mechanism to emphasize clinically important features. KD transfers these capabilities to the student model, which achieves higher accuracy than both standard U-Net and Mobile U-Net while significantly reducing computational complexity. This makes the framework suitable for real-time clinical deployment in resource-constrained settings.

4.9.5. Summary

Lightweight DL research in musculoskeletal disease detection and classification employs MobileNet variants, efficient U-Net architectures and KD.

4.10. Reproductive diseases

Reproductive diseases such as cervical cancer have benefited from lightweight and efficient DL models that enable accurate screening and diagnosis on portable, resource-constrained hardware. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 11.

4.10.1. Cervical cancer

Cervical cancer remains one of the leading causes of cancer-related deaths among women worldwide, particularly in low- and middle-income countries where access to regular screening is limited. Pap smear cytology and histopathological analysis are the primary diagnostic methods, but manual interpretation is labor-intensive, subjective, and requires specialized expertise. Lightweight DL models for cervical cancer detection and classification have emerged as efficient, scalable solutions that can improve screening accuracy and accessibility, particularly in resource-constrained clinical environments. Mehedi et al. [233] proposed CCanNet, a lightweight CNN designed for efficient classification of cervical cancer subtypes from pap smear images. The model incorporates squeeze blocks to reduce parameters and residual skip connections to mitigate performance degradation in deeper layers. Evaluated on the SipakMed dataset, CCanNet achieved 98.53% accuracy, outperforming several existing models while maintaining computational efficiency and interpretability. Chen et al. [234] intro-

duced a KD framework for cervical cell classification, transferring knowledge from larger models into compact backbones such as Xception and MobileNetV2. Their approach significantly reduced model size without sacrificing diagnostic accuracy, highlighting its potential for low-cost, AI-assisted cervical cancer screening on embedded devices. In a related effort, Chen et al. [235] developed a lightweight CNN framework with a hybrid loss function that combines cross-entropy, triplet loss, and label smoothing to improve feature discriminability. Using efficient backbones such as GhostNet and ShuffleNetV2, their method achieved high accuracy while minimizing computational cost and memory usage compared to conventional DNNs like DenseNet-121. Pacal [236] presented MaxCerVixT, a lightweight transformer-based architecture for cervical cancer detection from pap smear images. By integrating a MaxViT backbone with ConvNeXtV2 blocks and GRN-based MLP layers, MaxCerVixT reduced computational overhead while maintaining high accuracy, offering a strong balance between diagnostic performance and efficiency in resource-constrained settings.

Beyond classification, Kang and Li [237] proposed CerviSegNet-DistillPlus, an advanced framework for cervical cancer detection and segmentation. Building on DeepLabV3+, their model integrates KD, model pruning, and test-time augmentation. A multi-component loss function combining soft label distillation, feature map distillation, and segmentation loss enables efficient transfer from a teacher model to a lightweight student. Tested on Cx22, DTU/Herlev, and SIPaKMeD datasets, CerviSegNet-DistillPlus achieved dice coefficients up to 0.9892, surpassing comparable methods by up to 2% while significantly reducing parameters.

4.10.2. Summary

Lightweight DL research for reproductive disease detection and classification employs efficient backbone models such as Xception and ShuffleNetV2, KD and pruning.

4.11. Oral diseases

Oral diseases such as oral squamous cell carcinoma and dental periapical lesions have seen increasing use of lightweight and efficient DL models for automated diagnosis from histopathology and radiographic images. These approaches enable fast inference and edge deployment in clinics with limited computational resources. A consolidated summary of representative studies, modalities, datasets, and efficiency techniques is presented in Table 12.

4.11.1. Oral squamous cell carcinoma

Oral squamous cell carcinoma (OSCC) is the most common type of oral cancer, accounting for the majority of malignancies in the oral cavity. Early detection is critical for improving prognosis, yet diagnosis based on histopathological examination of mucosa tissue is time-consuming and dependent on expert interpretation. Lightweight DL models provide a pathway toward efficient, automated, and accessible OSCC diagnosis, particularly for deployment in real-time clinical and low-resource settings. Shabir et al. [238] proposed LWFDLTL, a lightweight

Table 10: Summary of lightweight and efficient DL for musculoskeletal disease detection.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Liu et al. [229]	Osteosarcoma	Microscopy	Modified MobileNet V3	✓	✗	✗	✗	✗	✗	Data collected at Ma'anshan People's Hospital of Anhui Province	Acc: 94.32%	Model size: 5.33 MB
Serrador et al. [230]	Vertebra segmentation	CT	3D U-Net	✓	✗	✗	✓	✗	✗	VerSe19, VerSe20	Dice: 84.46%, 84.33% respectively	0.37M parameters
Nguyen et al. [231]	Lumbar spondylosis	X-ray	MobileNetV2	✓	✗	✗	✓	✗	✗	Data collected at Kien Giang General Hospital and Trung Cang General Clinic	Acc: 91.0%	3.4M parameters
Qing & Qi [232]	Spinal disorders	X-ray	Mobile U-Net	✓	✗	✗	✓	✗	✗	SpineWeb16	Dice: 93.67%	Model size: 6.56 MB

Table 11: Summary of lightweight and efficient DL for cervical cancer detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Mehedi et al. [233]	Cervical cancer	Pap smear	CNN with squeeze blocks + residual skips	✓	✗	✗	✗	✗	✗	SipakMed	Acc: 98.53%	1.27M parameters
Chen et al. [234]	Cervical cancer	Pap smear	Xception	✓	✗	✗	✓	✗	✗	Herlev Pap smear	Acc: 72.25%	Model size: 88 MB
Chen et al. [235]	Cervical cancer	Pap smear	ShuffleNetV2	✓	✗	✗	✗	✗	✗	SipakMed	Acc: 96.18%	6× fewer parameters compared to Densenet-121
Pacal [236]	Cervical cancer	Pap smear	MaxViT + ConvNeXtV2 + GRN-MLP	✓	✗	✗	✗	✗	✗	SIPaKMeD and Mendeley LBC	Acc: 99.02%, 99.48% respectively	Inference time: 6 ms
Kang & Li [237]	Cervical cancer	Pap smear	Double swin-transformer	✗	✓	✗	✓	✗	✗	Cx22, DTU/Herlev, SIPaKMeD	Acc: 94%, 93.65%, 92.49% respectively	Model size: 29.8 MB

fusion deep transfer learning framework for histopathological OSCC diagnosis. Their approach combines EfficientNet-B0 with a custom shallow CNN, enabling the extraction of both deep hierarchical and shallow discriminative features. This fusion improves classification accuracy while maintaining low computational cost and fast inference. Designed for practical deployment, the model offers a clinically viable solution for automated OSCC screening and real-time decision support.

4.11.2. Dental periapical lesions

Dental periapical lesions, commonly caused by infections or trauma to the tooth pulp, are among the most frequent findings in dental radiographs. Accurate and timely detection is essential for guiding treatment decisions such as root canal therapy, but manual interpretation of radiographs is subject to variability and can be time-consuming. Lightweight DL models offer the potential to automate lesion detection in real time, enabling efficient and accessible dental diagnostics even in resource-limited clinical environments. Latke and Narawade [239] proposed a lightweight DL framework for detecting dental periapical lesions that combines Retinex-based image enhancement with a shallow CNN architecture. Their method improves contrast and illumination consistency in dental radiographs through Retinex preprocessing, which facilitates clearer visualization of lesion boundaries. The shallow CNN, designed with fewer layers and parameters, achieves efficient lesion classification with minimal computational cost. This combination allows for fast inference

and real-time deployment, making the framework suitable for integration into dental imaging workflows.

4.11.3. Summary

Lightweight DL research for oral disease detection and classification relies on compact CNN architectures and efficient backbone models such as EfficientNet.

4.12. Gastrointestinal diseases

Research on gastrointestinal diseases, including colon polyps and rectal cancer, has seen notable progress with lightweight and efficient DL models that enable real-time analysis and rapid decision-making on resource-constrained hardware. These models support rapid decision-making during endoscopy and preoperative planning from CT, while reducing computational and memory demands. A consolidated summary of representative studies, modalities, datasets, and efficiency techniques is presented in Table 13.

4.12.1. Colon polyps

Colon polyps are abnormal tissue growths in the colon that can progress into colorectal cancer if left undetected or untreated. Early and accurate detection is essential for effective screening and intervention, yet manual endoscopic analysis is time-consuming and prone to human error. Lightweight DL models offer a promising solution by enabling efficient, real-time classification of colon polyps during colonoscopy, particularly in resource-constrained clinical environments. Sharma et

Table 12: Summary of lightweight and efficient DL for oral disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Shabir et al. [238]	Oral squamous cell carcinoma	Histopath	EfficientNetB0 + shallow CNN	✓	✗	✗	✗	✗	✗	Histopathological imaging database for oral cancer analysis	Acc: 98.9%	Inference time: 56 ms
Latke & Narawade [239]	Dental periapical lesions	Dental Radiographs	Shallow CNN	✓	✗	✗	✗	✗	✗	Dataset consisting of intra-oral periapical radiographic images	Acc: 95.8%	Lightweight architecture

al. [240] proposed LPNet, a lightweight CNN specifically designed for colon polyp classification. Their approach integrates the discrete wavelet transform as a pooling strategy, replacing conventional pooling layers to better preserve essential spatial and frequency-domain features while reducing image dimensions. Combined with compact convolutional blocks, LPNet achieved significant reductions in parameter count and computational complexity, making it highly efficient for real-time inference. The model demonstrated strong classification accuracy while maintaining low memory requirements, supporting its deployment in portable diagnostic systems and point-of-care applications.

4.12.2. Rectal cancer

Rectal cancer is a major subtype of colorectal cancer, where accurate preoperative staging particularly T-staging is critical for determining treatment strategies such as surgery, chemotherapy, or radiotherapy. Manual staging from CT images requires expert radiologists and can be prone to variability, while conventional DL models for staging are often computationally intensive. Lightweight architectures offer a scalable and efficient solution for accurate, real-time staging in resource-constrained healthcare environments. Yan et al. [241] proposed LRCTNet, a lightweight DNN tailored for preoperative T-staging of rectal cancer from CT scans. Their framework leverages transfer learning from a pretrained swin transformer teacher and employs KD combining both hard-label and soft-label losses. To enhance efficiency, LRCTNet introduces two novel modules, LightFire and ResLightFire, which enable discriminative feature extraction while minimizing the number of parameters. Validated on clinical datasets, the model achieved strong staging accuracy while maintaining low computational overhead, demonstrating suitability for deployment in real-world, resource-limited clinical settings.

4.12.3. Summary

Lightweight DL research for gastrointestinal disease detection and classification relies on compact CNN architectures, including custom designs such as DWT-enhanced CNNs and Fire/ResLightFire-based models, and KD.

4.13. Others

A growing body of work targets diseases that fall outside the major organ-specific sections but are highly relevant for real-

world deployment of lightweight DL in healthcare. These include breast cancer, diabetes and diabetic foot ulcers, thoracic diseases, and multi-disease pipelines spanning heterogeneous modalities and settings. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 14.

4.13.1. Breast cancer

Breast cancer remains one of the leading causes of cancer-related mortality among women worldwide, where accurate and timely diagnosis is critical for effective treatment. Traditional diagnostic approaches, such as histopathology and digital breast imaging, require substantial expertise and computational resources, posing challenges for real-time deployment in resource-limited clinical settings. Lightweight and efficient DL architectures offer promising solutions for enabling accurate, on-device diagnostic support. Kumar et al. [242] presented MobiHisNet, an efficient CNN based on depthwise separable convolutions inspired by MobileNet, specifically designed for real-time breast cancer diagnosis on edge devices. PTQ to FP16 precision further reduced computational costs while preserving diagnostic accuracy. Similarly, Ogundokun et al. [243] introduced MobileNet-SVM, which leverages MobileNet for low-cost feature extraction coupled with an SVM classifier, achieving high diagnostic accuracy in IoMT-based imaging systems with minimal resource usage. Beyond compact CNN designs, Samala et al. [244] proposed an evolutionary pruning pathway, where transfer-learned deep CNNs undergo feature selection and random forest classification, followed by genetic algorithm-based pruning. This approach reduced neurons by 87%, parameters by 34%, and convolutional operations by 95%, while slightly improving AUC from 0.88 to 0.90. Choudhary et al. [245] also addressed efficiency by integrating structured filter pruning into transfer learning with VGG19, ResNet34, and ResNet50, achieving up to 92.07% accuracy while reducing FLOPs by 30–63%, thereby making pruned models suitable for point-of-care use.

More recently, Jiang et al. [246] developed a spatially-aware lightweight model that integrates multi-scale depthwise separable convolutions through a swift parallel convolution module and enhances local-global fusion with a spatially-aware attention module. This design reduced FLOPs and latency while maintaining strong performance across heterogeneous pathology images. Similarly, Modu et al. [247] applied KD from a large teacher model into a compact CNN student network,

Table 13: Summary of lightweight and efficient DL for gastrointestinal disease detection and classification.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Sharma et al. [240]	Colon polyps	Colonoscopy	CNN with DWT pooling	✓	✗	✗	✗	✗	✗	Kvasir-V1	Acc: 93.55%	3.2M parameters
Yan et al. [241]	Rectal cancer	CT	CNN consisting of LightFire and ResLightFire modules	✓	✗	✗	✓	✗	✗	Data collected at Xinqiao Hospital	Acc: 95.79%	0.407M parameters

incorporating PCA-based preprocessing and optimal distillation strategies to achieve high diagnostic accuracy with 87% fewer parameters, enabling deployment on embedded IoT devices. Complementary to architectural compression, Garifulla et al. [248] conducted a case study on CNN quantization for portable medical devices, applying half-precision, dynamic range, and integer quantization to models such as VGG16, ResNet34, and GoogLeNet. Their results showed near-original accuracy with up to 97% faster inference on heterogeneous accelerators, highlighting quantization’s role in energy-efficient, real-time diagnostics.

4.13.2. Diabetes and diabetic foot ulcer

Diabetes is a chronic metabolic disorder with rising global prevalence, often accompanied by severe complications such as diabetic foot ulcers (DFUs), which can lead to amputation if not detected early. Accurate, real-time diagnosis and monitoring remain challenging in resource-constrained clinical and IoT environments, where computationally efficient models are essential. Recent advances in lightweight DL architectures have shown strong potential for enabling continuous, scalable diabetes and DFU management. Yin et al. [249] introduced DiabDeep, a pervasive diabetes diagnosis framework that integrates wearable sensor data with a custom DiabNN architecture. Leveraging physiological, environmental, and demographic features, DiabNN employs a grow-and-prune synthesis paradigm to achieve high classification accuracy while reducing FLOPs by up to $454.5\times$ compared to conventional models. Designed for both server- and edge-side deployment, DiabDeep enables continuous, non-invasive diabetes detection and monitoring without disrupting daily life. For DFU detection, Amjad et al. [250] developed DFU-LWNet, a compact student model distilled from an InceptionV3 teacher. Incorporating MBConv blocks with SE for parameter efficiency, DFU-LWNet achieved 96.23% accuracy on a public DFU dataset and 93.13% on a thermal image dataset, with only 0.49M parameters. This design preserves high diagnostic accuracy while minimizing computational cost, enabling interpretable and fast deployment in real-world healthcare systems.

4.13.3. Thoracic diseases

Thoracic diseases such as pneumonia, TB, and pulmonary fibrosis remain leading causes of morbidity worldwide, where CXRs are the most common screening tool. However, accurate interpretation is time-consuming, expert-dependent, and computationally demanding when deploying conventional DL mod-

els in real-time or resource-constrained clinical environments. To address these challenges, lightweight CNN architectures enhanced with attention mechanisms have emerged as effective solutions for efficient and accurate thoracic disease detection. Sangeroki and Cenggoro [251] proposed a lightweight model that integrates ShuffleNetV2 with a modified CBAM. The ShuffleNetV2 backbone ensures computational efficiency, while the enhanced CBAM module improves diagnostic performance by directing the model’s focus toward disease-relevant regions in CXRs. This combination achieved a favorable balance of speed and accuracy, demonstrating strong potential for real-time thoracic disease screening in point-of-care and mobile health applications.

4.13.4. Multiple diseases

Medical AI research increasingly emphasizes multi-disease diagnosis across heterogeneous modalities including X-ray, CT, MRI, histopathology, auscultation, and sensor-based data reflecting real-world clinical demands. Deploying such systems in resource-constrained settings or IoT-enabled healthcare requires compact yet robust architectures that optimize the trade-off between accuracy, interpretability, and efficiency. Accordingly, recent approaches span lightweight backbone design, feature fusion, capsule networks, pruning, quantization, KD, and federated learning, collectively advancing scalable multi-disease AI. Asif et al. [252] introduced LiteFusionNet, a hybrid model combining MobileNet and MobileNetV2 via feature fusion to enhance representational diversity. By balancing depth and efficiency, it achieved up to 99% accuracy across five imaging modalities including brain MRI, dermatology, CT, CXR, and histopathology, consistently outperforming larger CNNs while maintaining a lightweight footprint. Zhou et al. [253] addressed the limited capacity of lightweight CNNs by proposing the Spatial-Channel Mixed Global–Local (SCM-GL) attention module, which decomposes feature maps into local and global contextual components through pixel–neighbor relationships. When integrated into backbones, SCM-GL improved discriminative learning across brain MRI, CXR, and osteosarcoma datasets, significantly boosting classification metrics such as recall, specificity, and F1-score, and outperforming conventional attention designs. Maheswari and Gopalakrishnan [254] developed SECNet, a multimodal framework combining SE blocks, capsule networks, and Woodpecker Mating Optimization for feature selection. Designed for X-ray, CT, and MRI fusion, SECNet achieved high efficiency while maintaining accuracy, demonstrating applicability in real-time diagno-

sis across modalities. Similarly, Yadav and Dhage [255] proposed TE-CapsNet, a streamlined capsule network preserving spatial hierarchies with fewer layers and pre-trained CNN integration. Tested on small datasets, it demonstrated a strong accuracy–efficiency balance, underscoring capsule networks’ utility in low-data regimes.

Fernandes and Yen [256] proposed DNNDeepeningPruning, a two-phase approach combining network growth via residual block addition with pruning to minimize FLOPs while maintaining diagnostic accuracy. Tested on medical imaging datasets, it produced compact, high-performing models. Shahid et al. [257] extended pruning to the federated setting with PruneFL, reducing communication overhead and maintaining accuracy on MedMNIST datasets, showing strong potential for privacy-preserving decentralized healthcare AI. Extending beyond static compression, Baccour et al. [258] proposed an RL–based dynamic pruning strategy guided by XAI. Their approach formulates online distributed inference as a non-linear integer programming problem, solved with RL to balance accuracy and resource usage adaptively. Applied to VGG-16 across multiple datasets including COVID-19, breast cancer, and DFU, the method significantly reduces computational and communication overhead while maintaining high accuracy, making it particularly suitable for distributed medical IoT devices with limited resources.

KD has become a pivotal strategy for building compact yet high-performing models. Qin et al. [259] proposed a distillation module transferring semantic region differences for boundary-preserving segmentation on LiTS and KiTS datasets. Gorade et al. [260] advanced this with hierarchical layer-selective feedback distillation, which explicitly identifies what and from where to distill, improving Dice scores for tumor segmentation. Wen et al. [261] designed boundary-guided KD to capture fine anatomical boundaries. Choi [262] combined KD with semi-supervised cross-teaching for abdominal organ segmentation, achieving strong Dice scores on FLARE. Sun et al. [263] proposed FKD-Med, aligning teacher soft labels and hard labels in a dual-output strategy, compressing student U-Nets up to 1027× smaller without accuracy loss on polyp and CXR segmentation. Gordienko et al. [264] generalized distillation with a bucket of models ensemble, showing that compact ResNet students can rival larger teachers across MedMNIST. Khurshid et al. [265] extended this with AssistDistil, combining pairwise distillation and latent boosting for compact segmentation models. Liao et al. [266] introduced joint feature soft label KD, transferring both intermediate representations and prediction distributions from a Res-Transformer U-Net to a ResU-Net student, yielding robustness across COVID-19, IDC, and stroke datasets.

Malibari [267] proposed equilibrium optimizer optimized lightweight automatic modulation classification network EOLWAMCNet, for chronic kidney and cardiovascular disease detection on edge devices. Singh et al. [268] expanded this for chronic liver and brain disease prediction. Alabbasy et al. [269] demonstrated distillation-based compression for fog/edge deployment in COVID-19 and malaria diagnosis, offering superior compression–speed trade-offs compared to pruning or

quantization. For cancer detection, Attallah et al. [270] fused ShuffleNet, MobileNet, and SqueezeNet features with PCA and Fast Walsh Hadamard Transform (FHWt)-DWT fusion, achieving 99.6% accuracy on lung and colon histopathology datasets. Hasan et al. [271] proposed a multi-scale CNN for histopathology with integrated explainability. Shahadat et al. [272] designed a 1D-CNN with SE blocks that achieved 100% accuracy on LC25000 with just 0.35M parameters and 6.4M FLOPs, setting a benchmark for extreme efficiency. Extending lightweight modeling to auscultation, Zhang et al. [273] built an AI-powered stethoscope with a 36K-parameter CNN and Best Discrepancy Forest classifier, achieving near-perfect real-time classification of cardiac and respiratory diseases on edge hardware.

4.13.5. Summary

Lightweight DL research in other diseases category predominantly relies on lightweight-by-design architectures such as depthwise separable convolutions, ShuffleNet/MobileNet variants, and streamlined U-Nets, while pruning, quantization, and KD are applied selectively to further reduce memory footprint and inference cost.

5. Lightweight and efficient DL in remote monitoring

Remote healthcare monitoring has emerged as a critical paradigm for preventive, continuous, and patient-centered care, particularly in aging populations and resource-constrained environments. By leveraging wearable sensors, mobile devices, and IoMT platforms, it enables real-time assessment of vital signs, daily activities, and mental health outside traditional clinical settings. However, the effectiveness of such systems depends on the efficiency of operating on low-power, memory-constrained hardware while ensuring reliability and scalability. Recent research has emphasized lightweight and efficient DL frameworks to achieve accurate yet computationally feasible models. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 15.

5.1. Human activity recognition and elderly care

Human activity recognition (HAR) is a cornerstone of modern healthcare monitoring for elderly care, fall detection, and assisted living. By using wearable sensors and edge devices, HAR systems enable continuous, unobtrusive, and real-time monitoring in resource-constrained environments. To achieve scalability and reliability in such settings, recent studies have emphasized lightweight DL architectures to balance accuracy with efficiency. Baltabay et al. [274] introduced a lightweight HAR framework that integrates biosensors, ECG, and inertial signals with single-board edge computers. Sensor data is converted into image-like representations and processed via CNNs, Random Forests, and hybrid LSTM–GRU models, with MobileNet-based transfer learning for comparison. Their study showed high accuracy and highlighted trade-offs between execution speed, parameter count, and memory footprint, showing

Table 14: Summary of lightweight and efficient DL for other diseases.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Kumar et al. [242]	Breast cancer	Histopath	MobileNet	✓	✗	✓	✗	✗	✗	BreakHis	Acc: 89.59%	Model size: 2.04 MB
Ogundokun et al. [243]	Breast cancer	Histopath	MobileNet + SVM	✓	✗	✗	✗	✗	✗	BreakHis	Acc: 91.3%	Lightweight architecture
Samala et al. [244]	Breast cancer	Mammography	CNN	✗	✓	✗	✗	✗	✗	Breast cancer data from Univ. of Michigan & MGH	AUC: 0.90±0.04	21.59M parameters
Choudhary et al. [245]	Breast cancer	Histopath	VGG/ResNet	✗	✓	✗	✗	✗	✗	IDC, BreakHis	Acc: up to 92.07%	30–63% FLOPs reduced
Jiang et al. [246]	Breast cancer	Histopath	Spatially-aware attention swift parallel convolution network	✓	✗	✗	✗	✗	✗	BreakHis, BACH	Acc: 99.27%, 99.39% respectively	0.359M parameters
Modu et al. [247]	Breast cancer	Omics/cytology	Shallow CNN	✓	✗	✗	✓	✗	✗	PBCNT, BCDB, WDBC	Acc: 99.3%	87% fewer parameters than teacher
Garifulla et al. [248]	Breast cancer	Ultrasound	GoogLeNet	✗	✗	✓	✗	✗	✗	Data collected at KNU Chilgok hospitals	Acc: 88%	Model size: 15.6 MB
Yin et al. [249]	Diabetes	Wearable Sensors	Hierarchical LSTM	✓	✓	✗	✗	✗	✗	Data from 52 participants	Acc: 95.3%	3.3K parameters
Amjad et al. [250]	DFU	Images	CNN with depth-wise separable convolutions	✓	✗	✗	✓	✗	✗	DFU data from Kaggle	Acc: 96.23%	0.49M parameters
Sangeroki & Cenggoro [251]	Thoracic diseases	CXR	ShuffleNetV2 + CBAM	✓	✗	✗	✗	✗	✗	CXR-14	AUC: 0.786141	Lightweight architecture
Asif et al. [252]	Multiple diseases	MRI/Skin/CT/X-ray/histology	MobileNet + MobileNetV2	✓	✗	✗	✗	✗	✗	Brain MRI, Monkeypox, COVID-CT, CXR, BreakHis	Acc: 97.33%, 91.11%, 99.00%, 98.15%, 92.11%	6.67M parameters
Zhou et al. [253]	Multiple diseases	CXR, Brain MR, Osteosarcoma histology	Spiking cortical model	✓	✗	✗	✗	✗	✗	CXR (Kaggle), brain MR (figshare), TCAI	Acc: 96.74%, 86%, 89.96%	1.71M parameters
Maheswari & Gopalakrishnan [254]	Multiple diseases	X-ray, CT, MRI	Squeeze Excitation Capsule Network	✓	✗	✗	✗	✗	✗	X-ray, NCCD-CT, BRATS-MRI	Acc: 98.8%, 99.5%, 99.4%	Reduced inference time vs SOTA
Yadav & Dhage [255]	Multiple diseases	X-ray, MRI	Time Efficient-CapsNet	✓	✗	✗	✗	✗	✗	COVID-19 CXR, Brain tumor (Kaggle)	Acc: 97.55%, 97.76%	Avg training time: 531 s, 674 s
Fernandes & Yen [256]	Multiple diseases	Dermoscopy, X-ray	DNN	✗	✓	✗	✗	✗	✗	ISIC-2016, CXR	Acc: 81.01%, 85.10%	FLOPs reduction: 83.27%, 62.7%
Shahid et al. [257]	Multiple diseases	CT	LeNet5	✓	✓	✗	✗	✗	✗	MedMNIST (OrganAMNIST, OrganCMNIST, OrganSMNIST)	Acc: 84.5%, 82.41%, 82.41% respectively	Model size: 0.1326 MB
Baccour et al. [258]	Multiple diseases	Multiple	VGG16	✗	✓	✗	✗	✗	✗	COVID-19, Breast cancer, DFU	R^2 : 0.98857, 0.9871, 0.9896	Reduced computational cost
Qin et al. [259]	Multiple diseases	CT	MobileNetV2	✓	✗	✗	✓	✗	✗	LiTS17, KiTS19	Dice: 0.595 ± 0.004, 0.684 ± 0.006	2.2M parameters
Gorade et al. [260]	Multiple diseases	CT	ResNet18	✓	✗	✗	✓	✗	✗	KiTS, LiTS	Dice: 52.18 ± 2.55%, 48.75 ± 2.23%	11.6M parameters
Wen et al. [261]	Multiple diseases	Xray, MRI, H&E, Fundoscopic	ESPNet	✓	✗	✗	✓	✗	✗	Montgomery, CHAOS, Glas, DRISHTI-GS	Dice: 93.9 ± 1.9 %, 86.0 ± 1.7%, 84.5 ± 3.9%, 83.7 ± 3.9%	0.3631M parameters
Choi [262]	Multiple diseases	CT	Residual-USE-Net	✓	✗	✗	✓	✗	✗	MICCAI FLARE 2022	Dice: 0.8520	5.2M parameters
Sun et al. [263]	Multiple diseases	CT, CXR	tinyUnet	✓	✗	✗	✓	✗	✗	CVC-ClinicDB, Chest-Xray	Acc: 91.23%, 95.7%	Up to 1027× smaller than teacher
Gordienko et al. [264]	Multiple diseases	Blood smear, dermoscopic, histological, retina fundus	ResNet8	✓	✗	✗	✓	✗	✗	MedMNIST (BloodMNIST, DermaMNIST, PathMNIST, RetinaMNIST)	Acc: 95.2%, 77.6%, 88.3%, 55% respectively	Efficient student
Khurshid et al. [265]	Multiple diseases	Iris, dermoscopic, X-ray	Lightweight Attention U-Net	✓	✗	✗	✓	✗	✗	IIITD cataract, HAM10000, PH2, Shenzhen, Montgomery	Mean Dice: 0.9336, 0.9270, 0.8957, 0.9590	50.99K parameters
Liao et al. [266]	Multiple diseases	X-ray, histopathology, CT	ResU-Net	✓	✗	✗	✓	✗	✗	COVID-19 Radiography, BreakHis, IDC, Brain Stroke	Acc: 91.2%, 89.7%, 88.1%, 80.2% respectively	8.032M parameters
Malibari [267]	Multiple diseases	Physiological data	EO-LWAMCNet	✓	✗	✗	✗	✗	✗	Heart disease, CKD	Acc: 93.5%, 94% respectively	Recognition speed: 10.5 tasks/s
Singh et al. [268]	Multiple diseases	Clinical, MRI	EO-LWAMCNet	✓	✗	✗	✗	✗	✗	Chronic Liver Disease, Brain Disease	Acc: 94.8%, 95% respectively	Lightweight architecture
Alabbasy et al. [269]	Multiple diseases	Xray, microscopic blood images	Lightweight DNNs	✓	✗	✗	✓	✗	✗	COVID-19, malaria (Kaggle)	Acc: 98.93%, 98.72% respectively	Model sizes: 245.76 KB, 156.15 KB respectively
Attallah et al. [270]	Multiple diseases	Histopath	ShuffleNet + MobileNet + SqueezeNet	✓	✗	✗	✗	✗	✗	LC25000	Acc: 99.6%	Lightweight architecture
Hasan et al. [271]	Multiple diseases	Histopathology	Lightweight multi-scale CNN	✓	✗	✗	✗	✗	✗	LC25000	Acc: 99.2%	1.1M parameters
Shahadat et al. [272]	Multiple diseases	Histopath	1D-CNN + SE layers	✓	✗	✗	✗	✗	✗	LC25000	Acc: 100%	0.35M parameters
Zhang et al. [273]	Multiple diseases	Audio	Lightweight CNN	✓	✗	✗	✗	✗	✗	ICBHI, heart sound	Acc: 99.94%	36K parameters

that Random Forests excel in speed and LSTM–GRU achieves compactness for real-time deployment. Yi et al. [275] proposed a hybrid approach that first applies statistical feature extraction to separate static from dynamic tasks. Static activities are classified with Random Forest, while dynamic activities are processed with a CNN compressed via IMP and float16 quantization. This design achieved F1 scores above 0.94 on UCI and WISDM datasets with large reductions in memory and computation.

KD has also been leveraged for compact HAR. Deng et al. [276] presented LHAR, a teacher–student framework where ensembles of DeepConvLSTM teachers transfer knowledge to a lightweight student with depthwise separable convolutions. Combined with multi-channel augmentation and class-balanced training, LHAR achieved strong accuracy and robustness across Opportunity and PAMAP2 datasets under cross-person settings. Fahim et al. [277] advanced this with a self-revision mechanism, where a distilled student retrain on its own outputs to refine features. Applied to accelerometer-based eldercare monitoring, it minimized memory and energy demands while improving accuracy and interpretability.

For safety-critical fall detection, Iswara et al. [278] developed a low-rank factorization–based DNN for wearable accelerometer data. By compressing the model through low-rank decomposition, they reduced parameters by 53.8% while retaining 97% accuracy, enabling real-time monitoring on-device.

5.2. Blood pressure monitoring

Blood pressure (BP) monitoring is essential for cardiovascular health, yet traditional cuff-based methods are impractical for continuous use. Lightweight DL has enabled portable, real-time BP estimation on edge devices. Sun et al. [279] introduced tinyML-based BP estimation from PPG signals by adapting CNNs including AlexNet, LeNet, SqueezeNet, ResNet and MobileNet with architecture tweaks, pruning, and quantization for resource-constrained hardware. Evaluated on thousands of ICU records and multiple edge platforms, the system delivered accuracy comparable to server-based solutions.

5.3. Motor assessment

Motor function assessment is a critical task in rehabilitation and clinical neurology, particularly for conditions such as CP, stroke, and neurodegenerative diseases. Traditional assessments based on clinician observation can be subjective and time-intensive, underscoring the need for automated, objective, and efficient systems. Vision-based approaches using skeletal data and temporal modeling have shown promise, but large graph-based architectures often remain computationally prohibitive for real-time or edge deployment. To address this, Zhao et al. [280] proposed a KD framework for lightweight vision-based motor assessment. Their method employs a spatial-temporal graph convolutional network as the teacher model, which represents the human skeleton as a graph and applies spatial graph convolutions to capture inter-joint relationships alongside temporal convolutions to model motion dynamics. A compact student model is then trained using both ground-truth

cross-entropy loss and mean-squared error alignment with the teacher’s outputs and intermediate features. This dual supervision enables the student to achieve competitive accuracy while significantly reducing computational cost and improving inference speed.

5.4. Emotion recognition

Emotion recognition is increasingly recognized as an important component of intelligent healthcare systems, enabling personalized interventions in mental health monitoring, stress management, and elderly care. Conventional multimodal frameworks often rely on high-dimensional physiological data such as EEG, ECG, and facial expressions, but their computational complexity and sensor requirements limit feasibility for real-world, resource-constrained deployment. Recent work has explored lightweight DL architectures to deliver practical, scalable solutions. Zhang et al. [281] proposed an LLM-enhanced multi-teacher KD framework for modality-incomplete emotion recognition. The approach trains a GRU-based student model that relies solely on easily obtainable galvanic skin response signals, extracting temporal features with low computational overhead. Knowledge is distilled from two complementary teacher networks: LLM that captures temporal EEG features with causal masking and confidence indicators, and a graph convolutional network that models spatial EEG dependencies. This dual-teacher distillation enables the lightweight student to integrate richer multimodal representations, even when operating with single-signal input. Evaluated on the DEAP and MAHNOB-HCI datasets, the framework outperformed existing baselines, demonstrating the potential of LLM-guided distillation for efficient, modality-reduced emotion recognition.

5.5. Indoor localization

Accurate indoor localisation is a key enabler for healthcare applications, such as elderly monitoring, rehabilitation support, and context-aware assisted living. Unlike GPS, which is unreliable indoors, healthcare IoT systems rely on wireless signals such as Bluetooth Low Energy (BLE) or Wi-Fi to track patient movement and activity. However, conventional DL models for localisation are often too resource-intensive for microcontroller-based edge devices. Recent advances in KD, and quantization provide pathways for developing compact, high-performing localisation models that can run directly on ultra-low-power hardware. Suwannaphong et al. [282] optimized TinyML-based indoor localisation by combining KD and quantization on transformer-based multihead dual convolutional self-attention and lightweight Mamba models. Their approach was evaluated on two challenging datasets: a BLE RSSI dataset collected from four residential houses for in-home tracking, and the large-scale UJIIndoorLoc Wi-Fi dataset covering multi-building, multi-floor environments. The framework produced compact yet accurate models suitable for deployment on microcontroller units with only 32–64 KB RAM.

5.6. Summary

Lightweight DL research in remote monitoring relies on lightweight-by-design architectures such as compact CNNs,

Table 15: Summary of lightweight and efficient DL for remote monitoring.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Baltabay et al. [274]	HAR	Wearables	CNN with sensor fusion	✓	✗	✗	✗	✗	✗	MHEALTH	Acc: 98%	58.3K parameters
Yi et al. [275]	HAR	Wearables	CNN	✗	✓	✓	✗	✗	✗	UCI HAR, WISDM	Acc: 95.89%, 95.1% respectively	Model size: 30.76 KB, 45.57 KB respectively
Deng et al. [276]	HAR	Wearables	CNN with depth-wise separable layers	✓	✗	✗	✓	✗	✗	Opportunity, PAMAP2	Acc: 81.01%, 99.32% respectively	Model size: 0.308 MB
Fahim et al. [277]	Elderly care	Wearables	Feed-forward neural network	✓	✗	✗	✓	✗	✗	GOTOV	Acc: 95%	Model size: 0.75 KB
Iswara et al. [278]	Fall detection	Wearables	Feed-forward neural network	✗	✗	✗	✗	✗	✓	Kaggle fall detection dataset	Acc: 97%	53.8% parameter reduction
Sun et al. [279]	BP monitoring	Wearables	ResNet20	✗	✓	✓	✗	✗	✗	MIMIC-IV	Acc: 3% drop than baseline	Model size reduction: 72%
Zhao et al. [280]	Motor assessment	Videos	Spatial-temporal graph convolutional network	✓	✗	✗	✓	✗	✗	Data of CP patients collected at Gillette Children’s Specialty Healthcare	Acc: upto 75.5%	Inference speed: upto 3× faster
Zhang et al. [281]	Emotion recognition	EEG and peripheral physiological signals	GRU	✓	✗	✗	✓	✗	✗	DEAP, MAHNOB-HCI	Acc: improved by 3.74%, 1.74% respectively	Compact student
Suwannaphong et al. [282]	Indoor localisation	Wireless RSSI	Mamba	✗	✗	✓	✓	✗	✗	BLE RSSI, UJIIndoorLoc	Acc: 93.47%	Model size: 13 KB

depthwise-separable CNNs, and GRUs, and compression strategies including pruning, quantization, low-rank factorization, and KD.

6. Lightweight and efficient DL for clinical decision-support

Clinical decision-support systems are central to modern healthcare, providing clinicians with evidence-based insights derived from EHRs, medical images, and multimodal patient data. These systems enhance diagnostic accuracy, enable personalized treatment planning, and improve patient outcomes. However, conventional DL models that drive many clinical decision-support applications often require substantial computational resources, hindering real-time use in hospital settings and limiting deployment in low-resource or point-of-care environments. Recent advances emphasize lightweight and efficient strategies such as pruning, quantization, and KD to reduce model complexity while preserving predictive performance. A consolidated summary of representative studies, their modalities, datasets, and efficiency techniques is presented in Table 16.

6.1. Clinical NLP

Clinical NLP powers applications such as automated clinical note analysis, adverse event detection, and predictive modeling from EHRs. While LLMs like BERT and BioBERT reach strong performance, their large parameter counts and inference costs challenge real-time decision support. Recent work therefore focuses on compression via KD and lightweight adaptations that retain accuracy while enabling practical deployment. Hasan et al. [283] proposed a distillation framework that combines ensemble learning with model compression. An ensemble of BERT-based models trained on MIMIC-III for mortality

and length-of-stay prediction was distilled into a compact DistilBERT student. The student retained 97% of the ensemble’s performance with a 70% parameter reduction and faster inference, demonstrating suitability for efficient, scalable decision support from complex patient notes. Du and Hu [284] introduced SqueezeBioBERT, a compact variant of BioBERT produced through transformer-aware distillation. Optimized for domain adaptation and efficiency, SqueezeBioBERT retained over 95% of its teacher’s performance across named entity recognition, relation extraction, and clinical question answering, while being 4.2× smaller. Using MIMIC-III for mortality prediction, Liu et al. [285] showed that condensing networks, pruning neurons, modifying LSTMs, and 8-bit quantization can jointly compress models and improve accuracy.

6.2. Medical visual question answering

Medical visual question answering (Med-VQA) combines medical imaging and clinical text so that systems can answer natural language queries about radiology and pathology images. Although multi-modal medical LLMs enable this capability, they are often too computationally expensive for real-time or resource-constrained clinical environments. Nathani et al. [286] proposed a distillation framework for multi-modal medical LLMs based on a Mixture-of-Experts (MoE) and Expert-of-Experts design. The MedMoE teacher integrates domain-specific experts for joint image–text processing. A compact student retains essential components like embeddings, self-attention, and early normalization while reducing expert sizes for efficiency. Training uses a composite loss with cross-entropy, KL-divergence distillation, and cosine embedding to transfer semantic alignment and representational richness. On Path-VQA, SLAKE, and VQA-RAD, the student achieved high accuracy and strong generalization across discriminative and generative tasks, highlighting the promise of distilled multi-

modal LLMs for scalable Med-VQA in resource-limited settings.

6.3. Summary

Lightweight DL research for clinical decision-support relies on compact transformer architectures such as DistilBERT, and SqueezeBioBERT, and KD and quantization.

7. Lightweight and efficient DL in image reconstruction and restoration

Medical image reconstruction and restoration are critical for improving diagnostic quality, reducing scan times, and enabling low-dose imaging. However, DL models designed for these tasks are often computationally heavy, limiting deployment where speed, efficiency, and hardware constraints are paramount. Recent work emphasizes lightweight architectures accelerate inference while maintaining high fidelity in reconstructed or restored images. A concise summary of representative approaches, datasets, and efficiency techniques is presented in Table 17.

Murugesan et al. [287] proposed KD-MRI, a KD framework tailored for MRI reconstruction. The method integrates attention-based feature distillation with imitation loss so a compact student mimics a high-capacity teacher built on a deep cascade of CNNs. Evaluated on cardiac, brain, and knee MRI under acceleration factors of 4 $\times$, 5 $\times$, and 8 $\times$, the distilled model on the knee dataset achieved a 65% parameter reduction, $\sim 2\times$ faster CPU inference, and $\sim 1.5\times$ faster GPU inference with negligible quality loss. Ablation analyses and comparisons to alternative distillation schemes demonstrated robustness, and the framework was further extended to MRI super-resolution while preserving accuracy and substantially reducing computation. These findings highlight how distillation-driven lightweight models can accelerate MRI workflows, reduce hardware burden, and enable real-time reconstruction in clinical practice.

7.1. Summary

Lightweight DL research in medical image reconstruction and restoration relies on compact CNN cascades optimized for efficiency, with KD used to transfer representational power from deeper teacher networks into lightweight student models.

8. Lightweight and efficient DL in genomics and precision medicine

Genomics and precision medicine are rapidly transforming healthcare by enabling the discovery of regulatory mechanisms, characterization of disease pathways, and development of personalized treatments. DL is effective for modeling DNA sequences, inferring gene regulatory networks, and predicting therapeutic responses, but these models often incur heavy computational costs that hinder clinical deployment. Recent research addresses this with pruning, quantization, and KD to reduce complexity while preserving accuracy. A concise summary of representative studies, modalities, datasets, and efficiency techniques is presented in Table 18.

8.1. DNA sequence analysis

Accurate modeling of DNA sequences underpins tasks such as motif discovery, transcription factor binding prediction, and regulatory mechanism analysis. To scale these tasks efficiently, lightweight approaches focus on pruning, self-distillation, and compact transformers. Luo et al. [288] introduced DeepPrune, a CNN-based framework for DNA–protein binding prediction that progressively prunes uninformative kernels. Built on the DeepBind architecture with convolutional motif scanning, global max pooling, and a novel mask layer, DeepPrune achieves both computational efficiency and biological interpretability. Validated on simulated motif-insertion datasets and 690 ENCODE ChIP-seq datasets, DeepPrune significantly improved motif inference and binding site prediction compared to baseline CNNs, demonstrating that pruning can enhance both accuracy and generalizability. To further improve efficiency, Yu et al. [289] proposed FinDNA, a self-distillation–based self-supervised framework that combines masked nucleotide modeling with contrastive learning across augmented sequence views. Using a ChordMixer teacher–student design, FinDNA captures both contextual and distributional properties of DNA while being highly compact requiring only 22% of DNABERT-2’s parameters and 5% of NT-500M-human’s. Despite this reduction, FinDNA outperformed larger models on 20 benchmark tasks, highlighting how parameter-efficient architectures can achieve SOTA accuracy while remaining cost-effective for practical genomic applications. Makhdoomi and Ghosh [290] developed TinyDNABERT, a compact transformer-based DNA language model adapted from BERT-Tiny and pre-trained on the human genome with 6-mer tokenization. With only 2 hidden layers, 128 hidden dimensions, and 2 attention heads, TinyDNABERT is over 100 $\times$ smaller than DNABERT yet achieves comparable performance on promoter prediction, splice site detection, and transcription factor binding site classification. Further optimized with LoRA applied to attention and feed-forward layers, the model drastically reduces training and inference costs, enabling deployment on modest GPUs and edge devices without sacrificing downstream accuracy.

8.2. Gene regulatory network inference

Inferring gene regulatory networks (GRNs) is fundamental for understanding transcriptional regulation, cellular differentiation, and disease mechanisms. However, GRN inference from high-dimensional expression data is computationally challenging, particularly when integrating prior biological knowledge or modeling causal relationships. Lightweight and efficient DL frameworks have recently emerged as powerful tools to address these challenges while maintaining scalability and biological interpretability. Peng et al. [291] proposed KINDLE, a three-stage knowledge-guided distillation framework for prior-free GRN inference. A hierarchical attention–based teacher model integrates temporal dynamics and prior constraints, and its insights are distilled into a compact student that operates solely on expression data. KINDLE, especially its Gaussian RBF variant, achieved state-of-the-art AUROC,

Table 16: Summary of lightweight and efficient DL for clinical decision-support.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Hasan et al. [283]	Clinical NLP	EHR	DistilBERT	✓	✗	✗	✓	✗	✗	MIMIC-III	Retains 97% of teacher’s performance	70% parameter reduction
Du & Hu [284]	Clinical NLP	Biomedical text	SqueezeBioBERT	✓	✗	✗	✓	✗	✗	BioASQ 4b, BioASQ 5b, BioASQ 6b	Retains 95% of teacher’s performance	4.2× smaller than teacher
Liu et al. [285]	Clinical NLP	EHR	Modified RNN	✗	✗	✓	✗	✗	✗	MIMIC-III	AUROC: 82%	90% reduction in memory usage
Nathani et al. [286]	Med-VQA	Multimodal	MoE (StableLM)	✓	✗	✗	✓	✗	✗	Path-VQA, SLAKE, VQA-RAD	Retains 97.12%, 97.67%, 97.57% of the teacher’s performance respectively	0.5B parameters

Table 17: Summary of lightweight and efficient DL for image reconstruction and restoration.

Study	Category	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
				Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Murugesan et al. [287]	MRI reconstruction	Cascade of CNN	✓	✗	✗	✓	✗	✗	ACDC, MRBrainS, knee MRI data	PSNR: 31.01 ± 3.27, 38.78 ± 1.24, 34.71 ± 3.44 respectively	49K parameters

AUPRC, and F1 scores on four benchmark mouse differentiation datasets, outperforming both expression-only and prior-guided methods. By enabling efficient inference without reliance on curated priors, KINDLE not only reduces computational cost but also facilitates the discovery of novel regulatory interactions. Complementing distillation-based approaches, Li et al. [292] introduced EfficientNet-resDDSC, a hybrid model for causal GRN inference from single-cell RNA-seq data. The framework combines EfficientNet-B0 with residual blocks and depthwise separable dilated convolutions to expand receptive fields while preserving computational efficiency. Tested across four single-cell datasets, the model demonstrated superior accuracy and robustness in noisy, high-dimensional settings. Its application to breast cancer GRNs further identified key regulators of BRCA1, underscoring the framework’s potential for both methodological advancement and translational impact in oncology.

8.3. Precision medicine

Precision medicine aims to tailor treatment strategies to individual patients by integrating clinical, imaging, and molecular data. DL models have shown promise for predictive oncology, but their large size and sensitivity to noise limit their use in clinical workflows where robustness, interpretability, and computational efficiency are essential. Lightweight DL frameworks are increasingly being explored to address these challenges and enable scalable deployment of predictive tools. Han et al. [293] proposed progressive distill (PD), a DL framework that integrates multi-iteration KD with model noise injection to enhance robustness in predicting clinical response to conversion therapy in advanced gastric cancer. The approach leverages repeated teacher–student distillation cycles to progressively refine the student model, while noise injection improves generalization to heterogeneous patient populations. Applied to a retrospective

cohort of 140 patients, PD significantly outperformed baseline CNNs, conventional distillation methods, and even clinician assessments, demonstrating its value as a non-invasive tool for personalized oncology treatment planning.

8.4. Summary

Lightweight DL research in genomics and precision medicine relies on compact CNNs, efficient transformer variants as well on selective use of KD and pruning.

9. Discussion

Lightweight and efficient DL has rapidly evolved from an optional optimization into a critical enabler of real-world healthcare AI. The surveyed literature shows that efficiency can be achieved through architectural design and model compression techniques. These approaches are not only technical achievements but also clinical necessities, since real-time decision-making and scalable deployment are essential in healthcare.

Architectural innovations such as MobileNet, EfficientNet, ShuffleNet, GhostNet, and MobileViT highlight how design principles like depthwise separable convolutions, compound scaling, channel shuffling, and hybrid CNN–transformer integration enable strong performance with reduced computational cost. The analysis of these models shows that efficiency and accuracy are not mutually exclusive but exist on a spectrum. Models like GhostNet and ShuffleNet provide excellent efficiency, while architectures such as EfficientNet-B7 and TinyViT deliver superior accuracy but at higher computational expense. This demonstrates that deployment choices must be context-driven, depending on whether latency, energy consumption, or accuracy is the highest priority in a given clinical scenario.

Table 18: Summary of lightweight and efficient DL in genomics and precision medicine.

Study	Category	Modality	Model	Lightweight by Design	Model Compression Techniques					Dataset(s)	Performance	Remarks
					Pruning	Quant.	KD	NAS	Low-Rank Fact.			
Luo et al. [288]	DNA sequence analysis	DNA-protein binding	CNN	✗	✓	✗	✗	✗	✗	TRANSFAC database, ChIP-seq ENCODE	AUC: 0.8849	Fewer kernels
Yu et al. [289]	DNA sequence analysis	DNA sequences	Self-supervised transformer based model	✓	✗	✗	✓	✗	✗	Genomic benchmarks, Genome Understanding Evaluation, MTcDNA	Acc: surpasses DNABERT-2 by 6.78% on an average on Genome Understanding Evaluation	22% parameters of DNABERT-2
Makhdoomi & Ghosh [290]	DNA sequence analysis	Human genome	TinyDNABERT	✓	✗	✗	✗	✗	✗	Genome Understanding Evaluation	Acc: 65.780%	100× smaller than DNABERT
Peng et al. [291]	GRN inference	Gene expression	Attention network	✓	✗	✗	✓	✗	✗	4 mouse differentiation datasets provided by BEELINE	F1: 52.9%, 73.1%, 76.3%, 87.5%	Compact student
Li et al. [292]	GRN inference	scRNA-seq	EfficientNet with residual blocks and depthwise separable dilated convolutions	✓	✗	✗	✗	✗	✗	scRNA-seq data	F1: 59.6%	Lightweight architecture
Han et al. [293]	Precision medicine	CT	EfficientNet-B7	✓	✗	✗	✓	✗	✗	Data of 140 gastric cancer patient cohort collected at The First Affiliated Hospital of Wenzhou Medical University	Acc: 85.71%	Lightweight architecture

Compression techniques complement lightweight architectural design by allowing large models to be adapted to resource-constrained environments. Pruning, quantization, KD, and low-rank factorization enable real-time inference on mobile and wearable devices. KD, in particular, provides a powerful mechanism for transferring knowledge from large teacher models to compact student models, preserving diagnostic accuracy even in highly constrained settings. NAS extends this paradigm by automating the discovery of efficient designs optimized for hardware and application-specific needs. These compression methods collectively expand the applicability of DL to scenarios where computational resources are scarce.

Figures 6 and 7 provide insight into how lightweight and efficient DL methods are distributed across healthcare domains and methodological strategies. Figure 6 shows that disease detection and classification dominate the literature, underscoring the central role of efficiency in diagnostic imaging and signal-based applications where computational latency directly affects clinical usability. The largest subfields include respiratory, cardiovascular, and dermatological diseases, which align with global disease prevalence and data availability. Meanwhile, smaller but growing portions of work address remote monitoring, genomics and precision medicine, and clinical decision support, demonstrating how efficiency is expanding into continuous and multimodal care. Figure 7 shows that hybrid approaches dominate the landscape, reflecting a clear preference for combining multiple efficiency techniques to jointly optimize accuracy and resource use. Lightweight-by-design alone remains the next most common strategy, whereas pruning, quantization, NAS, low-rank factorization and KD are rarely employed in isolation, underscoring the field’s movement toward integrated, multi-technique optimization.

In summary, lightweight and efficient DL is no longer an auxiliary optimization but a cornerstone of modern healthcare AI.

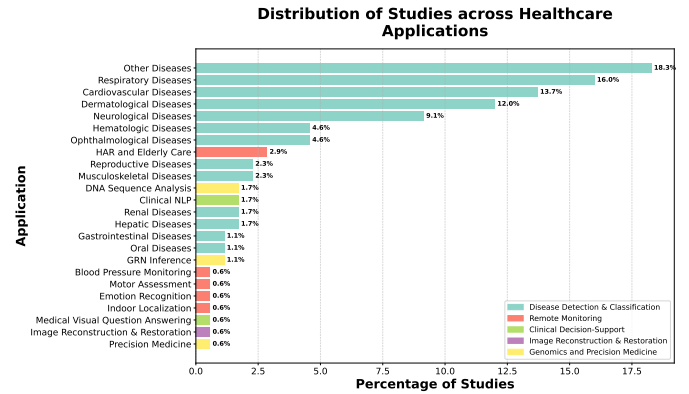


Figure 6: Distribution of studies across healthcare applications showing disease detection and classification as the predominant category, with respiratory, cardiovascular, and dermatological diseases representing the largest subcategories.

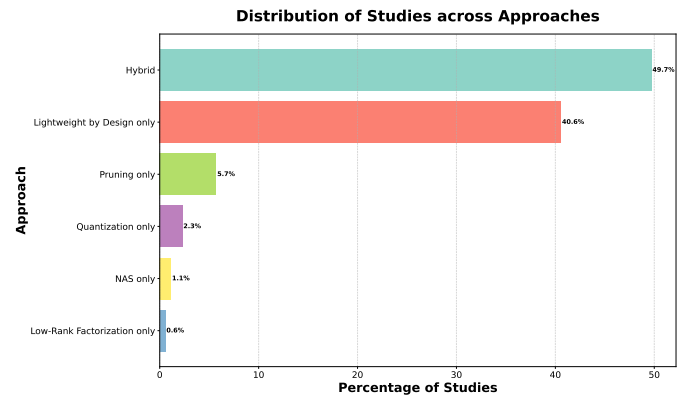


Figure 7: Distribution of studies across lightweight by design and compression approaches. Hybrid and lightweight design approaches dominate, while methods such as pruning, quantization, NAS, and low-rank factorization are rarely used alone.

By uniting architectural advances and model compression techniques, the field is enabling real-time, resource-conscious, and globally scalable systems.

10. Challenges and future directions

Although the rapid growth of lightweight and efficient DL has enabled significant progress in healthcare applications, deploying these models in real-world clinical environments introduces several challenges related to fairness, robustness, interpretability, and regulatory compliance. Addressing these issues requires not only methodological innovations but also careful consideration of clinical needs and hardware constraints.

10.1. Challenges

Key challenges include (1) fairness and generalizability, as compression can amplify performance gaps across hospitals, devices, and under-represented groups, raising concerns about equitable healthcare delivery; (2) robustness versus compactness, since smaller models often lose redundancy, making them more vulnerable to noise, distribution shifts, and adversarial inputs; (3) interpretability and clinical trust, as efficiency-first designs may reduce transparency, slowing clinician adoption and regulatory approval; (4) hardware–software mismatch, because FLOPs or parameter savings do not necessarily translate into real-world latency or energy gains without hardware-aware optimization; (5) sparse training support, as dynamic and train-time pruning face instability and lack of sparse-aware hardware or software; (6) quantization trade-offs, where QAT preserves accuracy but is compute-intensive, while PTQ is faster yet harder to calibrate; (7) distillation limitations, since teacher–student frameworks may transfer biases and struggle with architecture mismatches; and (8) clinical validation at scale, as many lightweight systems lack large-scale, multi-center validation across diverse populations and devices.

10.2. Future directions

Promising future directions include (1) hardware-aware design, emphasizing NAS and training pipelines optimized for latency and energy efficiency on target hardware; (2) compression stacks integrating KD, quantization, and low-rank factorization to achieve efficiency without major accuracy loss; (3) parameter-efficient fine-tuning approaches such as LoRA and QLoRA to adapt large biomedical models efficiently before distilling them into compact versions; (4) privacy-preserving learning through federated and on-device personalization for wearables and IoMT systems; (5) domain-informed modeling leveraging clinical and physics priors particularly in medical imaging reconstruction to improve accuracy under resource constraints; (6) evaluation standards that include energy and latency metrics alongside accuracy, with explainability and uncertainty quantification for regulatory acceptance; and (7) multi-center validation across diverse healthcare settings to establish generalizability and clinical trust.

11. Conclusion

Lightweight and efficient DL has become a cornerstone of modern healthcare AI, enabling scalable, real-time, and resource-conscious intelligence across clinical domains. This survey highlighted advances in architectural design such as MobileNet, EfficientNet, ShuffleNet, GhostNet, and hybrid CNN–transformer models alongside model compression methods including pruning, quantization, KD, low-rank factorization, and NAS. Together, these innovations balance accuracy, latency, and energy efficiency, making AI deployment feasible even in resource-limited environments. Applications in disease detection, remote monitoring, medical image reconstruction, and genomics demonstrate that efficiency is both a technical and clinical necessity. Moving forward, the field must integrate hardware-aware optimization, fairness, interpretability, and privacy-preserving design to realize the full potential of lightweight and globally accessible AI-driven healthcare. Although this survey provides comprehensive coverage of lightweight architectures and model-compression techniques, some areas remain beyond its current scope. These include compression of large biomedical foundation models, and the establishment of standardized efficiency benchmarks aligned with real-world clinical and hardware constraints.

Declaration of competing interest

The authors declare that they have no competing interests.

Data availability

No data was used for the research described in the article.

References

- [1] X. Chen, X. Wang, K. Zhang, K.-M. Fung, T. C. Thai, K. Moore, R. S. Mannel, H. Liu, B. Zheng, Y. Qiu, Recent advances and clinical applications of deep learning in medical image analysis, *Medical image analysis* 79 (2022) 102444.
- [2] H. Hussain, P. Tamizharasan, C. Rahul, Design possibilities and challenges of dnn models: a review on the perspective of end devices, *Artificial Intelligence Review* 55 (7) (2022) 5109–5167.
- [3] G. Menghani, Efficient deep learning: A survey on making deep learning models smaller, faster, and better, *ACM Computing Surveys* 55 (12) (2023) 1–37.
- [4] E. Barbierato, A. Gatti, Toward green ai: A methodological survey of the scientific literature, *IEEE Access* 12 (2024) 23989–24013.
- [5] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, H. Adam, *MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications* (2017).
- [6] K. Dong, C. Zhou, Y. Ruan, Y. Li, MobileNetV2 Model for Image Classification, in: *2020 2nd International Conference on Information Technology and Computer Application (ITCA)*, IEEE, Guangzhou, China, 2020, pp. 476–480.
- [7] B. Koonce, *Convolutional Neural Networks with Swift for Tensorflow: Image Recognition and Dataset Categorization*, Apress, Berkeley, CA, 2021.
- [8] D. Qin, C. Lechner, M. Delakis, M. Fornoni, S. Luo, F. Yang, W. Wang, C. Banbury, C. Ye, B. Akin, V. Aggarwal, T. Zhu, D. Moro, A. Howard, *MobileNetV4 – Universal Models for the Mobile Ecosystem* (2024).
- [9] M. Tan, Q. V. Le, *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks* (2019).

- [10] M. N. Ab Wahab, A. Nazir, A. T. Zhen Ren, M. H. Mohd Noor, M. F. Akbar, A. S. A. Mohamed, Efficientnet-Lite and Hybrid CNN-KNN Implementation for Facial Expression Recognition on Raspberry Pi, *IEEE Access* 9 (2021) 134065–134080.
- [11] X. Zhang, X. Zhou, M. Lin, J. Sun, ShuffleNet: An extremely efficient convolutional neural network for mobile devices, 2018, pp. 6848–6856.
- [12] N. Ma, X. Zhang, H.-T. Zheng, J. Sun, ShuffleNet V2: Practical Guidelines for Efficient CNN Architecture Design, in: V. Ferrari, M. Hebert, C. Sminchisescu, Y. Weiss (Eds.), *Computer Vision – ECCV 2018*, Vol. 11218, Springer International Publishing, Cham, 2018, pp. 122–138.
- [13] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, K. Keutzer, SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and < 0.5 MB model size, *arXiv preprint arXiv:1602.07360* (2016).
- [14] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, C. Xu, Ghostnet: More features from cheap operations, 2020, pp. 1580–1589.
- [15] G. Huang, S. Liu, L. Van der Maaten, K. Q. Weinberger, Condensenet: An efficient densenet using learned group convolutions, 2018, pp. 2752–2761.
- [16] K. Wu, J. Zhang, H. Peng, M. Liu, B. Xiao, J. Fu, L. Yuan, Tinyvit: Fast pretraining distillation for small vision transformers, *Springer*, 2022, pp. 68–85.
- [17] S. Mehta, M. Rastegari, Mobilevit: light-weight, general-purpose, and mobile-friendly vision transformer, *arXiv preprint arXiv:2110.02178* (2021).
- [18] M. Tan, B. Chen, R. Pang, V. Vasudevan, M. Sandler, A. Howard, Q. V. Le, Mnasnet: Platform-aware neural architecture search for mobile, 2019, pp. 2820–2828.
- [19] A. Gholami, K. Kwon, B. Wu, Z. Tai, X. Yue, P. Jin, S. Zhao, K. Keutzer, Squeezenext: Hardware-aware neural network design, 2018, pp. 1638–1647.
- [20] Z. Yan, X. Li, M. Li, W. Zuo, S. Shan, Shift-net: Image inpainting via deep feature rearrangement, 2018, pp. 1–17.
- [21] H. Chen, Y. Wang, C. Xu, B. Shi, C. Xu, Q. Tian, C. Xu, AdderNet: Do we really need multiplications in deep learning?, 2020, pp. 1468–1477.
- [22] P. K. A. Vasu, J. Gabriel, J. Zhu, O. Tuzel, A. Ranjan, Mobileone: An improved one millisecond mobile backbone, 2023, pp. 7907–7917.
- [23] H. Cheng, M. Zhang, J. Q. Shi, A survey on deep neural network pruning-taxonomy, comparison, analysis, and recommendations (2024). *arXiv:2308.06767*.
- [24] S. Han, H. Mao, W. J. Dally, Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding (2016). *arXiv:1510.00149*.
- [25] L. G. Mugnaini, C. T. Duarte, A. H. R. Costa, A. Jordao, Layer pruning with consensus: A triple-win solution (2024). *arXiv:2411.14345*.
- [26] E. Kurtic, D. Alistarh, Gmp*: Well-tuned gradual magnitude pruning can outperform most bert-pruning methods (2022). *arXiv:2210.06384*.
- [27] J. Back, N. Ahn, J. Kim, Magnitude attention-based dynamic pruning (2023). *arXiv:2306.05056*.
- [28] G. Rosenberg, J. K. Brubaker, M. J. A. Schuetz, E. Y. Zhu, S. Kadioğlu, S. E. Borujeni, H. G. Katzgraber, Scalable iterative pruning of large language and vision models using block coordinate descent (2024). *arXiv:2411.17796*.
- [29] W. Hu, P. Henderson, J. Cano, Ice-pruning: An iterative cost-efficient pruning pipeline for deep neural networks (2025). *arXiv:2505.07411*.
- [30] T. Hoefler, D. Alistarh, T. Ben-Nun, N. Dryden, A. Peste, Sparsity in deep learning: Pruning and growth for efficient inference and training in neural networks (2021). *arXiv:2102.00554*.
- [31] W. Kwon, S. Kim, M. W. Mahoney, J. Hassoun, K. Keutzer, A. Gholami, A fast post-training pruning framework for transformers (2022). *arXiv:2204.09656*.
- [32] I. Lazarevich, A. Kozlov, N. Malinin, Post-training deep neural network pruning via layer-wise calibration (2021). *arXiv:2104.15023*.
- [33] G. C. Marinó, A. Petrini, D. Malchiodi, M. Frasca, Deep neural networks compression: A comparative survey and choice recommendations, *Neurocomputing* 520 (2023) 152–170.
- [34] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, K. Keutzer, A survey of quantization methods for efficient neural network inference, *CoRR abs/2103.13630* (2021). *arXiv:2103.13630*.
- [35] O. Weng, Neural network quantization for efficient inference: A survey (2023). *arXiv:2112.06126*.
- [36] D. Liu, X. Liu, Ternary quantization: A survey (2023). *arXiv:2303.01505*.
- [37] K. Liu, Q. Zheng, K. Tao, Z. Li, H. Qin, W. Li, Y. Guo, X. Liu, L. Kong, G. Chen, Y. Zhang, X. Yang, Low-bit model quantization for deep neural networks: A survey (2025). *arXiv:2505.05530*.
- [38] T. Dettmers, M. Lewis, Y. Belkada, L. Zettlemoyer, Llm.int8(): 8-bit matrix multiplication for transformers at scale (2022). *arXiv:2208.07339*.
- [39] P. Yin, J. Lyu, S. Zhang, S. Osher, Y. Qi, J. Xin, Understanding straight-through estimator in training activation quantized neural nets (2019). *arXiv:1903.05662*.
- [40] Y. Bai, Y.-X. Wang, E. Liberty, Proxquant: Quantized neural networks via proximal operators (2019). *arXiv:1810.00861*.
- [41] M. Nagel, R. A. Amjad, M. van Baalen, C. Louizos, T. Blankevoort, Up or down? adaptive rounding for post-training quantization (2020). *arXiv:2004.10568*.
- [42] S. Jung, C. Son, S. Lee, J. Son, Y. Kwak, J.-J. Han, S. J. Hwang, C. Choi, Learning to quantize deep networks by optimizing quantization intervals with task loss (2018). *arXiv:1808.05779*.
- [43] Z. Wu, S. Wang, J. Zhang, J. Chen, Y. Wang, Fima-q: Post-training quantization for vision transformers by fisher information matrix approximation (2025). *arXiv:2506.11543*.
- [44] A. Moslemi, A. Briskina, Z. Dang, J. Li, A survey on knowledge distillation: Recent advancements, *Machine Learning with Applications* 18 (2024) 100605.
- [45] A. M. Mansourian, R. Ahmadi, M. Ghafouri, A. M. Babaei, E. B. Golezani, Z. Y. Ghamchi, V. Ramezani, A. Taherian, K. Dinashi, A. Miri, S. Kasaei, A comprehensive survey on knowledge distillation (2025). *arXiv:2503.12067*.
- [46] G. Hinton, O. Vinyals, J. Dean, Distilling the knowledge in a neural network (2015). *arXiv:1503.02531*.
- [47] J. Guo, M. Chen, Y. Hu, C. Zhu, X. He, D. Cai, Reducing the teacher-student gap via spherical knowledge distillation (2021). *arXiv:2010.07485*.
- [48] B. B. Sau, V. N. Balasubramanian, Deep model compression: Distilling knowledge from noisy teachers (2016). *arXiv:1610.09650*.
- [49] M. Yuan, B. Lang, F. Quan, Student-friendly knowledge distillation (2023). *arXiv:2305.10893*.
- [50] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, Y. Bengio, Fitnets: Hints for thin deep nets (2015). *arXiv:1412.6550*.
- [51] S. Zagoruyko, N. Komodakis, Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer (2017). *arXiv:1612.03928*.
- [52] Z. Huang, N. Wang, Like what you like: Knowledge distill via neuron selectivity transfer (2017). *arXiv:1707.01219*.
- [53] B. Heo, M. Lee, S. Yun, J. Y. Choi, Knowledge transfer via distillation of activation boundaries formed by hidden neurons (2018). *arXiv:1811.03233*.
- [54] B. Heo, J. Kim, S. Yun, H. Park, N. Kwak, J. Y. Choi, A comprehensive overhaul of feature distillation (2019). *arXiv:1904.01866*.
- [55] C. Shu, Y. Liu, J. Gao, Z. Yan, C. Shen, Channel-wise knowledge distillation for dense prediction (2021). *arXiv:2011.13256*.
- [56] Y. Chen, Y. Xian, A. S. Koepke, Y. Shan, Z. Akata, Distilling audio-visual knowledge by compositional contrastive learning (2021). *arXiv:2104.10955*.
- [57] D. Chen, J.-P. Mei, Y. Zhang, C. Wang, Y. Feng, C. Chen, Cross-layer distillation with semantic calibration (2021). *arXiv:2012.03236*.
- [58] Z. Yang, Z. Li, M. Shao, D. Shi, Z. Yuan, C. Yuan, Masked generative distillation (2022). *arXiv:2205.01529*.
- [59] T. Huang, Y. Zhang, M. Zheng, S. You, F. Wang, C. Qian, C. Xu, Knowledge diffusion for distillation (2023). *arXiv:2305.15712*.
- [60] A. M. Mansourian, A. Jalali, R. Ahmadi, S. Kasaei, Attention-guided feature distillation for semantic segmentation (2025). *arXiv:2403.05451*.
- [61] J. Yim, D. Joo, J. Bae, J. Kim, A gift from knowledge distillation: Fast optimization, network minimization and transfer learning, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 7130–7138.
- [62] S. You, C. Xu, C. Xu, D. Tao, Learning from multiple teacher networks, in: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '17, Association for

- Computing Machinery, New York, NY, USA, 2017, p. 1285–1294.
- [63] C. Zhang, Y. Peng, Better and faster: Knowledge transfer from multiple self-supervised learning tasks via graph distillation for video classification (2018). [arXiv:1804.10069](#).
- [64] J. Xie, B. Shuai, J.-F. Hu, J. Lin, W.-S. Zheng, Improving fast segmentation with teacher-student learning (2018). [arXiv:1810.08476](#).
- [65] F. Tung, G. Mori, Similarity-preserving knowledge distillation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1365–1374.
- [66] L. Liu, Q. Huang, S. Lin, H. Xie, B. Wang, X. Chang, X. Liang, Exploring inter-channel correlation for diversity-preserved knowledge distillation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8271–8280.
- [67] Y. Wang, W. Zhou, T. Jiang, X. Bai, Y. Xu, Intra-class feature variation distillation for semantic segmentation, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part VII 16, Springer, 2020, pp. 346–362.
- [68] L. Song, X. Gong, H. Zhou, J. Chen, Q. Zhang, D. Doermann, J. Yuan, Exploring the knowledge transferred by response-based teacher-student distillation, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 2704–2713.
- [69] Y. Feng, X. Sun, W. Diao, J. Li, X. Gao, Double similarity distillation for semantic image segmentation, IEEE Transactions on Image Processing 30 (2021) 5363–5376.
- [70] C. Wang, Z. Wang, D. Chen, S. Zhou, Y. Feng, C. Chen, Online adversarial knowledge distillation for graph neural networks, Expert Systems with Applications 237 (2024) 121671.
- [71] Q. Guo, X. Wang, Y. Wu, Z. Yu, D. Liang, X. Hu, P. Luo, Online knowledge distillation via collaborative learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 11020–11029.
- [72] X. Zhu, S. Gong, et al., Knowledge distillation by on-the-fly native ensemble, Advances in Neural Information Processing Systems 31 (2018).
- [73] T. Furlanello, Z. Lipton, M. Tschannen, L. Itti, A. Anandkumar, Born again neural networks, in: International Conference on Machine Learning, PMLR, 2018, pp. 1607–1616.
- [74] L. Zhang, J. Song, A. Gao, J. Chen, C. Bao, K. Ma, Be your own teacher: Improve the performance of convolutional neural networks via self distillation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 3713–3722.
- [75] X. Lan, X. Zhu, S. Gong, Self-referenced deep learning, in: Computer Vision–ACCV 2018: 14th Asian Conference on Computer Vision, Perth, Australia, December 2–6, 2018, Revised Selected Papers, Part II 14, Springer, 2019, pp. 284–300.
- [76] T.-B. Xu, C.-L. Liu, Data-distortion guided self-distillation for deep neural networks, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 33, 2019, pp. 5565–5572.
- [77] L. Zhang, C. Bao, K. Ma, Self-distillation: Towards efficient and compact neural networks, IEEE Transactions on Pattern Analysis and Machine Intelligence 44 (8) (2021) 4388–4403.
- [78] H. Mobahi, M. Farajtabar, P. Bartlett, Self-distillation amplifies regularization in hilbert space, Advances in Neural Information Processing Systems 33 (2020) 3351–3361.
- [79] Z. Zhang, M. Sabuncu, Self-distillation as instance-specific label smoothing, Advances in Neural Information Processing Systems 33 (2020) 2184–2195.
- [80] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial nets, Advances in Neural Information Processing Systems 27 (2014).
- [81] H. Chen, Y. Wang, C. Xu, Z. Yang, C. Liu, B. Shi, C. Xu, C. Xu, Q. Tian, Data-free learning of student networks, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 3514–3522.
- [82] J. Ye, Y. Ji, X. Wang, X. Gao, M. Song, Data-free knowledge amalgamation via group-stack dual-gan, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12516–12525.
- [83] G. Fang, J. Song, C. Shen, X. Wang, D. Chen, M. Song, Data-free adversarial distillation, [arXiv preprint arXiv:1912.11006](#) (2019).
- [84] Z. Xu, Y.-C. Hsu, J. Huang, Training shallow and thin networks for acceleration via knowledge distillation with conditional adversarial networks, [arXiv preprint arXiv:1709.00513](#) (2017).
- [85] J. Liu, Y. Chen, K. Liu, Exploiting the ground-truth: An adversarial imitation based knowledge distillation approach for event detection, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 33, 2019, pp. 6754–6761.
- [86] T. Fukuda, M. Suzuki, G. Kurata, S. Thomas, J. Cui, B. Ramabhadran, Efficient knowledge distillation from an ensemble of teachers, in: Interspeech, 2017, pp. 3697–3701.
- [87] Y. e. Lin, S. Yin, Y. Ding, X. Liang, Atmkd: adaptive temperature guided multi-teacher knowledge distillation, Multimedia Systems 30 (5) (2024) 292.
- [88] X. Cheng, Z. Zhang, W. Weng, W. Yu, J. Zhou, De-mkd: Decoupled multi-teacher knowledge distillation based on entropy, Mathematics 12 (11) (2024) 1672.
- [89] S. Gupta, J. Hoffman, J. Malik, Cross modal distillation for supervision transfer, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 2827–2836.
- [90] N. C. Garcia, P. Morerio, V. Murino, Modality distillation with multiple stream networks for action recognition, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 103–118.
- [91] A. Andonian, S. Chen, R. Hamid, Robust cross-modal representation learning with progressive self-distillation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16430–16441.
- [92] S. Lee, B. C. Song, Graph-based knowledge distillation by multi-head attention network (2019). [arXiv:1907.02226](#).
- [93] Q. Wang, W. Yu, L. Che, C. Liu, Z. Zhang, J. Gong, P. Chen, Similarity knowledge distillation with calibrated mask, in: ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024, pp. 5770–5774.
- [94] T. Huang, Y. Zhang, S. You, F. Wang, C. Qian, J. Cao, C. Xu, Masked distillation with receptive tokens (2023). [arXiv:2205.14589](#).
- [95] D. Liu, Y. Zhu, Z. Liu, Y. Liu, C. Han, J. Tian, R. Li, W. Yi, A survey of model compression techniques: past, present, and future, Frontiers in Robotics and AI Volume 12 - 2025 (2025).
- [96] G. Cai, J. Li, X. Liu, Z. Chen, H. Zhang, Learning and compressing: Low-rank matrix factorization for deep neural network compression, Applied Sciences 13 (4) (2023).
- [97] K. Cherukuri, A. Lala, Low-rank matrix approximation for neural network compression (2025). [arXiv:2504.20078](#).
- [98] Z. Zhang, The singular value decomposition, applications and beyond (2015). [arXiv:1510.08532](#).
- [99] K. Cherukuri, A. Lala, Low-rank matrix approximation for neural network compression (2025). [arXiv:2504.20078](#).
- [100] B. Zhang, D. Cheng, Y. Zhang, F. Liu, J. Tian, Lossless model compression via joint low-rank factorization optimization (2024). [arXiv:2412.06867](#).
- [101] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, Lora: Low-rank adaptation of large language models (2021). [arXiv:2106.09685](#).
- [102] T. Dettmers, A. Pagnoni, A. Holtzman, L. Zettlemoyer, Qlora: Efficient finetuning of quantized llms (2023). [arXiv:2305.14314](#).
- [103] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, M.-H. Chen, Dora: Weight-decomposed low-rank adaptation (2024). [arXiv:2402.09353](#).
- [104] M. Valipour, M. Rezagholizadeh, I. Kobayev, A. Ghodsi, Dylora: Parameter efficient tuning of pre-trained models using dynamic search-free low-rank adaptation (2023). [arXiv:2210.07558](#).
- [105] Q. Zhang, M. Chen, A. Bukharin, N. Karampatziakis, P. He, Y. Cheng, W. Chen, T. Zhao, Adalora: Adaptive budget allocation for parameter-efficient fine-tuning (2023). [arXiv:2303.10512](#).
- [106] C. White, M. Safari, R. Sukthankar, B. Ru, T. Elsken, A. Zela, D. Dey, F. Hutter, Neural architecture search: Insights from 1000 papers (2023). [arXiv:2301.08727](#).
- [107] H. Hu, J. Langford, R. Caruana, S. Mukherjee, E. Horvitz, D. Dey, Efficient forward architecture search (2019). [arXiv:1905.13360](#).
- [108] J. Shen, M. Khodak, A. Talwalkar, Efficient architecture search for diverse tasks (2022). [arXiv:2204.07554](#).
- [109] H. Benmezi, K. E. Maghraoui, H. Ouarnoughi, S. Niar, M. Wistuba, N. Wang, A comprehensive survey on hardware-aware neural architecture search (2021). [arXiv:2101.09336](#).

- [110] B. Baker, O. Gupta, N. Naik, R. Raskar, Designing neural network architectures using reinforcement learning (2017). [arXiv:1611.02167](#).
- [111] K. Kandasamy, W. Neiswanger, J. Schneider, B. Póczos, E. Xing, Neural architecture search with bayesian optimisation and optimal transport (2019). [arXiv:1802.07191](#).
- [112] H. Cai, L. Zhu, S. Han, Proxylessnas: Direct neural architecture search on target task and hardware (2019). [arXiv:1812.00332](#).
- [113] B. Zoph, V. Vasudevan, J. Shlens, Q. V. Le, Learning transferable architectures for scalable image recognition (2018). [arXiv:1707.07012](#).
- [114] X. Wan, B. Ru, P. M. Esperança, Z. Li, On redundancy and diversity in cell-based neural architecture search (2022). [arXiv:2203.08887](#).
- [115] Y. Yang, C. Wang, L. Gong, X. Zhou, Fpnet: Customized convolutional neural network for fpga platforms, in: 2019 International Conference on Field-Programmable Technology (ICFPT), 2019, pp. 399–402.
- [116] J. Lin, W.-M. Chen, Y. Lin, J. Cohn, C. Gan, S. Han, Mcunet: Tiny deep learning on iot devices (2020). [arXiv:2007.10319](#).
- [117] B. Wu, X. Dai, P. Zhang, Y. Wang, F. Sun, Y. Wu, Y. Tian, P. Vajda, Y. Jia, K. Keutzer, Fbnet: Hardware-aware efficient convnet design via differentiable neural architecture search (2019). [arXiv:1812.03443](#).
- [118] A. Abdusalomov, S. Mirzakhilov, S. Umirzakova, I. Kalandarov, D. Mirzaaxmedov, A. Meliboev, Y. Im Cho, Optimized lightweight architecture for coronary artery disease classification in medical imaging, *Diagnostics* 15 (4) (2025) 446.
- [119] H. Zhang, F. Tian, Y. Tan, L. Shen, J. Liu, J. Liu, K. Qian, Y. Han, G. Su, B. Hu, et al., An ai-assisted all-in-one integrated coronary artery disease diagnosis system using a portable heart sound sensor with an on-board executable lightweight model, *IEEE Transactions on Mobile Computing* (2025).
- [120] T. Iqbal, A. Khalid, I. Ullah, Explaining decisions of a light-weight deep neural network for real-time coronary artery disease classification in magnetic resonance imaging, *Journal of Real-Time Image Processing* 21 (2) (2024) 31.
- [121] N. Alamatsaz, L. Tabatabaei, M. Yazdchi, H. Payan, N. Alamatsaz, F. Nasimi, A lightweight hybrid cnn-lstm explainable model for ecg-based arrhythmia detection, *Biomedical Signal Processing and Control* 90 (2024) 105884.
- [122] L. Meng, W. Tan, J. Ma, R. Wang, X. Yin, Y. Zhang, Enhancing dynamic ecg heartbeat classification with lightweight transformer model, *Artificial Intelligence in medicine* 124 (2022) 102236.
- [123] F. Jiang, J. Xiao, L. Liu, C. Wang, Dceten: A lightweight ecg automatic classification network based on transformer model, *Digital Communications and Networks* (2024).
- [124] J. Wong, J. Nerbonne, Q. Zhang, Ultra-efficient edge cardiac disease detection towards real-time precision health, *IEEE Access* 12 (2023) 9940–9951.
- [125] X. An, S. Shi, Q. Wang, Y. Yu, Q. Liu, Research on a lightweight arrhythmia classification model based on knowledge distillation for wearable single-lead ecg monitoring systems, *Sensors (Basel, Switzerland)* 24 (24) (2024) 7896.
- [126] A. Faraone, H. Sigurthorsdottir, R. Delgado-Gonzalo, Atrial fibrillation detection on low-power wearables using knowledge distillation, in: 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2021, pp. 6795–6799.
- [127] Y. Wang, L. Sun, Energy-efficient dynamic sensor time series classification for edge health devices, *Computer Methods and Programs in Biomedicine* 254 (2024) 108268.
- [128] L. Xiaolin, R. C. Panicker, B. Cardiff, D. John, Multistage pruning of cnn based ecg classifiers for edge devices, in: 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2021, pp. 1965–1968.
- [129] M. Gu, Y. Zhang, Y. Wen, G. Ai, H. Zhang, P. Wang, G. Wang, A lightweight convolutional neural network hardware implementation for wearable heart rate anomaly detection, *Computers in Biology and Medicine* 155 (2023) 106623.
- [130] H. D. M. Ribeiro, A. Arnold, J. P. Howard, M. J. Shun-Shin, Y. Zhang, D. P. Francis, P. B. Lim, Z. Whinnett, M. Zolgharni, Ecg-based real-time arrhythmia monitoring using quantized deep neural networks: A feasibility study, *Computers in Biology and Medicine* 143 (2022) 105249.
- [131] J. Lu, D. Liu, X. Cheng, L. Wei, A. Hu, X. Zou, An efficient unstructured sparse convolutional neural network accelerator for wearable ecg classification device, *IEEE Transactions on Circuits and Systems I: Regular Papers* 69 (11) (2022) 4572–4582.
- [132] S. Ran, X. Yang, M. Liu, Y. Zhang, C. Cheng, H. Zhu, Y. Yuan, Homecare-oriented ecg diagnosis with large-scale deep neural network for continuous monitoring on embedded devices, *IEEE Transactions on Instrumentation and Measurement* 71 (2022) 1–13.
- [133] S. B. Song, J. W. Nam, J. H. Kim, Nas-ppg: Ppg-based heart rate estimation using neural architecture search, *IEEE Sensors Journal* 21 (13) (2021) 14941–14949.
- [134] S. Mukhopadhyay, S. Dey, A. Ghose, P. Singh, P. Dasgupta, Generating tiny deep neural networks for ecg classification on micro-controllers, in: 2023 IEEE International Conference on Pervasive Computing and Communications Workshops, IEEE, 2023, pp. 392–397.
- [135] P. Lu, Y. Gao, H. Xi, Y. Zhang, C. Gao, B. Zhou, H. Zhang, L. Chen, X. Mao, Kecnet: a light neural network for arrhythmia classification based on knowledge reinforcement, *Journal of Healthcare Engineering* 2021 (1) (2021) 6684954.
- [136] Y. Cao, T. Wei, B. Zhang, N. Lin, J. J. Rodrigues, J. Li, D. Zhang, Ml-net: Multi-channel lightweight network for detecting myocardial infarction, *IEEE Journal of Biomedical and Health Informatics* 25 (10) (2021) 3721–3731.
- [137] B. B. Borah, K. Das, U. Hazarika, S. Roy, A deep knowledge distillation healthcare framework for detection of multi-label myocardial infarction from multi-lead ecg signals, *SN Computer Science* 6 (1) (2024) 14.
- [138] S. B. Shuvo, S. N. Ali, S. I. Swapnil, M. S. Al-Rakhami, A. Gumaei, Cardioxnet: A novel lightweight deep learning framework for cardiovascular disease classification using heart sound recordings, *IEEE Access* 9 (2021) 36955–36967.
- [139] L. Zhu, W. Qiu, Y. Ma, F. Tian, M. Sun, Z. Wang, K. Qian, B. Hu, Y. Yamamoto, B. W. Schuller, Lepcnet: A lightweight end-to-end pcg classification neural network model for wearable devices, *IEEE Transactions on Instrumentation and Measurement* 73 (2023) 1–11.
- [140] Y. Al-Issa, A. M. Alqudah, A lightweight hybrid deep learning system for cardiac valvular disease classification, *Scientific Reports* 12 (1) (2022) 14297.
- [141] S. Sreeram, B. Kainz, Sculpting efficiency: Pruning medical imaging models for on-device inference, *arXiv preprint arXiv:2309.05090* (2023).
- [142] H. I. Hussein, A. O. Mohammed, M. M. Hassan, R. J. Mstafa, Lightweight deep cnn-based models for early detection of covid-19 patients from chest x-ray images, *Expert Systems with Applications* 223 (2023) 119900.
- [143] S. Karakanis, G. Leontidis, Lightweight deep learning models for detecting covid-19 from chest x-ray images, *Computers in biology and medicine* 130 (2021) 104181.
- [144] A. M. Alqudah, S. Qazan, A. Alqudah, Automated systems for detection of covid-19 using chest x-ray images and lightweight convolutional neural networks (2020).
- [145] S. R. Nayak, J. Nayak, U. Sinha, V. Arora, U. Ghosh, S. C. Satapathy, An automated lightweight deep neural network for diagnosis of covid-19 from chest x-ray images, *Arabian journal for science and engineering* 48 (8) (2023) 11085–11102.
- [146] N. Sriwiboon, Efficient and lightweight cnn model for covid-19 diagnosis from ct and x-ray images using customized pruning and quantization techniques, *Neural Computing and Applications* 37 (18) (2025) 13059–13078.
- [147] A. BabaAhmadi, S. Khalafi, M. ShariatPanahi, M. Ayati, Designing an improved deep learning-based model for covid-19 recognition in chest x-ray images: a knowledge distillation approach, *Iran Journal of Computer Science* 7 (2) (2024) 177–187.
- [148] X. He, S. Wang, G. Ying, J. Zhang, X. Chu, Efficient multi-objective evolutionary 3d neural architecture search for covid-19 detection with chest ct scans, *arXiv preprint arXiv:2101.10667* (2021).
- [149] S. Hassantabar, N. Stefano, V. Ghanakota, A. Ferrari, G. N. Nicola, R. Bruno, I. R. Marino, K. Hamidouche, N. K. Jha, Coviddeep: Sars-cov-2/covid-19 test based on wearable medical sensors and efficient neural networks, *IEEE Transactions on Consumer Electronics* 67 (4) (2021) 244–256.
- [150] S. B. Nettur, S. Karpurapu, U. Nettur, L. S. Gajja, S. Myneni, A. Dusi, L. Posham, Lightweight weighted average ensemble model for pneumonia detection in chest x-ray images, *arXiv preprint arXiv:2501.16249*

- (2025).
- [151] R. Soni, J. Guan, G. Avinash, V. R. Saripalli, Hmc: A hybrid reinforcement learning based model compression for healthcare applications, in: 2019 IEEE 15th International Conference on Automation Science and Engineering (CASE), IEEE, 2019, pp. 146–151.
 - [152] E. C. Too, D. G. Mwathi, L. K. Gitonga, P. Mwaka, S. Kinyori, An x-ray image-based pruned dense convolution neural network for tuberculosis detection, *Computer Methods and Programs in Biomedicine Update* 6 (2024) 100169.
 - [153] J. Tian, Y. Zhang, J. Lei, C. Sun, G. Hu, Lightweight classification network for pulmonary tuberculosis based on ct images, *Journal of Artificial Intelligence and Technology* 3 (1) (2023) 25–31.
 - [154] S. Agarwal, K. Arya, Y. K. Meena, Cnn-o-elmnet: Optimized lightweight and generalized model for lung disease classification and severity assessment, *IEEE Transactions on Medical Imaging* (2024).
 - [155] Z. Gao, Y. Tian, S.-C. Lin, J. Lin, A ct image classification network framework for lung tumors based on pre-trained mobilenetv2 model and transfer learning, and its application and market analysis in the medical field, *arXiv preprint arXiv:2501.04996* (2025).
 - [156] A. S. Sakr, Automatic detection of various types of lung cancer based on histopathological images using a lightweight end-to-end cnn approach, in: 2022 20th International Conference on Language Engineering (ES-OLEC), Vol. 20, IEEE, 2022, pp. 141–146.
 - [157] M. A. Pavel, R. Islam, S. B. Babor, R. Mehadi, R. Khan, Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant, *Plos one* 19 (11) (2024) e0306441.
 - [158] S. S. Alif, N. A. Promise, F. A. Abid, A. N. Zereen, Dcsnet: A lightweight knowledge distillation-based model with explainable ai for lung cancer diagnosis from histopathological images, *arXiv preprint arXiv:2505.09334* (2025).
 - [159] Z. Zheng, H. Yao, C. Lin, K. Huang, L. Chen, Z. Shao, H. Zhou, G. Zhao, Kd_convnext: knowledge distillation-based image classification of lung tumor surgical specimen sections, *Frontiers in Genetics* 14 (2023) 1254435.
 - [160] S. Khan, F. Siddiqui, M. A. Ahad, Bridging efficiency and interpretability: Explainable ai for multi-classification of pulmonary diseases utilizing modified lightweight cnns, *Image and Vision Computing* 158 (2025) 105553.
 - [161] T. Sanida, M. Dasygenis, A novel lightweight cnn for chest x-ray-based lung disease identification on heterogeneous embedded system, *Applied Intelligence* 54 (6) (2024) 4756–4780.
 - [162] S. Asif, M. Zhao, F. Tang, Y. Zhu, Lwse: a lightweight stacked ensemble model for accurate detection of multiple chest infectious diseases including covid-19, *Multimedia Tools and Applications* 83 (8) (2024) 23967–24003.
 - [163] M. A. Asham, A. A. Al-Shargabi, R. Al-Sabir, I. Meftah, A lightweight deep learning model with knowledge distillation for pulmonary diseases detection in chest x-rays, *Multimedia Tools and Applications* (2024) 1–29.
 - [164] M. M. Kabir, M. Mridha, A. Rahman, M. A. Hamid, M. M. Monowar, Detection of covid-19, pneumonia, and tuberculosis from radiographs using ai-driven knowledge distillation, *Heliyon* 10 (5) (2024).
 - [165] B. S. N. Galih, L. Novamizanti, F. Akhyar, Identification of lung disease via x-ray images using knowledge distillation and vision transformer, in: 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA), IEEE, 2024, pp. 1234–1238.
 - [166] R. Aldamani, D. A. Abuhani, T. Shanableh, Lungvision: X-ray imagery classification for on-edge diagnosis applications, *Algorithms* 17 (7) (2024) 280.
 - [167] T. Wanasinghe, S. Bandara, S. Madusanka, D. Meedeniya, M. Bandara, I. D. L. T. Díez, Lung sound classification with multi-feature integration utilizing lightweight cnn model, *IEEE Access* 12 (2024) 21262–21276.
 - [168] X. Lin, H. Cheng, Y. Lu, H. Luo, H. Li, Y. Qian, L. Zhou, L. Zhang, M. Wang, Contactless sleep apnea detection in snoring signals using hybrid deep neural networks targeted for embedded hardware platform with real-time applications, *Biomedical Signal Processing and Control* 77 (2022) 103765.
 - [169] J. Acharya, A. Basu, Deep neural network for respiratory sound classification in wearable devices enabled by patient specific model tuning, *IEEE transactions on biomedical circuits and systems* 14 (3) (2020) 535–544.
 - [170] L. Ma, F. Zhang, End-to-end predictive intelligence diagnosis in brain tumor using lightweight neural network, *Applied Soft Computing* 111 (2021) 107666.
 - [171] S. U. R. Khan, M. Zhao, Y. Li, Detection of mri brain tumor using residual skip block based modified mobilenet model, *Cluster Computing* 28 (4) (2025) 248.
 - [172] I. Pacal, O. Celik, B. Bayram, A. Cunha, Enhancing efficientnetv2 with global and efficient channel attention mechanisms for accurate mri-based brain tumor classification, *Cluster Computing* 27 (8) (2024) 11187–11212.
 - [173] S. Muksimova, S. Umirzakova, S. Mardieva, N. Iskhakova, M. Sultanov, Y. Im Cho, A lightweight attention-driven yolov5m model for improved brain tumor detection, *Computers in Biology and Medicine* 188 (2025) 109893.
 - [174] M. Sivakumar, S. Padmapriya, Improving efficiency of brain tumor classification models using pruning techniques, *Current Medical Imaging* 20 (1) (2024) e15734056303076.
 - [175] R. Anantathanavit, F. H. Raswa, T. Thaipisutikul, J.-C. Wang, Lightweight brain tumor diagnosis via knowledge distillation, in: 2024 International Conference on Multimedia Analysis and Pattern Recognition (MAPR), IEEE, 2024, pp. 1–6.
 - [176] X. Shi, Y. Li, J. Cheng, J. Bai, G. Zhao, Y.-W. Chen, Knowledge distillation using segment anything to u-net model for lightweight high accuracy medical image segmentation, in: 2024 IEEE 13th Global Conference on Consumer Electronics (GCCE), IEEE, 2024, pp. 1073–1076.
 - [177] Z. Zheng, G. Kang, Model compression with nas and knowledge distillation for medical image segmentation, in: 2021 4th International Conference on Data Science and Information Technology, 2021, pp. 173–176.
 - [178] D. Wu, J. Tao, Z. Qin, R. A. Mumtaz, J. Qin, L. Yu, J. Courtney, Lightnet: a novel lightweight convolutional network for brain tumor segmentation in healthcare, *IEEE Journal of Biomedical and Health Informatics* (2023).
 - [179] A. A. A. El-Latif, S. A. Chelloug, M. Alabdulhafith, M. Hammad, Accurate detection of alzheimer’s disease using lightweight deep learning model on mri data, *Diagnostics* 13 (7) (2023) 1216.
 - [180] M.-K. Yi, S. O. Hwang, Detection of freezing of gait in parkinson’s disease using a lightweight attention-based deep learning model with pruning and dynamic quantization techniques on wearable devices, Available at SSRN 4915041.
 - [181] A. A. J. Karim, K. H. M. Asad, M. G. R. Alam, Larger models yield better results? streamlined severity classification of adhd-related concerns using bert-based knowledge distillation, *PLoS One* 20 (2) (2025) e0315829.
 - [182] F. Tempel, E. A. F. Ihlen, I. Strümke, Lightweight neural architecture search for cerebral palsy detection, in: 2025 IEEE Symposium on Computational Intelligence in Health and Medicine Companion (CIHM Companion), IEEE, 2025, pp. 1–5.
 - [183] S. Qiu, W. Wang, H. Jiao, Lightseizurenet: A lightweight deep learning model for real-time epileptic seizure detection, *IEEE Journal of Biomedical and Health Informatics* 27 (4) (2022) 1845–1856.
 - [184] S. Baghersalimi, A. Amirshahi, F. Forooghifar, T. Teijeiro, A. Amini-far, D. Atienza, Many-to-one knowledge distillation of real-time epileptic seizure detection for low-power wearable internet of things systems, *arXiv preprint arXiv:2208.00885* (2022).
 - [185] H.-S. Choi, Electromyogram (emg) signal classification based on lightweight neural network with fpgas for wearable application, *Electronics* 12 (6) (2023) 1398.
 - [186] V. Karthikeyan, M. N. Kishore, S. Sajin, End-to-end light-weighted deep-learning model for abnormality classification in kidney ct images, *International Journal of Imaging Systems and Technology* 34 (1) (2024) e23022.
 - [187] S. Asif, M. Zhao, X. Chen, Y. Zhu, Stonenet: An efficient lightweight model based on depthwise separable convolutions for kidney stone detection from ct images, *interdiscip sci.* 2023 dec; 15 (4): 633–652.
 - [188] H. Cho, I. Song, J. Jang, Y. Yoo, A lightweight deep learning network on a system-on-chip for wearable ultrasound bladder volume measurement systems: Preliminary study, *Bioengineering* 10 (5) (2023) 525.
 - [189] S. Chakrabarti, U. S. Rao, Lightweight neural network for smart diagnosis of cholangiocarcinoma using histopathological images, *Scientific Reports* 13 (1) (2023) 18854.
 - [190] Y. Xu, Y. Wang, Q. Chen, H. Hu, H. Huang, L. Lin, Y.-W. Chen, J. Li,

- H. Lin, Fpga oriented lightweight deep learning inference for liver cancer segmentation, in: 2024 IEEE International Symposium on Biomedical Imaging (ISBI), IEEE, 2024, pp. 1–5.
- [191] P. Xu, K. Kim, J. Koh, D. Wu, Y. R. Lee, S. Y. Park, W. Y. Tak, H. Liu, Q. Li, Efficient knowledge distillation for liver ct segmentation using growing assistant network, *Physics in Medicine & Biology* 66 (23) (2021) 235005.
- [192] A. Batool, Y.-C. Byun, Lightweight efficientnetb3 model based on depthwise separable convolutions for enhancing classification of leukemia white blood cell images, *IEEE access* 11 (2023) 37203–37215.
- [193] P. K. Das, B. Nayak, S. Meher, A lightweight deep learning system for automatic detection of blood cancer, *Measurement* 191 (2022) 110762.
- [194] K. D. Prakash, J. Khan, K. Kim, Lightweight and efficient yolov8 with residual attention mechanism for precise leukemia detection and classification, *IEEE Access* (2024).
- [195] S. Khan, M. Sajjad, N. Abbas, J. Escorcia-Gutierrez, M. Gamarra, K. Muhammad, Efficient leukocytes detection and classification in microscopic blood images using convolutional neural network coupled with a dual attention network, *Computers in Biology and Medicine* 174 (2024) 108146.
- [196] A. Hosseini, M. A. Eshraghi, T. Taami, H. Sadeghsalehi, Z. Hoseinzadeh, M. Ghaderzadeh, M. Rafiee, A mobile application based on efficient lightweight cnn model for classification of b-all cancer from non-cancerous cells: A design and implementation study, *Informatics in Medicine Unlocked* 39 (2023) 101244.
- [197] K. Datta Gupta, D. K. Sharma, S. Ahmed, H. Gupta, D. Gupta, C.-H. Hsu, A novel lightweight deep learning-based histopathological image classification model for iomt, *Neural Processing Letters* 55 (1) (2023) 205–228.
- [198] L. Mei, S. Jin, T. Huang, H. Peng, W. Zha, J. He, S. Zhang, C. Xu, W. Yang, H. Shen, et al., High-accuracy and high-throughput reactive lymphocyte identification using lightweight neural networks, *Biomedical Signal Processing and Control* 97 (2024) 106722.
- [199] S. B. Nettur, S. Karpurapu, U. Nettur, L. S. Gajja, S. Myneni, A. Dusi, L. Posham, UltralightsqueezeNet: A deep learning architecture for malaria classification with up to 54x fewer trainable parameters for resource constrained devices, *IEEE Access* (2025).
- [200] K. Aurangzeb, R. S. Alharthi, S. I. Haider, M. Alhussein, An efficient and light weight deep learning model for accurate retinal vessels segmentation, *IEEE Access* 11 (2022) 23107–23118.
- [201] D. Jha, S. Ali, N. K. Tomar, H. D. Johansen, D. Johansen, J. Rittscher, M. A. Riegler, P. Halvorsen, Real-time polyp detection, localization and segmentation in colonoscopy using deep learning, *Ieee Access* 9 (2021) 40496–40510.
- [202] A. Sunija, S. Kar, S. Gayathri, V. P. Gopi, P. Palanisamy, Octnet: A lightweight cnn for retinal disease classification from optical coherence tomography images, *Computer methods and programs in biomedicine* 200 (2021) 105877.
- [203] M. Bhandari, T. B. Shahi, A. Neupane, Evaluating retinal disease diagnosis with an interpretable lightweight cnn model resistant to adversarial attacks, *Journal of Imaging* 9 (10) (2023) 219.
- [204] H. Gavel, C. Azad, Retinal disease detection for low resource devices through pruned and quantized cnns, in: 2025 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS), IEEE, 2025, pp. 1–6.
- [205] A. Al Shafi, M. H. Shawon, N. Shahid, R. Rahman, R. Khan, Enhancing diabetic retinopathy detection through transformer based knowledge distillation and explainable ai, in: 2024 International Joint Conference on Neural Networks (IJCNN), IEEE, 2024, pp. 1–8.
- [206] N. Islam, M. M. H. Jony, E. Hasan, S. Sutradhar, A. Rahman, M. M. Islam, Toward lightweight diabetic retinopathy classification: A knowledge distillation approach for resource-constrained settings, *Applied Sciences* 13 (22) (2023) 12397.
- [207] X. Huang, C. Yao, F. Xu, L. Chen, H. Wang, X. Chen, J. Ye, Y. Wang, Mac-resnet: Knowledge distillation based lightweight multiscale-attention-crop-resnet for eyelid tumors detection and classification, *Journal of Personalized Medicine* 13 (1) (2022) 89.
- [208] A. R. Baig, Q. Abbas, R. Almakki, M. E. Ibrahim, L. AlSuwaidan, A. E. Ahmed, Light-dermo: A lightweight pretrained convolution neural network for the diagnosis of multiclass skin lesions, *Diagnostics* 13 (3) (2023) 385.
- [209] S. Asif, S. U. R. Khan, K. Amjad, M. Awais, Skinc-net: an efficient lightweight deep learning model for multiclass skin lesion classification in dermoscopic images, *Multimedia Tools and Applications* (2024) 1–27.
- [210] L. Hoang, S.-H. Lee, E.-J. Lee, K.-R. Kwon, Multiclass skin lesion classification using a novel lightweight deep learning framework for smart healthcare, *Applied Sciences* 12 (5) (2022) 2677.
- [211] S. Gamil, F. Zeng, M. Alrifayy, M. Asim, N. Ahmad, An efficient adaboost algorithm for enhancing skin cancer detection and classification, *Algorithms* 17 (8) (2024) 353.
- [212] M. N. Alche, D. Acevedo, M. Mejail, Efficientarl: Improving skin cancer diagnoses by combining lightweight attention on efficientnet, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 3354–3360.
- [213] Y. Ding, Z. Yi, M. Li, J. Long, S. Lei, Y. Guo, P. Fan, C. Zuo, Y. Wang, Hi-mvit: A lightweight model for explainable skin disease classification based on modified mobilevit, *Digital Health* 9 (2023) 20552076231207197.
- [214] C. Yuan, D. Zhao, S. S. Agaian, Ucm-net: A lightweight and efficient solution for skin lesion segmentation using mlp and cnn, *Biomedical Signal Processing and Control* 96 (2024) 106573.
- [215] P. Thaker, A. Asif, Knowledge distillation for enabling efficient ai-based skin cancer detection in resource-constrained environments, in: International Conference on Medical Imaging and Computer-Aided Diagnosis, Springer, 2024, pp. 77–84.
- [216] S. Saha, M. M. Hemal, M. Z. A. Eidmum, M. F. Mridha, Knowledge distillation approach for skin cancer classification on lightweight deep learning model, *Healthcare Technology Letters* 12 (1) (2025) e12120.
- [217] N. Islam, K. M. Hasib, F. A. Joti, A. Karim, S. Azam, Leveraging knowledge distillation for lightweight skin cancer classification: Balancing accuracy and computational efficiency, *arXiv preprint arXiv:2406.17051* (2024).
- [218] M. R. Kabir, R. H. Borshon, M. K. Wasi, R. M. Sultan, A. Hosain, R. Khan, Skin cancer detection using lightweight model souping and ensembling knowledge distillation for memory-constrained devices, *Intelligence-Based Medicine* 10 (2024) 100176.
- [219] M. S. Khan, K. N. Alam, A. R. Dhruva, H. Zunair, N. Mohammed, Knowledge distillation approach towards melanoma detection, *Computers in Biology and Medicine* 146 (2022) 105581.
- [220] Y. Wang, Y. Wang, J. Cai, T. K. Lee, C. Miao, Z. J. Wang, Ssd-kd: A self-supervised diverse knowledge distillation method for lightweight skin lesion classification using dermoscopic images, *Medical Image Analysis* 84 (2023) 102693.
- [221] X. Zeng, Z. Ji, H. Zhang, R. Chen, Q. Liao, J. Wang, T. Lyu, L. Zhao, Dsp-kd: dual-stage progressive knowledge distillation for skin disease classification, *Bioengineering* 11 (1) (2024) 70.
- [222] Z. Zhang, B. Lu, Efficient skin lesion segmentation with boundary distillation, *Medical & Biological Engineering & Computing* 62 (9) (2024) 2703–2716.
- [223] S. Medhat, H. Abdel-Galil, A. E. Aboutabl, H. Saleh, Iterative magnitude pruning-based light-version of alexnet for skin cancer classification, *Neural Computing and Applications* 36 (3) (2024) 1413–1428.
- [224] K. Xiang, L. Peng, H. Yang, M. Li, Z. Cao, S. Jiang, G. Qu, A novel weight pruning strategy for light weight neural networks with application to the diagnosis of skin disease, *Applied Soft Computing* 111 (2021) 107707.
- [225] M. Hussain, M. Fiza, A. Khalil, A. A. Siyal, F. A. Dharejo, W. Hyder, A. Guzzo, M. Krichen, G. Fortino, Transfer learning-based quantized deep learning models for nail melanoma classification, *Neural Computing and Applications* 35 (30) (2023) 22163–22178.
- [226] A. M. Almars, Deepgenmon: A novel framework for monkeypox classification integrating lightweight attention-based deep learning and a genetic algorithm, *Diagnostics* 15 (2) (2025) 130.
- [227] D. Biswas, J. Tešić, Binarydnet53: a lightweight binarized cnn for monkeypox virus image classification, *Signal, Image and Video Processing* 18 (10) (2024) 7107–7118.
- [228] A. D. Raha, M. Gain, S. K. Saha, M. S. Rahman, A. Adhikary, R. Debnath, A. K. Bairagi, S. Biswas, Resource-aware monkeypox diagnosis: Leveraging high-capacity and lightweight models with knowledge distillation, in: 2025 17th International Conference on COMMunication Systems and NETWORKS (COMSNETS), IEEE, 2025, pp. 72–77.

- [229] Q. Liu, X. She, Q. Xia, Ai based diagnostics product design for osteosarcoma cells microscopy imaging of bone cancer patients using camobilet v3, *Journal of Bone Oncology* 49 (2024) 100644.
- [230] L. Serrador, F. P. Villani, S. Moccia, C. P. Santos, Knowledge distillation on individual vertebrae segmentation exploiting 3d u-net, *Computerized Medical Imaging and Graphics* 113 (2024) 102350.
- [231] M.-K. Nguyen, V.-T. Huynh, T.-G. Thi Vo, M.-T. Tran, Knowledge distillation for lumbar spine x-ray classification, in: *International Symposium on Information and Communication Technology*, Springer, 2024, pp. 329–342.
- [232] D. Qing, L. Qi, Research on lightweight spine x-ray image segmentation algorithm based on knowledge distillation, in: *Proceedings of the 2024 4th International Conference on Bioinformatics and Intelligent Computing*, 2024, pp. 142–146.
- [233] M. H. K. Mehedi, M. Khandaker, S. Ara, M. A. Alam, M. Mridha, Z. Aung, A lightweight deep learning method to identify different types of cervical cancer, *Scientific Reports* 14 (1) (2024) 29446.
- [234] W. Chen, L. Gao, X. Li, W. Shen, Lightweight convolutional neural network with knowledge distillation for cervical cells classification, *Biomedical Signal Processing and Control* 71 (2022) 103177.
- [235] W. Chen, W. Shen, L. Gao, X. Li, Hybrid loss-constrained lightweight convolutional neural networks for cervical cell classification, *Sensors* 22 (9) (2022) 3272.
- [236] I. Pacal, Maxcervix: A novel lightweight vision transformer-based approach for precise cervical cancer detection, *Knowledge-Based Systems* 289 (2024) 111482.
- [237] J. Kang, N. Li, Cervisegnet-distillplus: an efficient knowledge distillation model for enhancing early detection of cervical cancer pathology, *IEEE Access* 12 (2024) 85134–85149.
- [238] A. Shabir, K. T. Ahmed, M. Mujahid, K. Kanwal, Lwfdtl: lightweight fusion deep transfer learning for oral squamous cell carcinoma diagnosis using histopathological oral mucosa, *Multimedia Tools and Applications* (2024) 1–25.
- [239] V. Latke, V. Narawade, Detection of dental periapical lesions using retinex based image enhancement and lightweight deep learning model, *Image and Vision Computing* 146 (2024) 105016.
- [240] P. Sharma, D. Das, A. Gautam, B. K. Balabantaray, Lpnet: A lightweight cnn with discrete wavelet pooling strategies for colon polyps classification, *International Journal of Imaging Systems and Technology* 33 (2) (2023) 495–510.
- [241] J. Yan, P. Liu, T. Xiong, M. Han, Q. Jia, Y. Gao, Lrcnet: A lightweight rectal cancer t-staging network based on knowledge distillation via a pretrained swin transformer, *Biomedical Signal Processing and Control* 105 (2025) 107696.
- [242] A. Kumar, A. Sharma, V. Bharti, A. K. Singh, S. K. Singh, S. Saxena, Mobihisnet: A lightweight cnn in mobile edge computing for histopathological image classification, *IEEE Internet of Things Journal* 8 (24) (2021) 17778–17789.
- [243] R. O. Ogundokun, S. Misra, A. O. Akinrotimi, H. Ogul, Mobilenet-svm: A lightweight deep transfer learning model to diagnose bch scans for iomt-based imaging sensors, *Sensors* 23 (2) (2023) 656.
- [244] R. K. Samala, H.-P. Chan, L. M. Hadjiiski, M. A. Helvie, C. Richter, K. Cha, Evolutionary pruning of transfer learned deep convolutional neural networks for breast cancer diagnosis in digital breast tomosynthesis, *Physics in Medicine & Biology* 63 (9) (2018) 095005.
- [245] T. Choudhary, V. Mishra, A. Goswami, J. Sarangapani, A transfer learning with structured filter pruning approach for improved breast cancer classification on point-of-care devices, *Computers in Biology and Medicine* 134 (2021) 104432.
- [246] L. Jiang, C. Zhang, H. Zhang, H. Cao, A lightweight spatially-aware classification model for breast cancer pathology images, *Biocybernetics and Biomedical Engineering* 44 (3) (2024) 586–608.
- [247] F. Modu, R. Prasad, F. Aliyu, Lightweight cnn for resource-constrained bcd system using knowledge distillation, *IEEE Access* (2025).
- [248] M. Garifulla, J. Shin, C. Kim, W. H. Kim, H. J. Kim, J. Kim, S. Hong, A case study of quantizing convolutional neural networks for fast disease diagnosis on portable medical devices, *Sensors* 22 (1) (2021) 219.
- [249] H. Yin, B. Mukadam, X. Dai, N. K. Jha, Diabdeep: Pervasive diabetes diagnosis based on wearable medical sensors and efficient neural networks, *IEEE Transactions on Emerging Topics in Computing* 9 (3) (2019) 1139–1150.
- [250] K. Amjad, S. Asif, Z. Waheed, Y. Guo, A novel lightweight deep learning framework with knowledge distillation for efficient diabetic foot ulcer detection, *Applied Soft Computing* 167 (2024) 112296.
- [251] B. A. Sangeroki, T. W. Cenggoro, A fast and accurate model of thoracic disease detection by integrating attention mechanism to a lightweight convolutional neural network, *Procedia Computer Science* 179 (2021) 112–118.
- [252] S. Asif, Q.-u. Ain, R. Al-Sabri, M. Abdullah, Litefusionnet: boosting the performance for medical image classification with an intelligent and lightweight feature fusion network, *Journal of Computational Science* 80 (2024) 102324.
- [253] Q. Zhou, Z. Huang, M. Ding, X. Zhang, Medical image classification using light-weight cnn with spiking cortical model based attention module, *IEEE Journal of Biomedical and Health Informatics* 27 (4) (2023) 1991–2002.
- [254] G. Maheswari, S. Gopalakrishnan, A smart multimodal framework based on squeeze excitation capsule network (secnet) model for disease diagnosis using dissimilar medical images, *International Journal of Information Technology* 17 (1) (2025) 49–67.
- [255] S. Yadav, S. Dhage, Te-capsnet: time efficient capsule network for automatic disease classification from medical images, *Multimedia tools and applications* 83 (16) (2024) 49389–49418.
- [256] F. E. Fernandes, G. G. Yen, Automatic searching and pruning of deep neural networks for medical imaging diagnostic, *IEEE Transactions on Neural Networks and Learning Systems* 32 (12) (2020) 5664–5674.
- [257] A. Shahid, M. M. Arshad, F. Azam, M. W. Anwar, I. Shafi, Improving federated learning efficiency with pruneff: A healthcare perspective, in: *Proceedings of the 2024 13th International Conference on Computing and Pattern Recognition*, 2024, pp. 346–352.
- [258] E. Baccour, A. Erbad, A. Mohamed, M. Hamdi, M. Guizani, Reinforcement learning-based dynamic pruning for distributed inference via explainable ai in healthcare iot systems, *Future Generation Computer Systems* 155 (2024) 1–17.
- [259] D. Qin, J.-J. Bu, Z. Liu, X. Shen, S. Zhou, J.-J. Gu, Z.-H. Wang, L. Wu, H.-F. Dai, Efficient medical image segmentation based on knowledge distillation, *IEEE Transactions on Medical Imaging* 40 (12) (2021) 3820–3831.
- [260] V. Gorade, S. Mittal, D. Jha, U. Bagci, Rethinking intermediate layers design in knowledge distillation for kidney and liver tumor segmentation, in: *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, IEEE, 2024, pp. 1–6.
- [261] Y. Wen, L. Chen, S. Xi, Y. Deng, X. Tang, C. Zhou, Towards efficient medical image segmentation via boundary-guided knowledge distillation, in: *2021 IEEE International Conference on Multimedia and Expo (ICME)*, IEEE Computer Society, 2021, pp. 1–6.
- [262] J. W. Choi, Knowledge distillation from cross teaching teachers for efficient semi-supervised abdominal organ segmentation in ct, in: *MICCAI Challenge on Fast and Low-Resource Semi-supervised Abdominal Organ Segmentation*, Springer, 2022, pp. 101–115.
- [263] G. Sun, H. Shu, F. Shao, T. Racharak, W. Kong, Y. Pan, J. Dong, S. Wang, L.-M. Nguyen, J. Xin, Fkd-med: Privacy-aware, communication-optimized medical image segmentation via federated learning and model lightweighting through knowledge distillation, *Ieee Access* 12 (2024) 33687–33704.
- [264] Y. Gordienko, M. Shulha, Y. Kochura, O. Rokovyi, O. Alienin, V. Taran, S. Stirenko, Ensemble knowledge distillation for edge intelligence in medical applications, in: *Advancements in Knowledge Distillation: Towards New Horizons of Intelligent Systems*, Springer, 2023, pp. 135–168.
- [265] M. Khurshid, Y. Akhter, M. Vatsa, R. Singh, Assistdistil for medical image segmentation, *Biomedical Signal Processing and Control* 97 (2024) 106568.
- [266] Z. Liao, Q. Dong, Y. Ge, W. Liu, H. Chen, Y. Song, Knowledge distillation of attention and residual u-net: Transfer from deep to shallow models for medical image classification, in: *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*, Springer, 2023, pp. 162–173.
- [267] A. A. Malibari, An efficient iot-artificial intelligence-based disease prediction using lightweight cnn in healthcare system, *Measurement: Sensors* 26 (2023) 100695.
- [268] M. P. Singh, J. Singh, V. Ravi, T. J. Alahmadi, B. D. Shivahare, M. Di-

- wakar, M. Tayal, P. Singh, et al., A healthcare system employing lightweight cnn for disease prediction with artificial intelligence, *The Open Public Health Journal* 17 (1) (2024).
- [269] F. M. Alabbasy, A. S. Abohamama, M. F. Alrahmawy, Compressing medical deep neural network models for edge devices using knowledge distillation, *Journal of King Saud University-Computer and Information Sciences* 35 (7) (2023) 101616.
- [270] O. Attallah, M. F. Aslan, K. Sabanci, A framework for lung and colon cancer diagnosis via lightweight deep learning models and transformation methods, *Diagnostics* 12 (12) (2022) 2926.
- [271] M. A. Hasan, F. Haque, S. R. Sabuj, H. Sarker, M. O. F. Goni, F. Rahman, M. M. Rashid, An end-to-end lightweight multi-scale cnn for the classification of lung and colon cancer with xai integration, *Technologies* 12 (4) (2024) 56.
- [272] N. Shahadat, R. Lama, A. Nguyen, Lung and colon cancer detection using a deep ai model, *Cancers* 16 (22) (2024) 3879.
- [273] M. Zhang, M. Li, L. Guo, J. Liu, A low-cost ai-empowered stethoscope and a lightweight model for detecting cardiac and respiratory diseases from lung and heart auscultation sounds, *Sensors* 23 (5) (2023) 2591.
- [274] M. Baltabay, A. Yazici, M. Sterling, E. Ever, Designing efficient and lightweight deep learning models for healthcare analysis, *Neural Processing Letters* 55 (6) (2023) 6947–6977.
- [275] M.-K. Yi, W.-K. Lee, S. O. Hwang, A human activity recognition method based on lightweight feature extraction combined with pruned and quantized cnn for wearable device, *IEEE Transactions on Consumer Electronics* 69 (3) (2023) 657–670.
- [276] S. Deng, J. Chen, D. Teng, C. Yang, D. Chen, T. Jia, H. Wang, Lhar: Lightweight human activity recognition on knowledge distillation, *IEEE Journal of Biomedical and Health Informatics* (2023).
- [277] M. Fahim, S. A. Kazmi, V. Sharma, H. Shin, T. Q. Duong, Edge intelligence: A deep distilled model for wearables to enable proactive eldercare, *IEEE Transactions on Artificial Intelligence* (2025).
- [278] G. N. Iswara, A. G. Putrada, H. H. Nuha, Low-rank factorization for edge computing in fall detection with wearable sensors, in: *2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)*, IEEE, 2024, pp. 549–554.
- [279] B. Sun, S. Bayes, A. M. Abotaleb, M. Hassan, The case for tinyml in healthcare: Cnns for real-time on-edge blood pressure estimation, in: *Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing*, 2023, pp. 629–638.
- [280] P. Zhao, Z. Shen, M. Alencastre-Miranda, D. Whiteman, J. Gervas-Arruga, H. I. Krebs, Distilling knowledge in vision-based human motor assessment for improved accuracy and running efficiency, in: *2024 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, IEEE, 2024, pp. 826–831.
- [281] Y. Zhang, H. Liu, Y. Xiao, M. Amoon, D. Zhang, D. Wang, S. Yang, C. Quek, Llm-enhanced multi-teacher knowledge distillation for modality-incomplete emotion recognition in daily healthcare, *IEEE Journal of Biomedical and Health Informatics* (2024).
- [282] T. Suwannaphong, F. Jovan, I. Craddock, R. McConville, Optimising tinyml with quantization and distillation of transformer and mamba models for indoor localisation on edge devices, *Scientific Reports* 15 (1) (2025) 10081.
- [283] M. J. Hasan, F. Rahman, N. Mohammed, Distilling the knowledge of clinical outcome predictions in large language models for resource constrained healthcare systems, Available at SSRN 4591013 (2023).
- [284] H. G. Du, Y. Hu, Squeezebiobert: Biobert distillation for healthcare natural language processing, in: *International Conference on Computational Data and Social Networks*, Springer, 2020, pp. 193–201.
- [285] D. Liu, N. Sepulveda, M. Zheng, Artificial neural networks condensation: A strategy to facilitate adaption of machine learning in medical settings by reducing computational burden, *arXiv preprint arXiv:1812.09659* (2018).
- [286] M. Nathani, R. Soni, R. Mishra, Knowledge distillation in mixture of experts for multi-modal medical llms, in: *2024 IEEE International Conference on Big Data (BigData)*, IEEE, 2024, pp. 4367–4373.
- [287] B. Murugesan, S. Vijayarangan, K. Sarveswaran, K. Ram, M. Sivaprakasam, Kd-mri: A knowledge distillation framework for image reconstruction and image restoration in mri workflow, in: *Medical imaging with deep learning*, PMLR, 2020, pp. 515–526.
- [288] X. Luo, W. Chi, M. Deng, Deepprune: Learning efficient and interpretable convolutional networks through weight pruning for predicting dna-protein binding, *Frontiers in genetics* 10 (2019) 1145.
- [289] T. Yu, L. Cheng, R. Khalitov, E. B. Olsson, Z. Yang, Self-distillation improves self-supervised learning for dna sequence inference, *Neural Networks* 183 (2025) 106978.
- [290] F. Makhdooni, N. Ghosh, Tinydnabert: An efficient and optimized bert model for dna-language in genome, in: *2024 IEEE 21st India Council International Conference (INDICON)*, IEEE, 2024, pp. 1–6.
- [291] R. Peng, Y. Lu, Q. Sun, Y. Lu, C. Zhang, Z. Liu, J. Wang, Kindle: Knowledge-guided distillation for prior-free gene regulatory network inference, *arXiv preprint arXiv:2505.09664* (2025).
- [292] A. Li, M. Li, R. Fei, S. Mallik, B. Hu, Y. Yu, Efficientnet-resddsc: A hybrid deep learning model integrating residual blocks and dilated convolutions for inferring gene causality in single-cell data, *Interdisciplinary Sciences: Computational Life Sciences* 17 (1) (2025) 166–184.
- [293] S. Han, T. Zhang, W. Deng, S. Han, H. Wu, B. Jiang, W. Xie, Y. Chen, T. Deng, X. Wen, et al., Deep learning progressive distill for predicting clinical response to conversion therapy from preoperative ct images of advanced gastric cancer patients, *Scientific Reports* 15 (1) (2025) 17092.